# Class05: Data Visualization with GGPLOT

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**Class Work:**

## Title Here

### Subtitle

###Subtitle

**Stuff in bold** not in bold *something italics*

Indent text

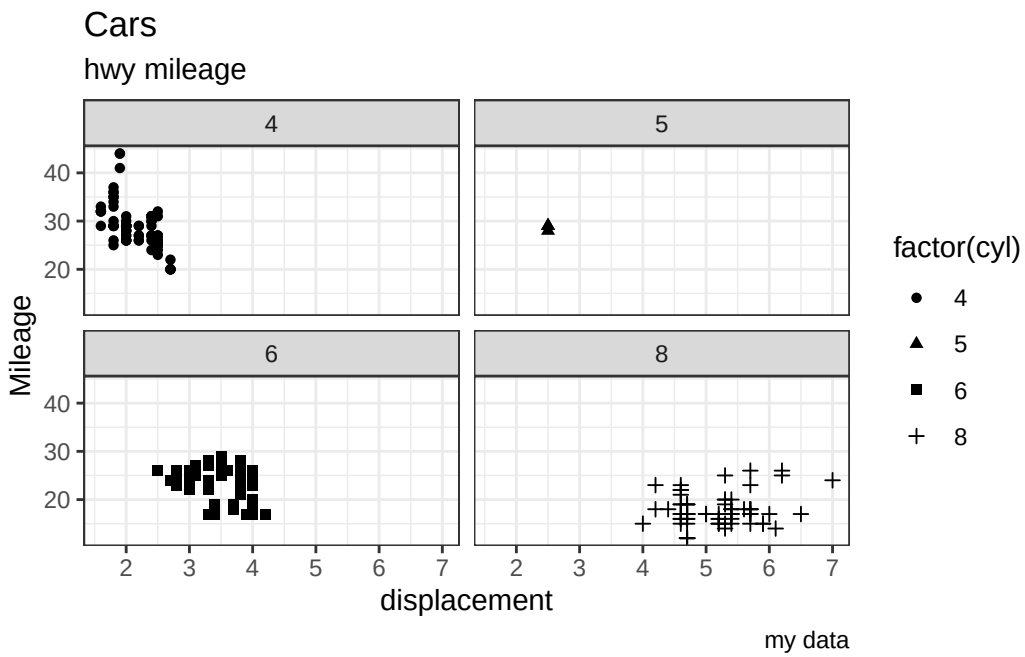
```
#Generating numbers from 1-5 and adding 1 to each number
x<-1:5
x+1
```

```
[1] 2 3 4 5 6
```

Plot from class:

```
#Generating scatter plot using the mpg dataset
library(ggplot2)
ggplot(data=mpg) +
  #Setting variables for axes and grouping
  aes(x=displ, y=hwy, shape=factor(cyl)) +
  #Adding points to the plot
  geom_point() +
  #Split plot into panels by cylinder number
  facet_wrap(~cyl) +
  #Generating simple theme
```

```
  theme_bw() +
  #Generating labels and titles
  labs(title="Cars", subtitle= "hwy mileage",
       caption="my data", x="displacement", y="Mileage")
```




**Q1. For which phases is data visualization important for our scientific workflows?**

All of the above

**Q2. TRUE or FALSE? The ggplot2 package comes already installed with R?**

FALSE

**Part 5**

**Q. Which plot types are typically NOT used to compare distributions of numeric variables?**

Network graphs

**Q. Which statement about data visualization with ggplot2 is incorrect?**

ggplot2 is the only way to create plots in R

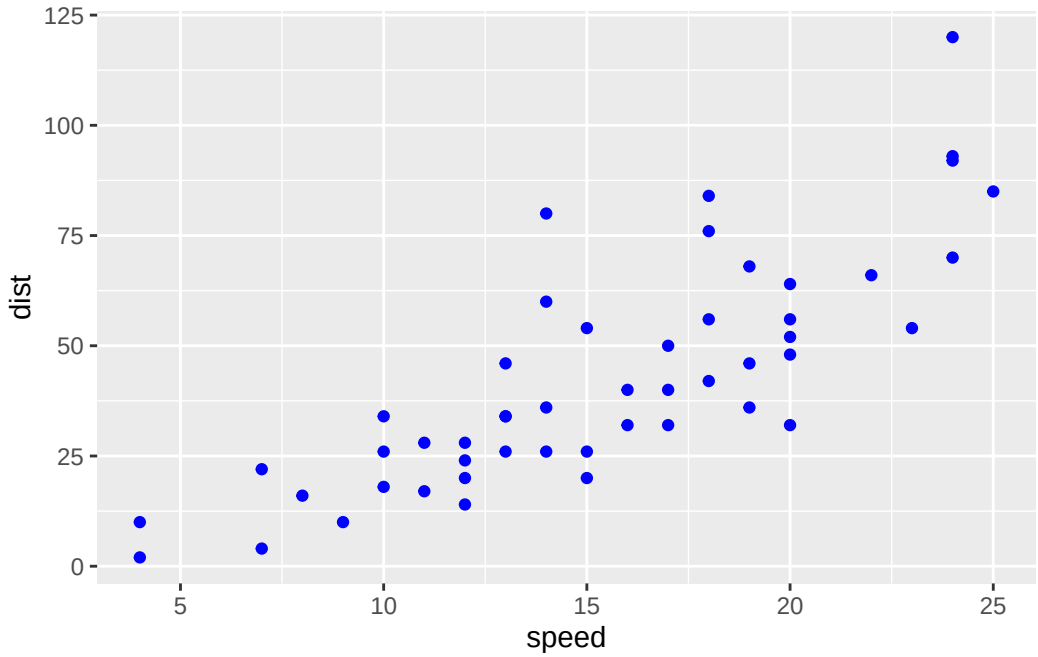
**Part 6. Creating Scatter Plots**

**Q. Which geometric layer should be used to create scatter plots in ggplot2?**

geom_point()

**Generating scatter plot for part 6**

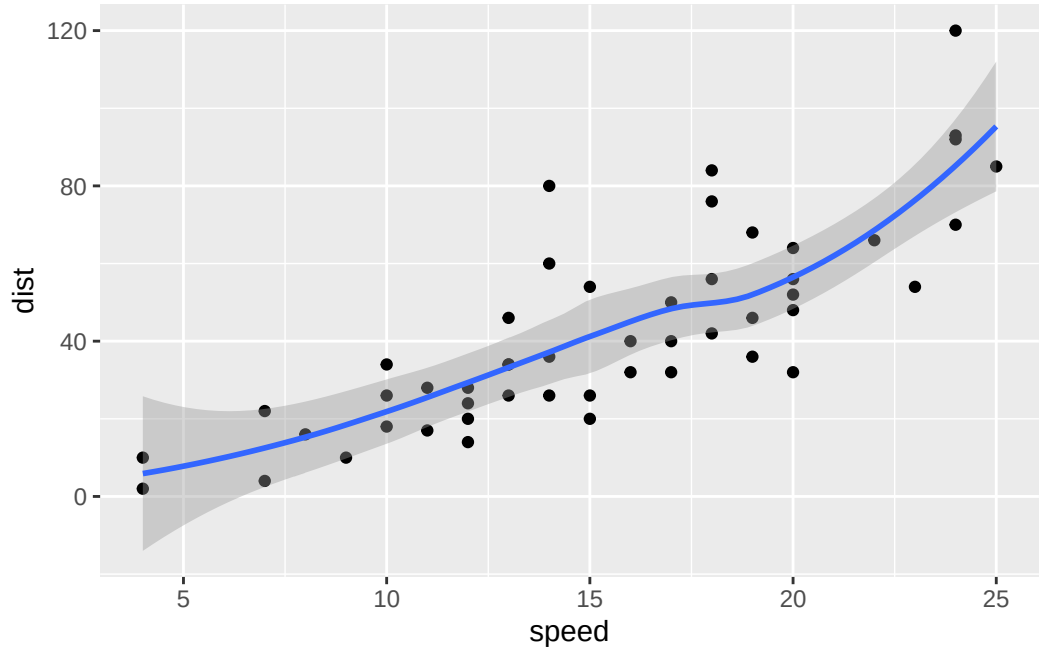
```
#Loading ggplot2 for creating plots
library(ggplot2)
#Create a scatter plot of the cars dataset
ggplot(cars)+
  #Define which variables used for x and y axes and making points blue
  aes(x=speed,y=dist)+geom_point(col="Blue")
```




**Q. In your own RStudio can you add a trend line layer to help show the relationship between the plot variables with the geom_smooth() function?**

```
#Generating a trend line layer to show relationship between the plot variables
library(ggplot2)
ggplot(cars) +
  aes(x = speed, y = dist) +
  geom_point() +
  geom_smooth()
```

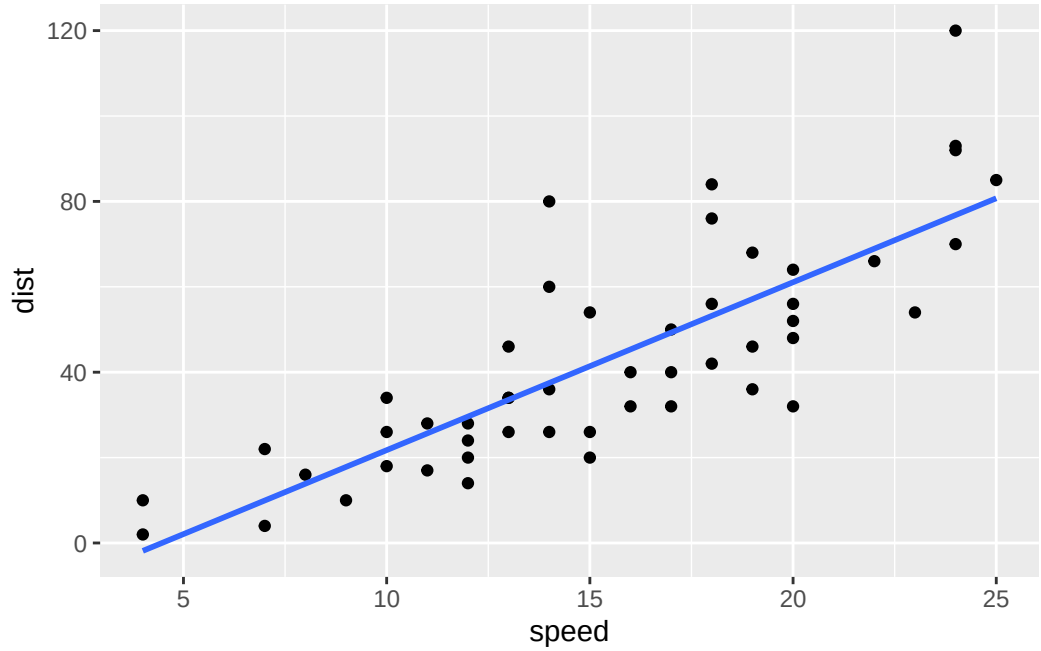
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




**Q. Argue with geom_smooth() to add a straight line from a linear model without the shaded standard error region?**

```
#Generating scatter plot of cars data with linear trend line
library(ggplot2)
ggplot(cars) +
  aes(x = speed, y = dist) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```

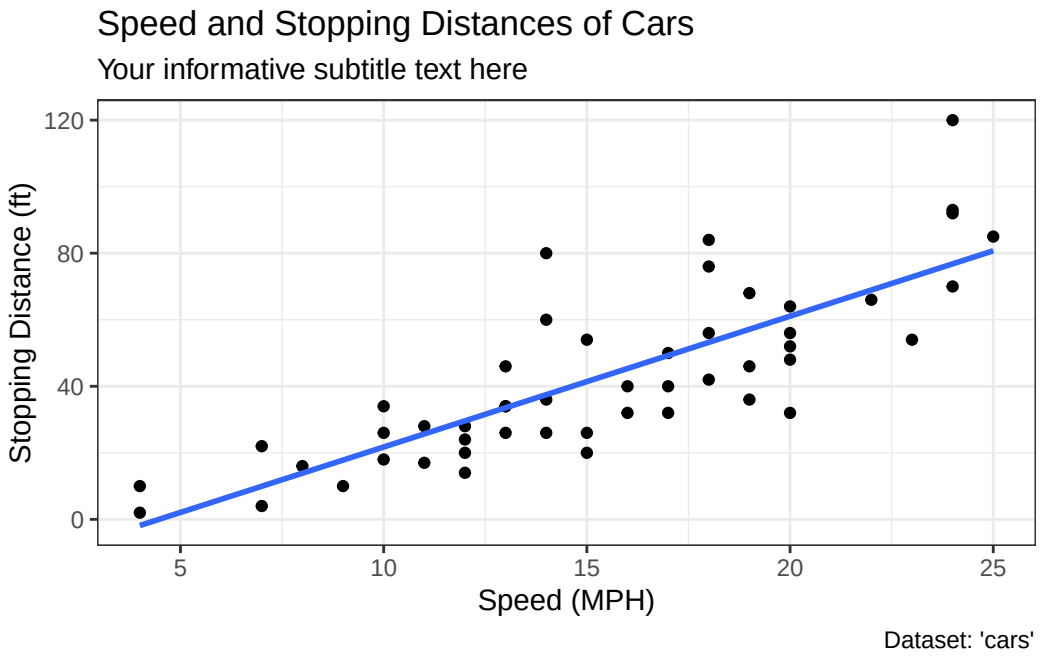
```
`geom_smooth()` using formula = 'y ~ x'
```




**Q. Can you finish this plot by adding various label annotations with the labs() function and changing the plot look to a more conservative "black & white" theme by adding the theme_bw() function:**

```
#Adding various label annotations and changing the plot to black and white theme
library(ggplot2)
ggplot(cars) +
  aes(x = speed, y = dist) +
  geom_point() +
  labs(title="Speed and Stopping Distances of Cars",
       x="Speed (MPH)",
       y="Stopping Distance (ft)",
       subtitle = "Your informative subtitle text here",
       caption="Dataset: 'cars'") +
  geom_smooth(method="lm", se=FALSE) +
  theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'
```




**Adding more plot aesthetics through aes()**

```
#Loading in gene expression dataset from online file
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
#Read the data into R and preview it
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

**Q. Use the nrow() function to find out how many genes are in this dataset. What is your answer?**

```
#Counting total number of genes in the dataset
nrow(genes)
```

```
[1] 5196
```

**Q. Use the colnames() function and the ncol() function on the genes data frame to find out what the column names are (we will need these later) and how many columns there are. How many columns did you find?**

```
#Displaying column names in the dataset
colnames(genes)
```

```
[1] "Gene"       "Condition1" "Condition2" "State"
```

```
#Counting how many columns are present
ncol(genes)
```

```
[1] 4
```

**Q. Use the table() function on the State column of this data.frame to find out how many 'up' regulated genes there are. What is your answer?**

```
#Counting how many genes are in each category
table(genes$State)
```

```
      down unchanging         up
        72       4997        127
```

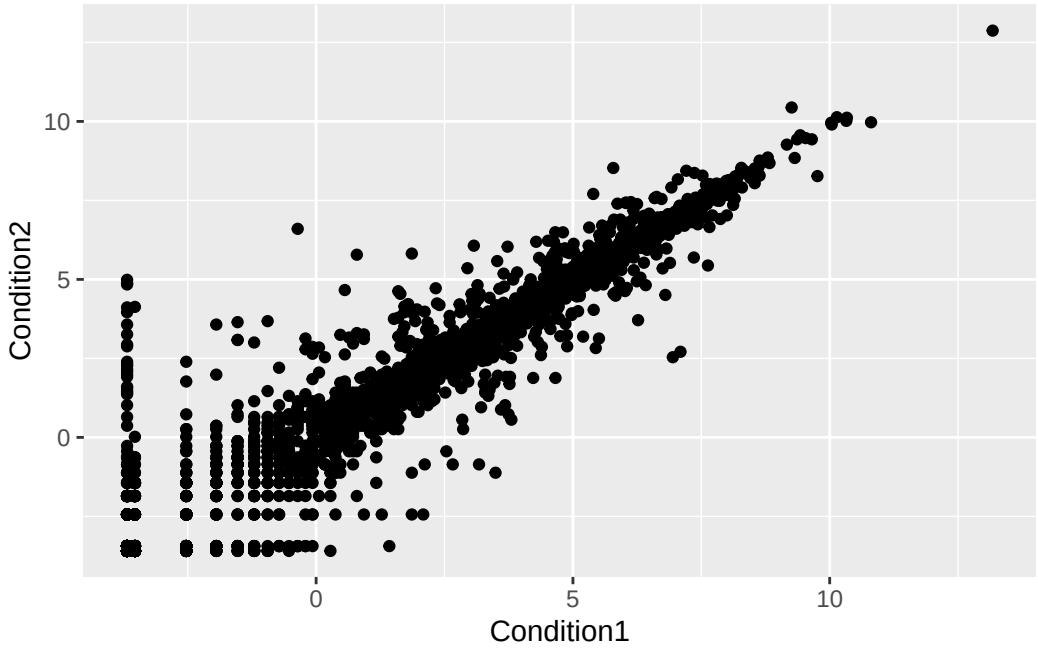
**Q. Using your values above and 2 significant figures. What fraction of total genes is up-regulated in this dataset?**

```
#Calculating the percentage of each gene category
round( table(genes$State)/nrow(genes) * 100, 2)
```

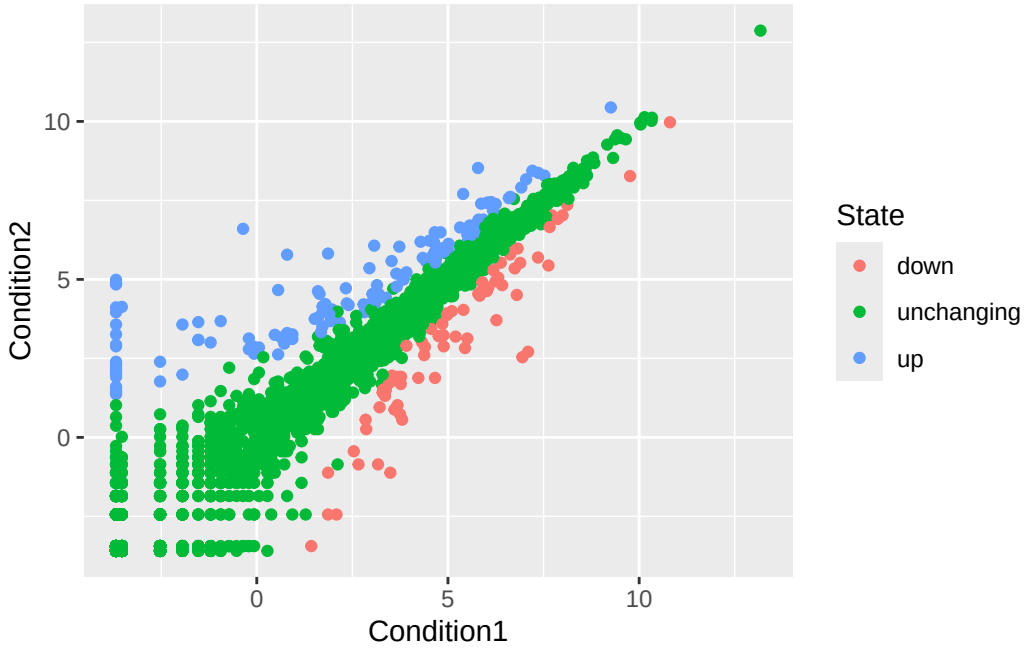
```
      down unchanging         up
      1.39      96.17       2.44
```

**Q. Complete the code below to produce the following plot**

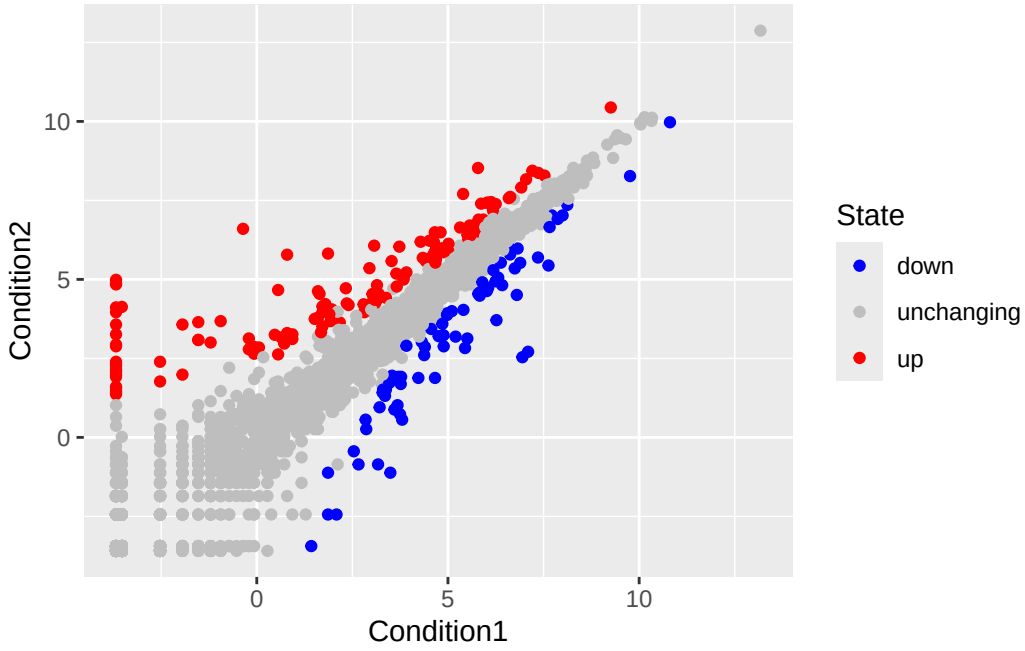
```
#Generating scatter plot of genes data set comparing Condition 1 and Condition 2
ggplot(genes) +
  aes(x=Condition1, y=Condition2) +
  geom_point()
```




```
#Adding State column to see the difference in expression values between conditions
ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point()
```

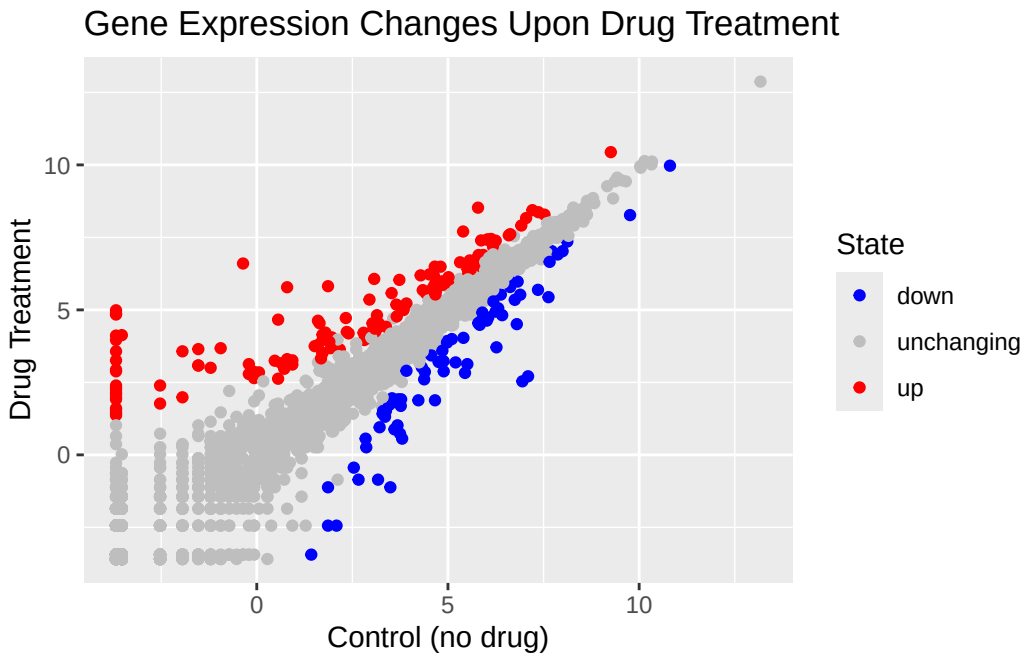



```
#Adding another layer to explicitly specify color scale
ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point() +
  scale_colour_manual( values=c("blue", "gray", "red"))
```




**Q. Nice, now add some plot annotations to the p object with the labs() function**

```
#Adding plot annotations using labs()
ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point() +
  scale_colour_manual( values=c("blue", "gray", "red")) +
  labs(title="Gene Expression Changes Upon Drug Treatment",
       x="Control (no drug)",
       y="Drug Treatment")
```




## Part 7. Going Further

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
```

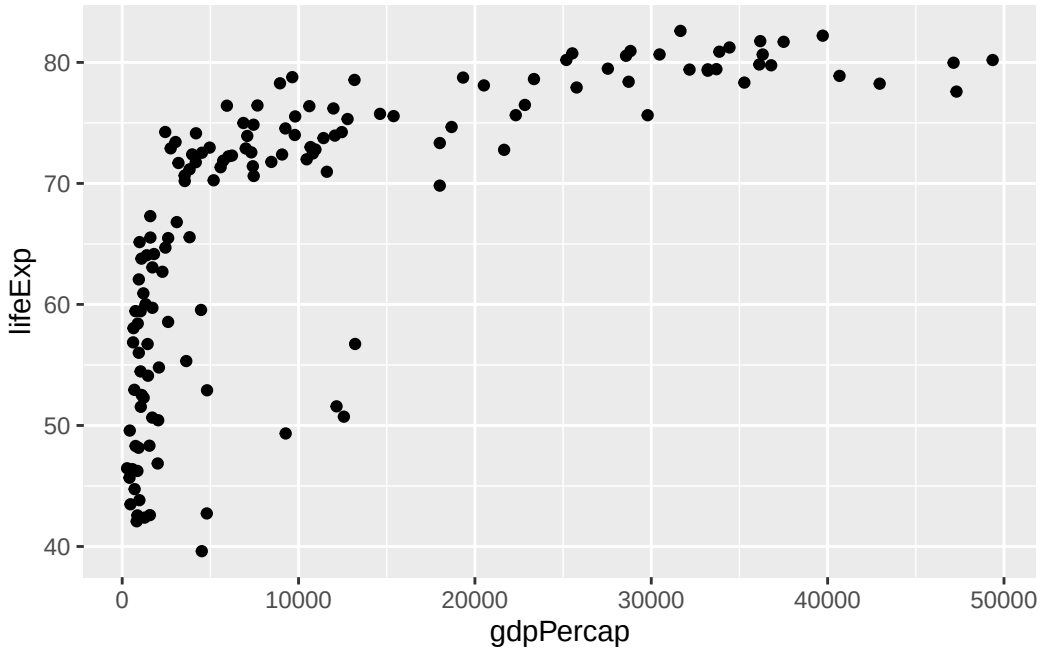
```
# install.packages("dplyr")
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

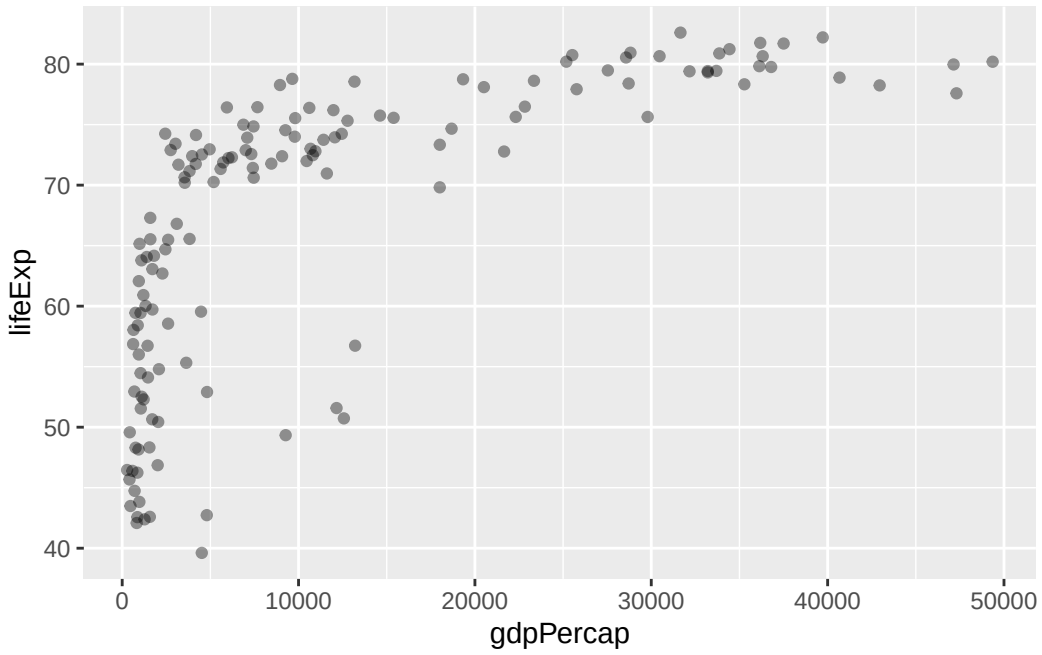
```
gapminder_2007 <- gapminder %>% filter(year==2007)
```

**Q. Complete the code below to produce a first basic scater plot of this gapminder_2007 dataset:**

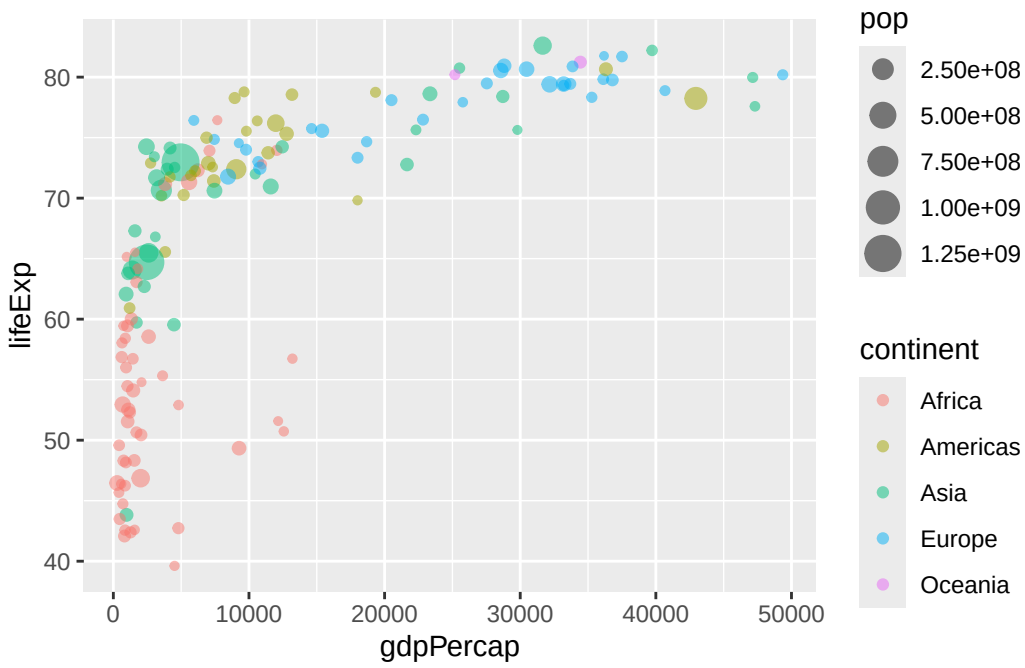
```
#Generating scatter plot of gapminder_2007 dataset
ggplot(gapminder_2007) +
  aes(x=gdpPercap, y=lifeExp) +
  geom_point()
```



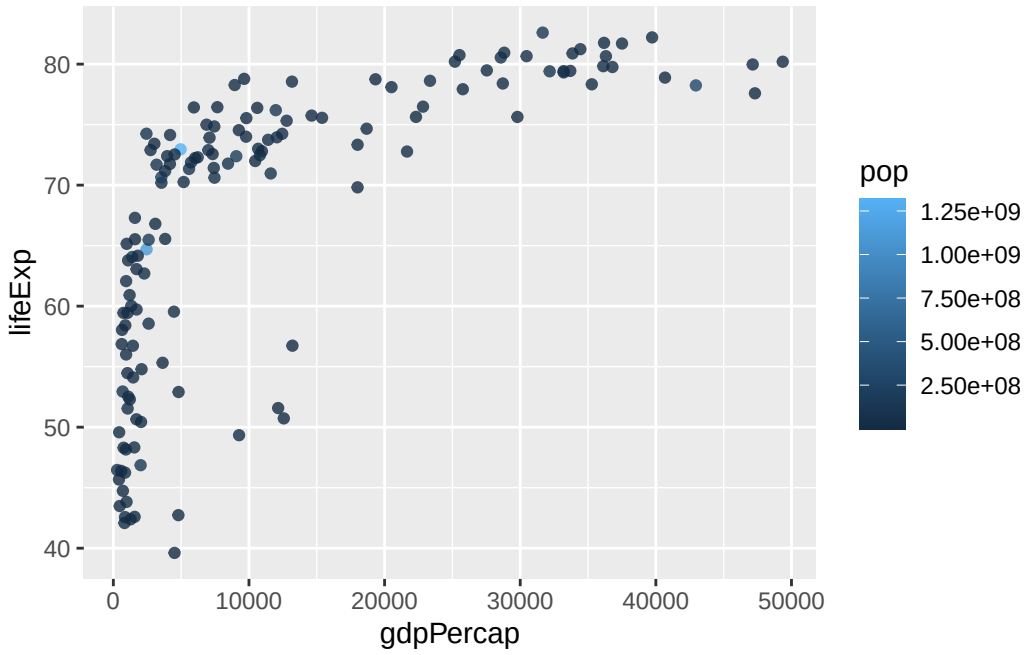

```
#Making the points slightly more transparent with alpha=0.4
ggplot(gapminder_2007) +
  aes(x=gdpPercap, y=lifeExp) +
  geom_point(alpha=0.4)
```



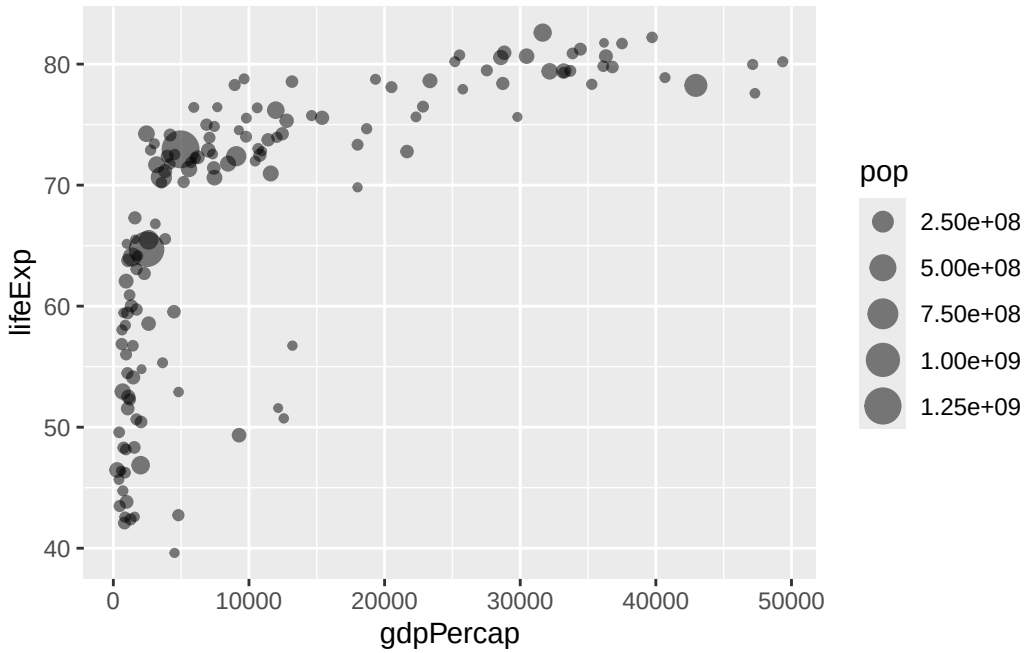

```
#Mapping the continent variable to the point color aesthetic and the population pop
#through the point size argument to aes() to obtain plot with 4 different variables
ggplot(gapminder_2007) +
  aes(x=gdpPercap, y=lifeExp, color=continent, size=pop) +
  geom_point(alpha=0.5)
```



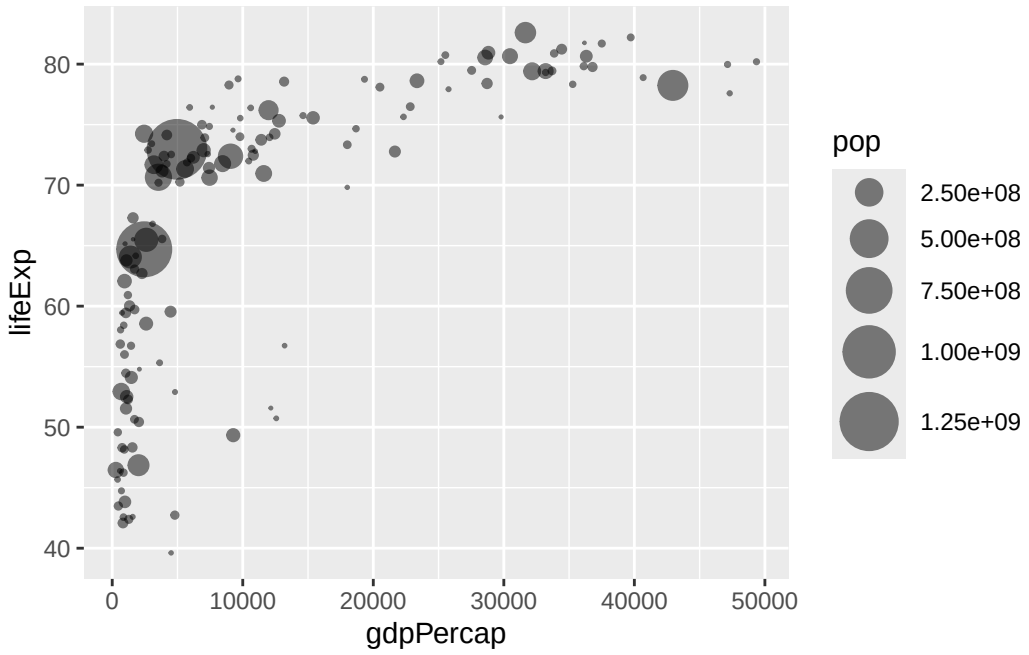

```
#Generating plot and color the point by the numeric variable population pop
ggplot(gapminder_2007) +
  aes(x = gdpPercap, y = lifeExp, color = pop) +
  geom_point(alpha=0.8)
```




```
#Plotting GDP per capita vs. the life expectancy and set the point size
#based on the population of each country
ggplot(gapminder_2007) +
  aes(x = gdpPercap, y = lifeExp, size = pop) +
  geom_point(alpha=0.5)
```

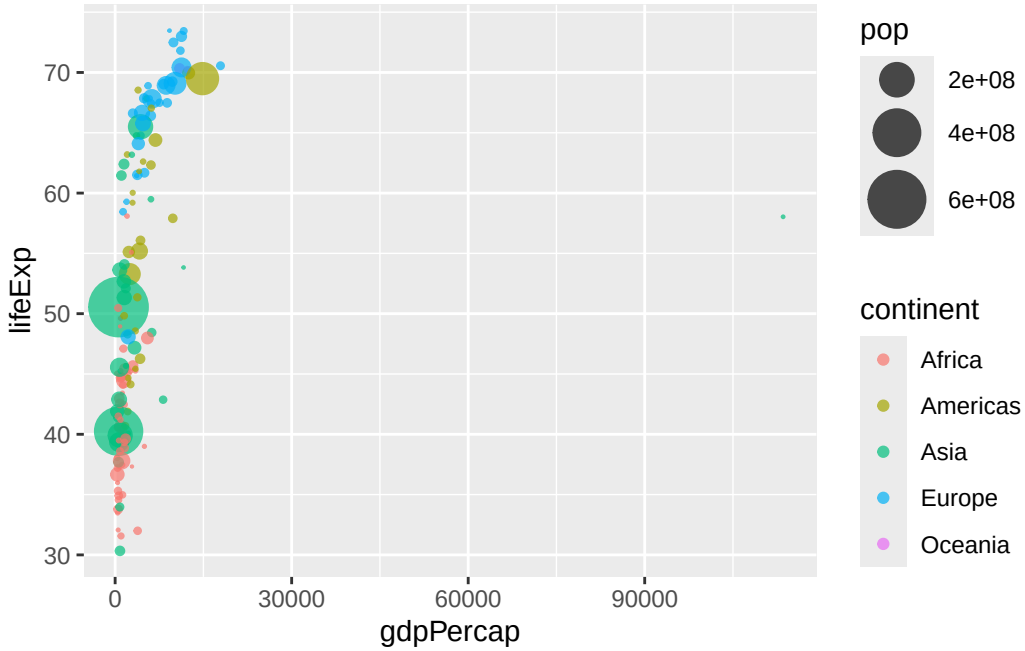



```
#Reflect the actual population differences by the point size and
#using the scale_size_area() function
ggplot(gapminder_2007) +
  geom_point(aes(x = gdpPercap, y = lifeExp,
                 size = pop), alpha=0.5) +
  scale_size_area(max_size = 10)
```



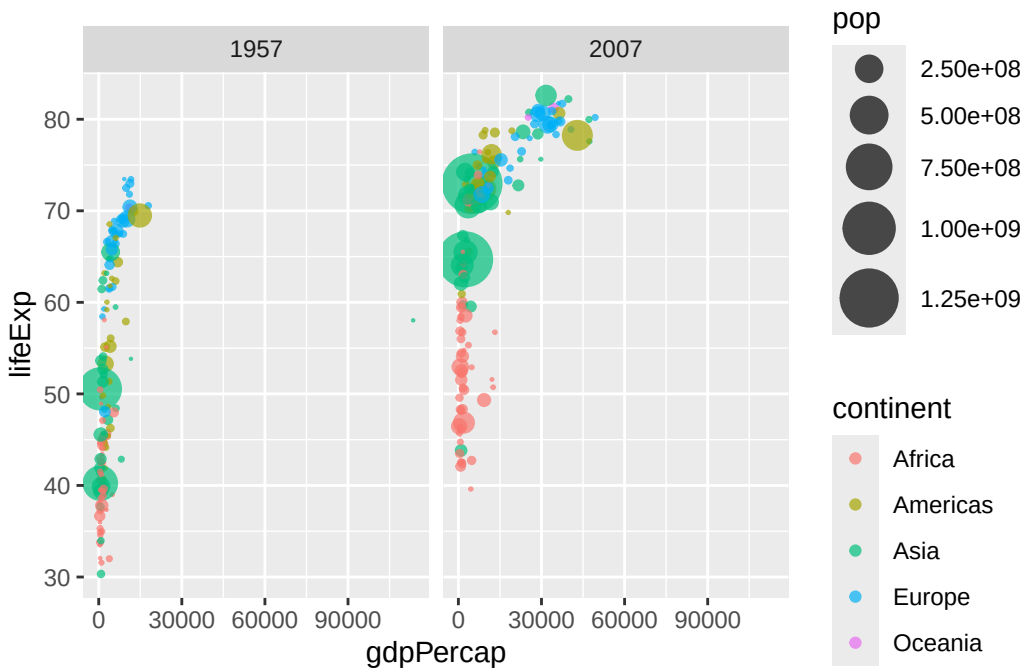

**Q. Can you adapt the code you have learned thus far to reproduce our gapminder scatter plot for the year 1957? What do you notice about this plot is it easy to compare with the one for 2007?**

```
#Generating gapminder scatter plot for year 1957
gapminder_1957 <- gapminder %>% filter(year==1957)

ggplot(gapminder_1957) +
  aes(x = gdpPercap, y = lifeExp, color=continent,
                size = pop) +
  geom_point(alpha=0.7) +
  scale_size_area(max_size = 10)
```




**Q. Do the same steps above but include 1957 and 2007 in your input dataset for ggplot(). You should now include the layer facet_wrap(~year) to produce the following plot:**

```
#Generating scatter plot to include both 1957 and 2007

gapminder_1957 <- gapminder %>% filter(year==1957 | year==2007)

ggplot(gapminder_1957) +
  geom_point(aes(x = gdpPercap, y = lifeExp, color=continent,
                 size = pop), alpha=0.7) +
  scale_size_area(max_size = 10) +
  facet_wrap(~year)
```

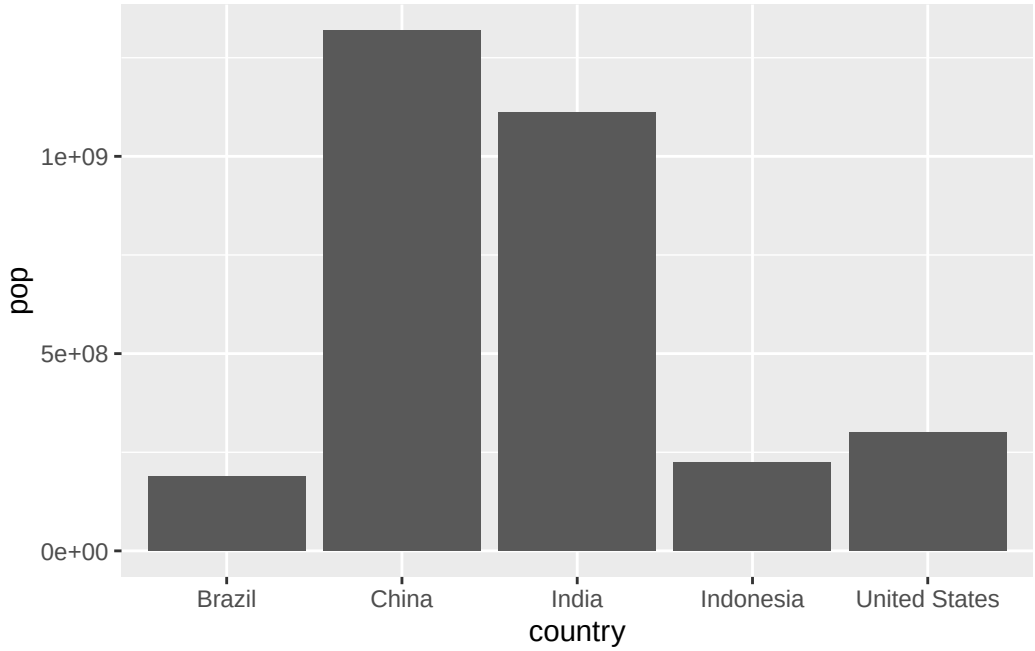



**Part 8: Bar Charts**

```
#Generating the number of people in the five biggest countries by populatiion in 2007
gapminder_top5 <- gapminder %>%
  filter(year==2007) %>%
  arrange(desc(pop)) %>%
  top_n(5, pop)

gapminder_top5
```

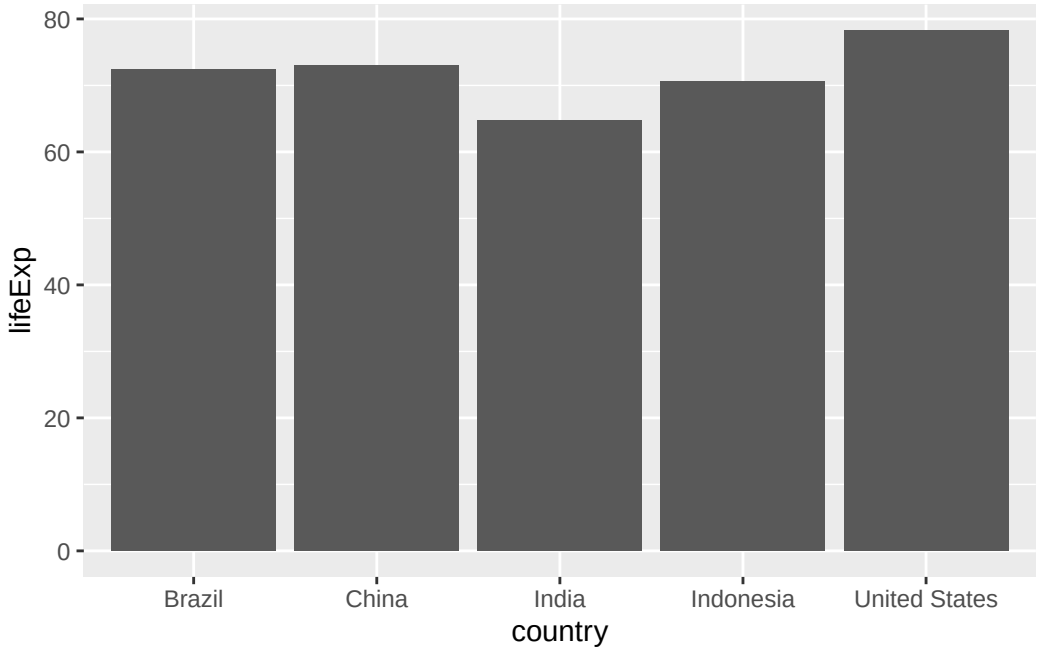
```
        country continent year lifeExp        pop gdpPercap
1         China      Asia 2007  72.961 1318683096  4959.115
2         India      Asia 2007  64.698 1110396331  2452.210
3 United States  Americas 2007  78.242  301139947 42951.653
4     Indonesia      Asia 2007  70.650  223547000  3540.652
5        Brazil  Americas 2007  72.390  190010647  9065.801
```

```
#Generating bar chart with gapminder_top5 dataset
ggplot(gapminder_top5) +
  geom_col(aes(x = country, y = pop))
```

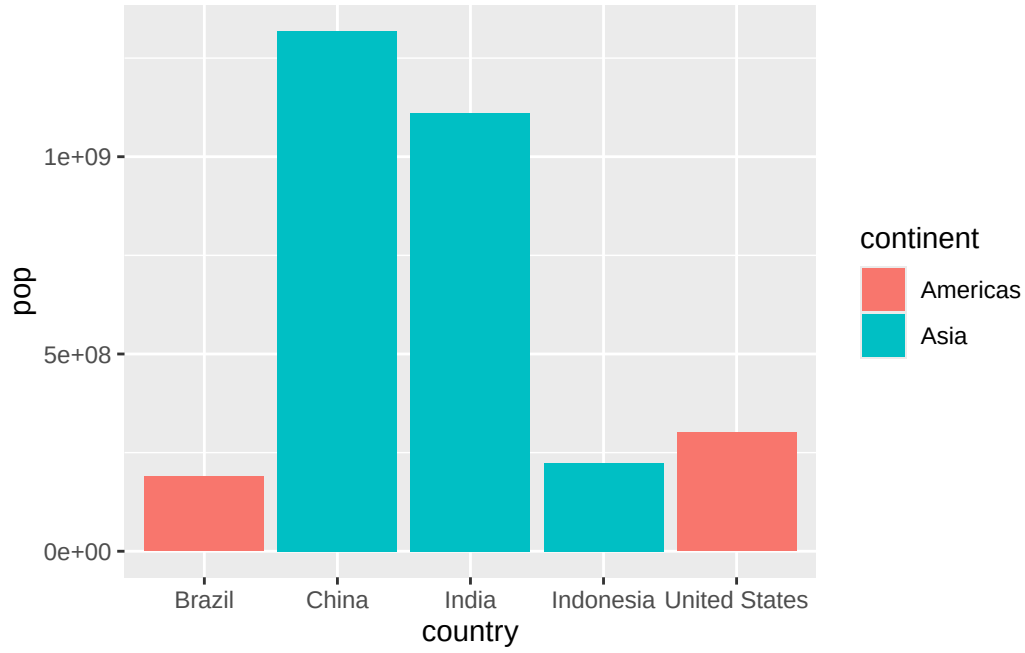



**Q Create a bar chart showing the life expectancy of the five biggest countries by population in 2007.**

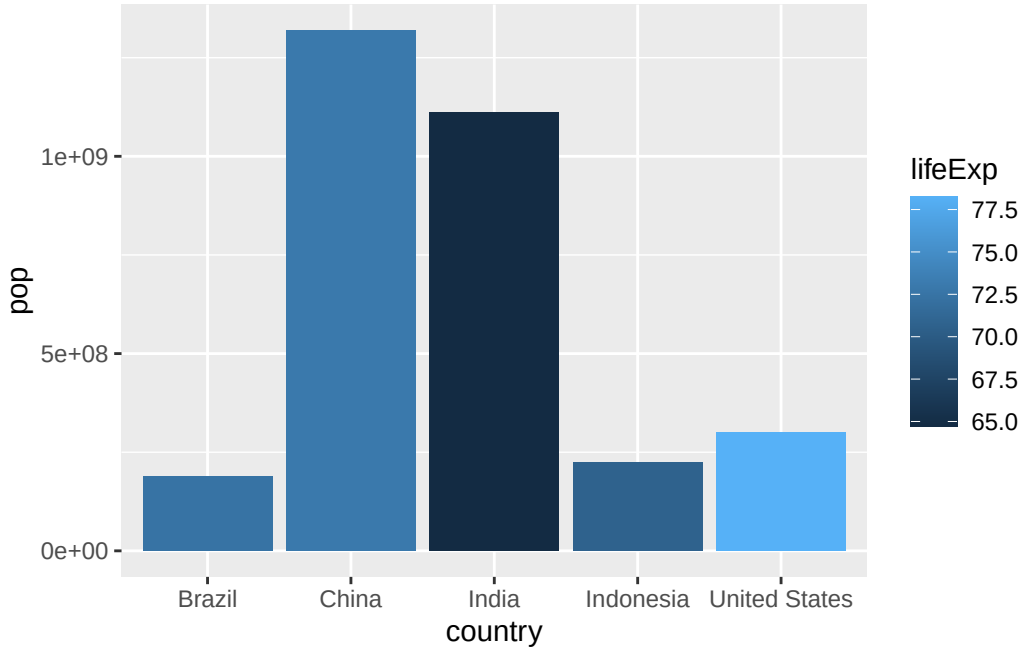
```
# Generating bar chart with gapminder_top5 dataset
ggplot(gapminder_top5) +
  geom_col(aes(x = country, y = lifeExp))
```




```
#Filling the bars with color
ggplot(gapminder_top5) +
  geom_col(aes(x = country, y = pop, fill = continent))
```

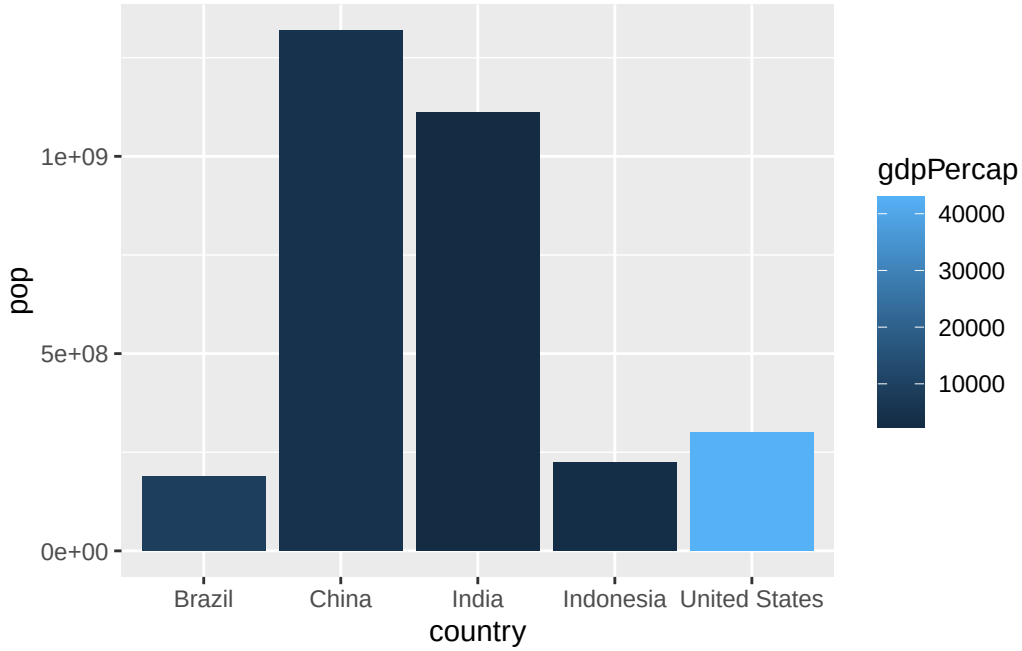


```
#Using numeric variable (life expectancy) to fill color
ggplot(gapminder_top5) +
  geom_col(aes(x = country, y = pop, fill = lifeExp))
```

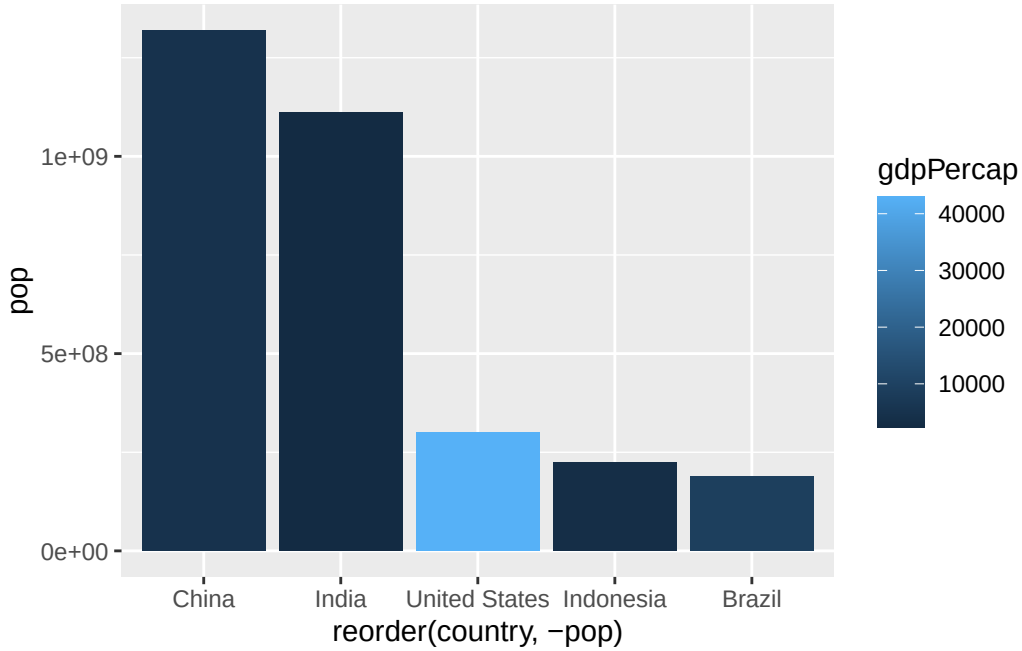



**Q. Plot population size by country. Create a bar chart showing the population (in millions) of the five biggest countries by population in 2007.**

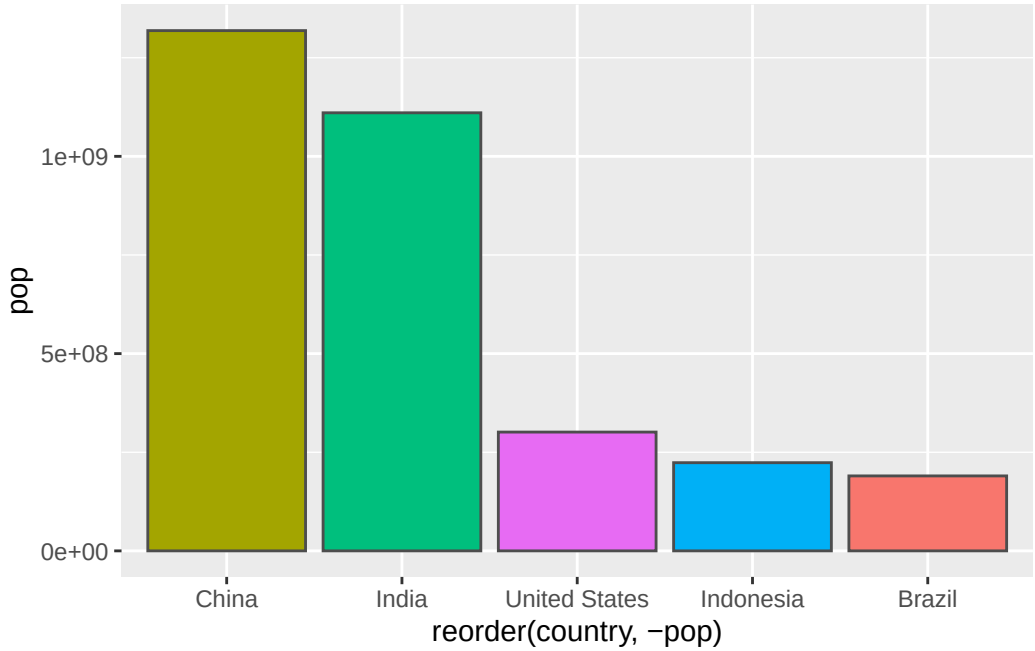
```
#Generating bar chart showing population (in millons) of the five biggest countries by popula
ggplot(gapminder_top5) +
  aes(x=country, y=pop, fill=gdpPercap) +
  geom_col()
```




```
#Changing the order of the bars
ggplot(gapminder_top5) +
  aes(x=reorder(country, -pop), y=pop, fill=gdpPercap) +
  geom_col()
```




```
#Filling just by country
ggplot(gapminder_top5) +
  aes(x=reorder(country, -pop), y=pop, fill=country) +
  geom_col(col="gray30") +
  guides(fill="none")
```

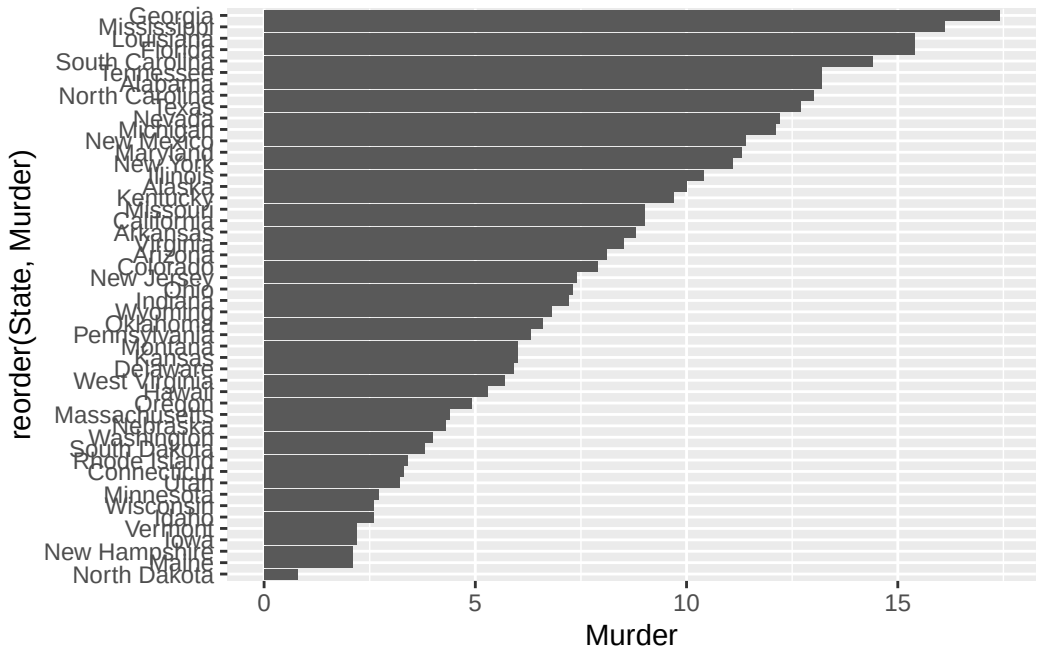



### Flipping bar charts

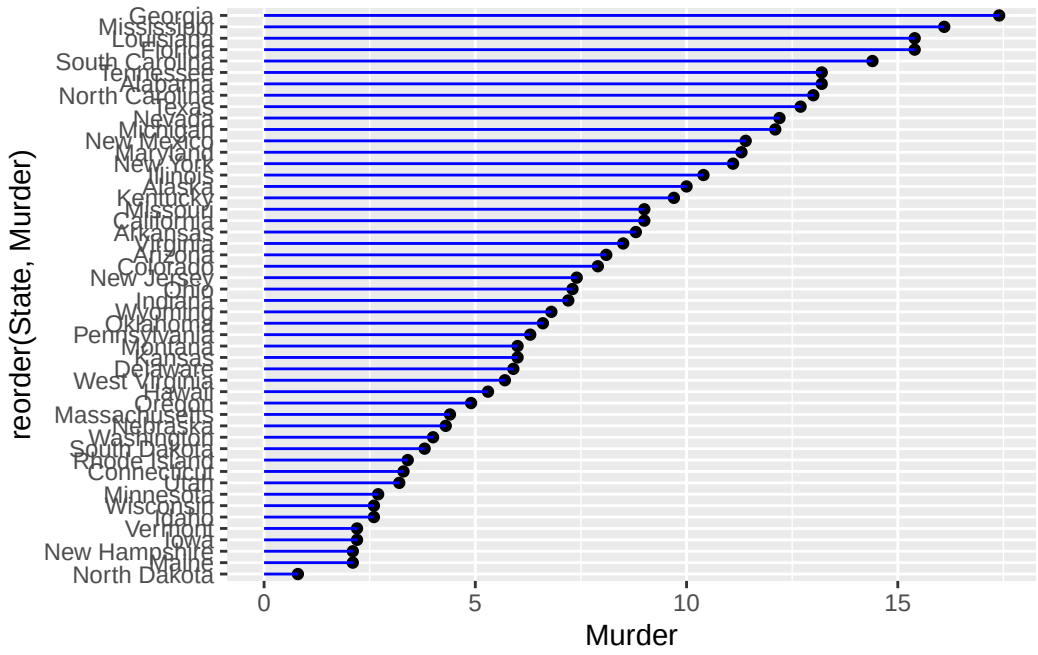
```
#Loading dataset USArrests
head(USArrests)
```

```
           Murder Assault UrbanPop Rape
Alabama      13.2     236       58 21.2
Alaska       10.0     263       48 44.5
Arizona       8.1     294       80 31.0
Arkansas      8.8     190       50 19.5
California    9.0     276       91 40.6
Colorado      7.9     204       78 38.7
```

```
#Generating bar chart of USArrests
USArrests$State <- rownames(USArrests)
ggplot(USArrests) +
  aes(x=reorder(State,Murder), y=Murder) +
  geom_col() +
  coord_flip()
```




```
#Plotting USArrests by using points and line segments
ggplot(USArrests) +
  #Mapping variables
  aes(x=reorder(State,Murder), y=Murder) +
  geom_point() +
  #Adding verticle line segments from 0 up to each state's murder rate
  geom_segment(aes(x=State,
                   xend=State,
                   y=0,
                   yend=Murder), color="blue") +
  #Flip the axes so states are on the y=axis
  coord_flip()
```




**Part 9. Extensions: Animation**

```
library(gapminder)
```

```
Attaching package: 'gapminder'
```

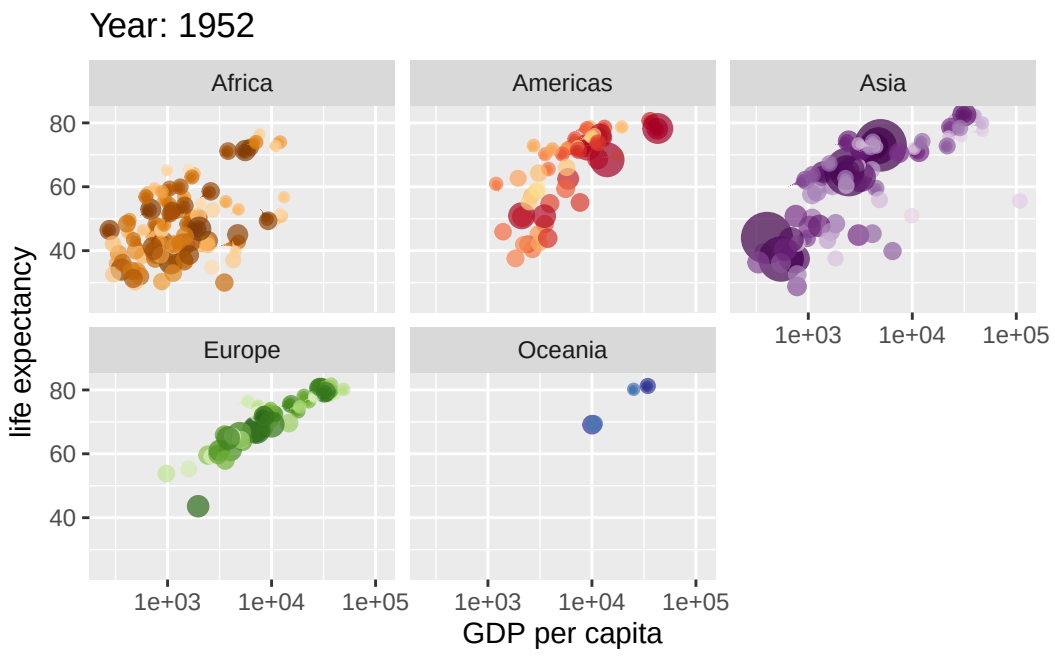
```
The following object is masked _by_ '.GlobalEnv':
```

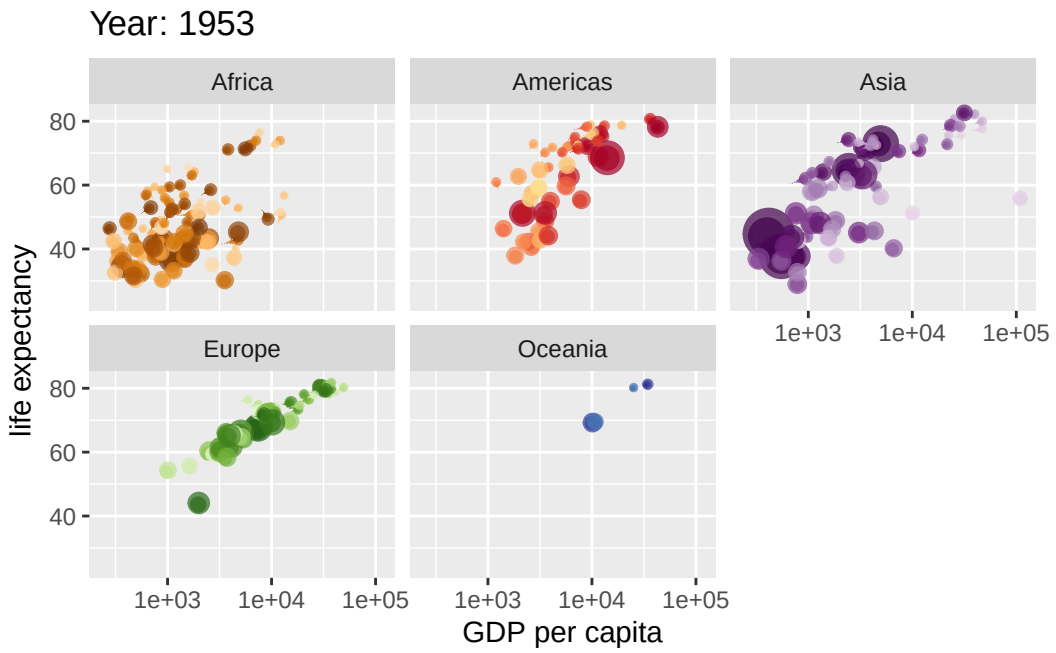
```
    gapminder
```

```
library(gganimate)

# Setup nice regular ggplot of the gapminder data
ggplot(gapminder, aes(gdpPercap, lifeExp, size = pop, colour = country)) +
  geom_point(alpha = 0.7, show.legend = FALSE) +
  scale_colour_manual(values = country_colors) +
  scale_size(range = c(2, 12)) +
  scale_x_log10() +
  # Facet by continent
  facet_wrap(~continent) +
  # Here comes the gganimate specific bits
```

```
  labs(title = 'Year: {frame_time}', x = 'GDP per capita', y = 'life expectancy') +
  transition_time(year) +
  shadow_wake(wake_length = 0.1, alpha = FALSE)
```

```
Warning in formals(fun): argument is not a function
```

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Asia
Europe
Oceania
life expectancy
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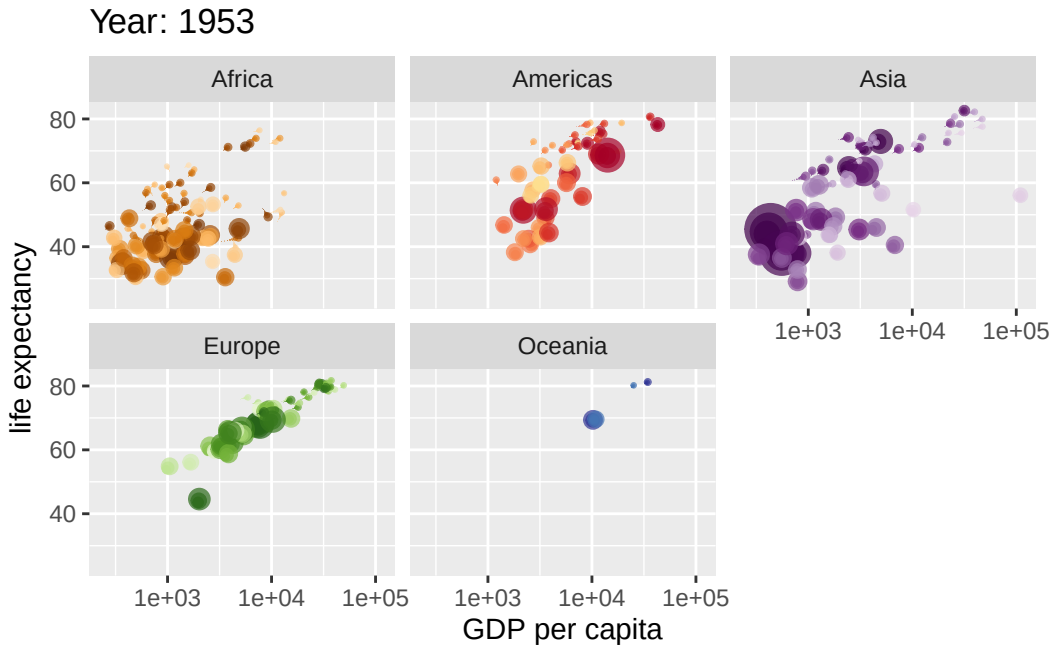
60
40
1e+03
1e+04
1e+05
GDP per capita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Asia
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Oceania
life expectancy
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1e+03
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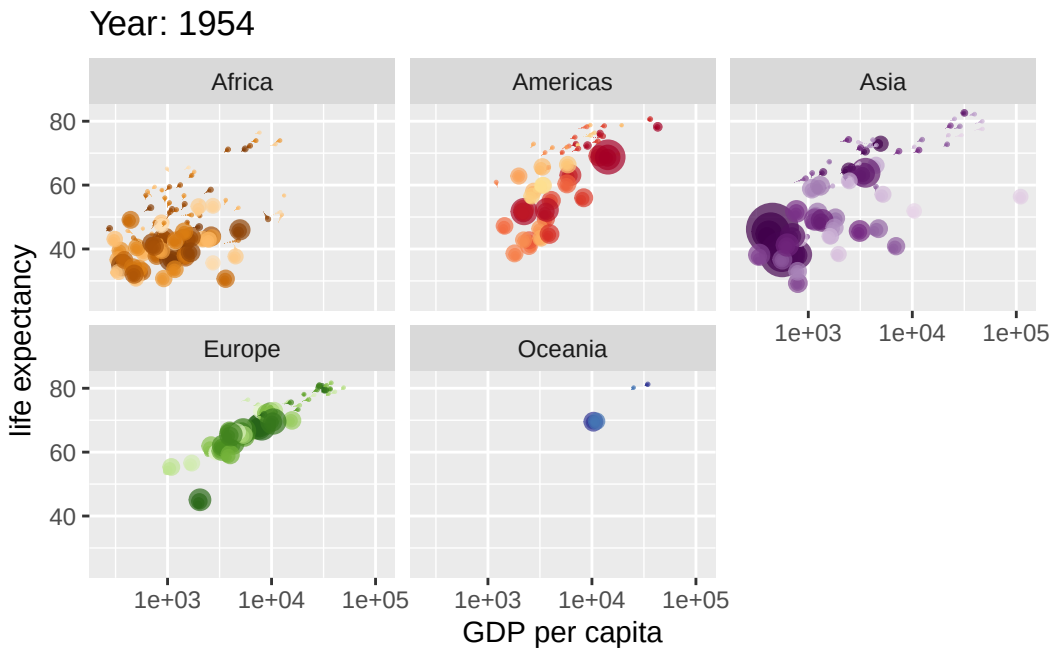
1e+05
GDP per capita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Asia
Europe
Oceania
life expectancy
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40
1e+03
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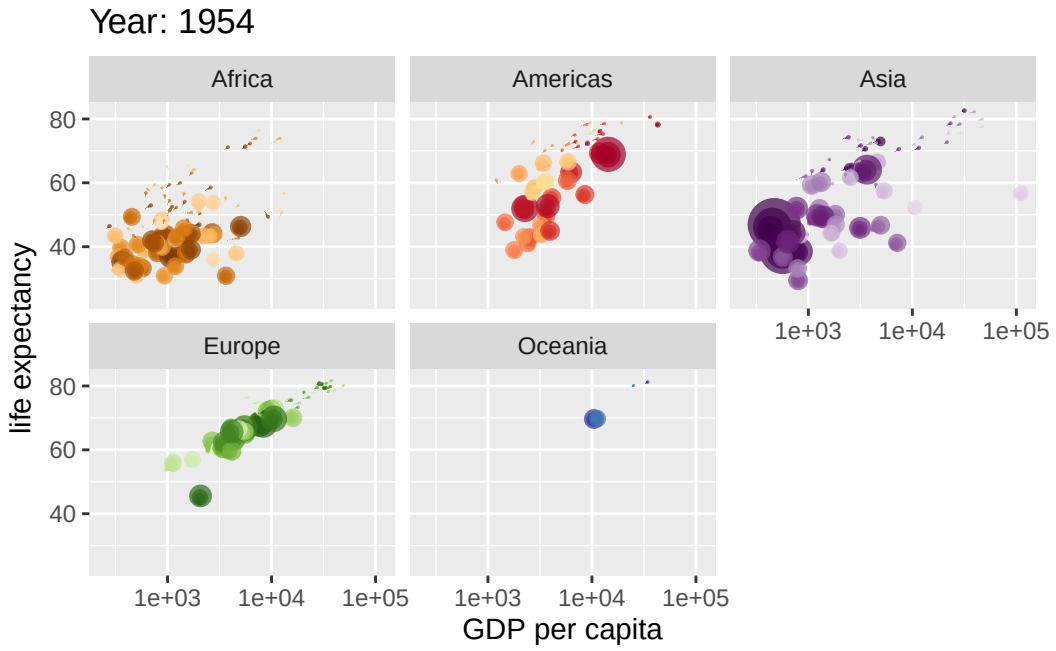
1e+05
GDP per capita


Year: 1954
Africa
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Europe
Oceania
life expectancy
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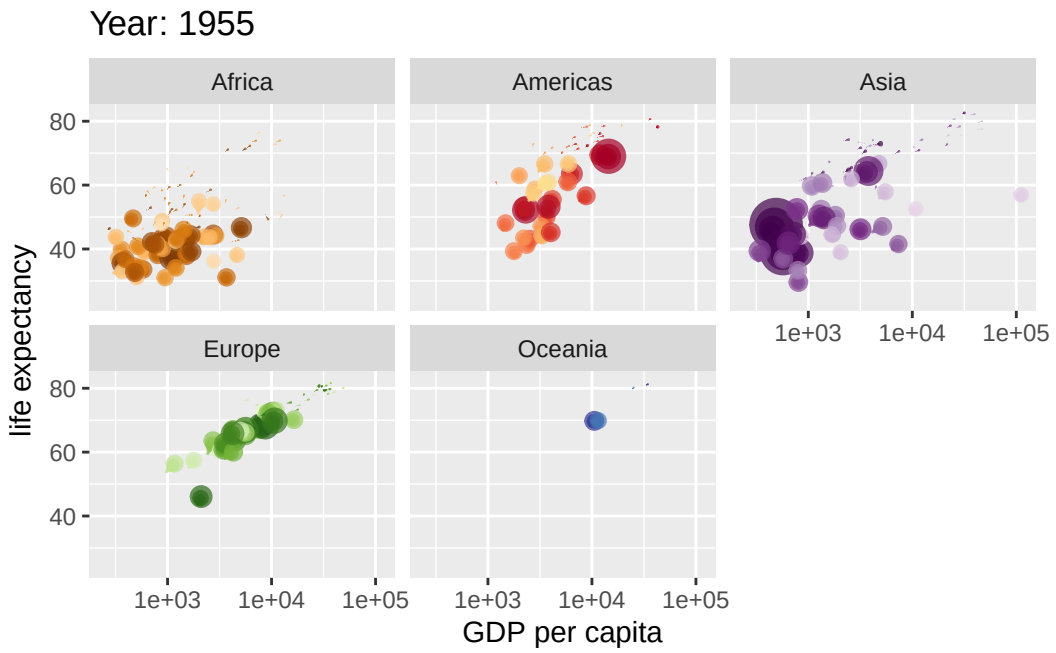
1e+03
1e+04
1e+05
GDP per capita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Africa
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Oceania
life expectancy
80
60
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1e+03
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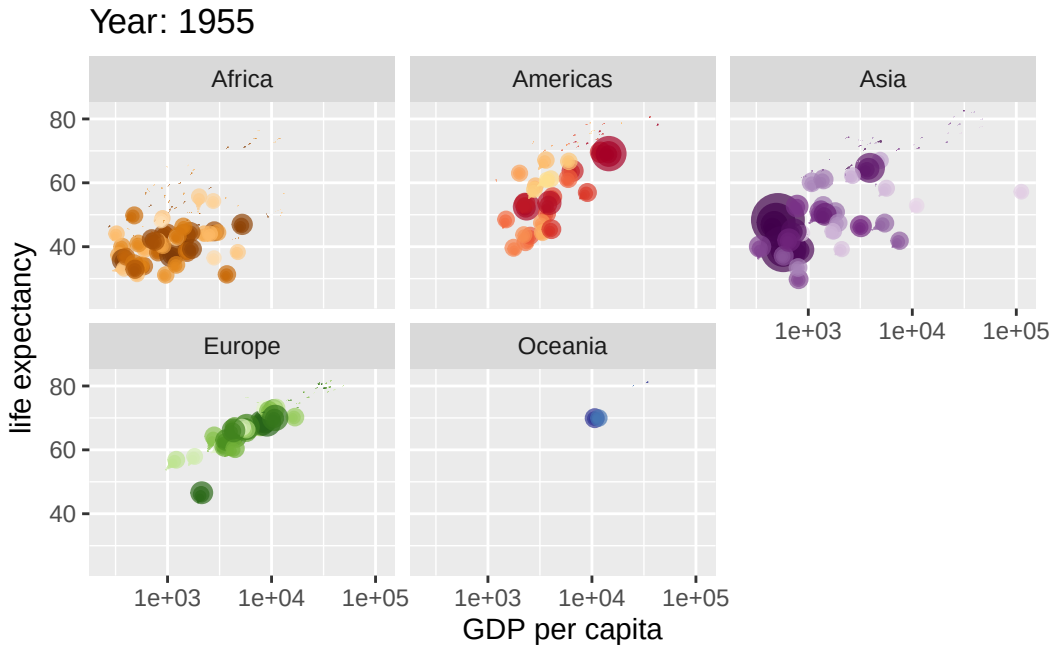
1e+05
GDP per capita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life expectancy
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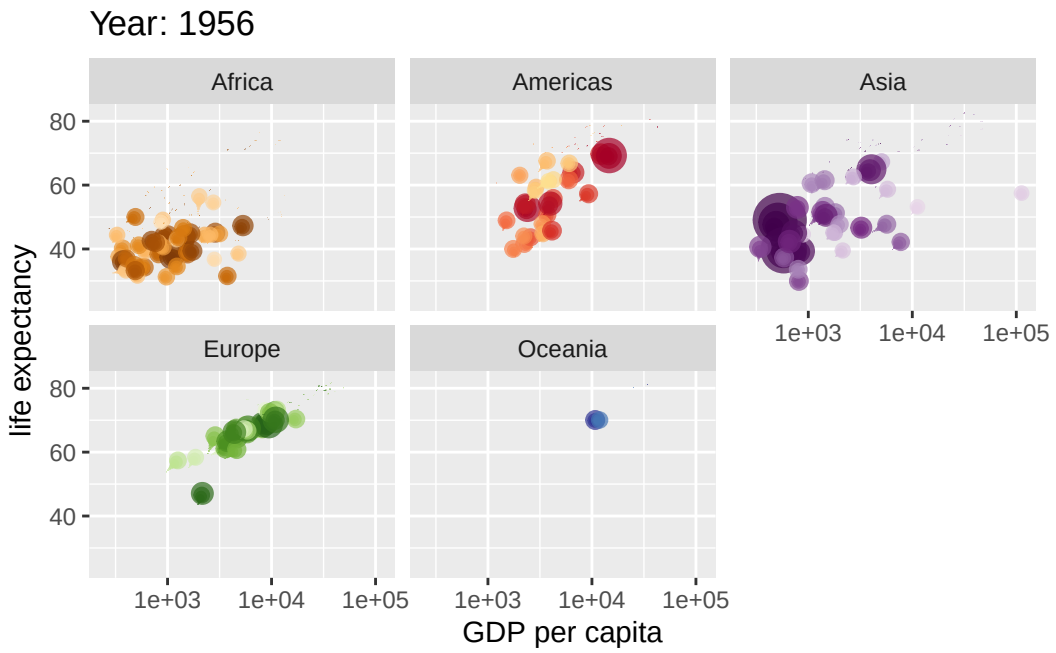
GDP per capita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life expectancy
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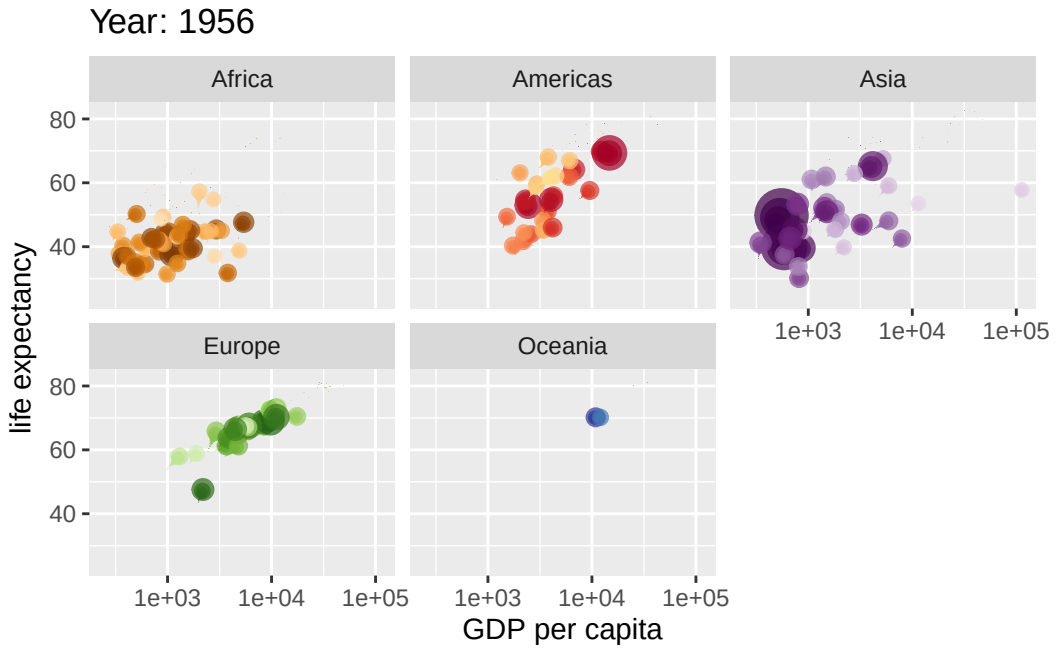
1e+05
GDP per capita


Year: 1956
Africa
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Oceania
life expectancy
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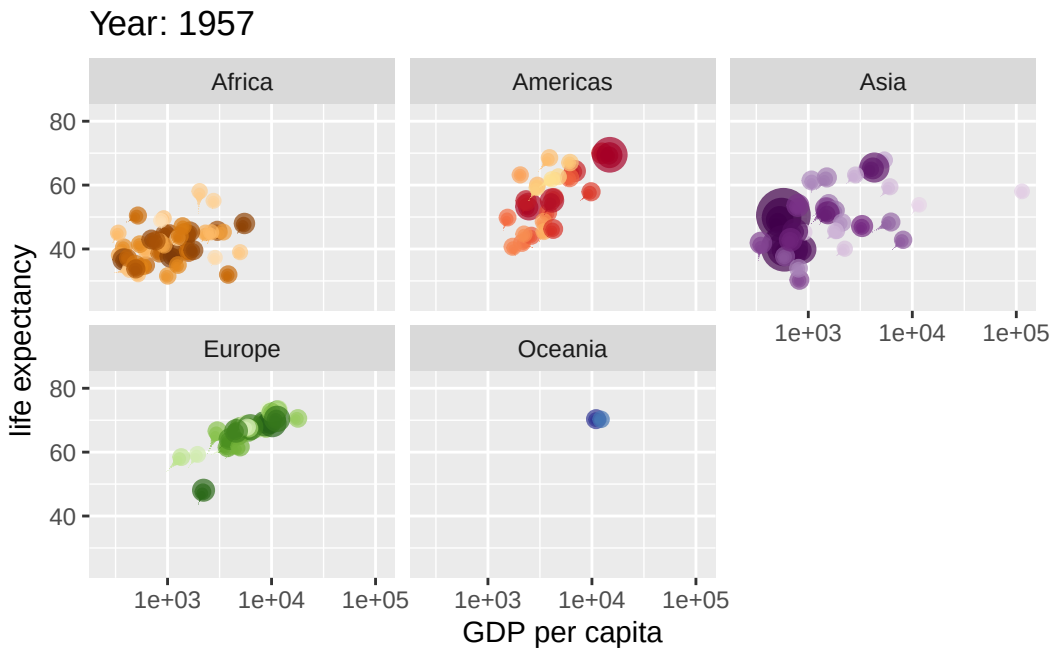
1e+05
GDP per capita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life expectancy
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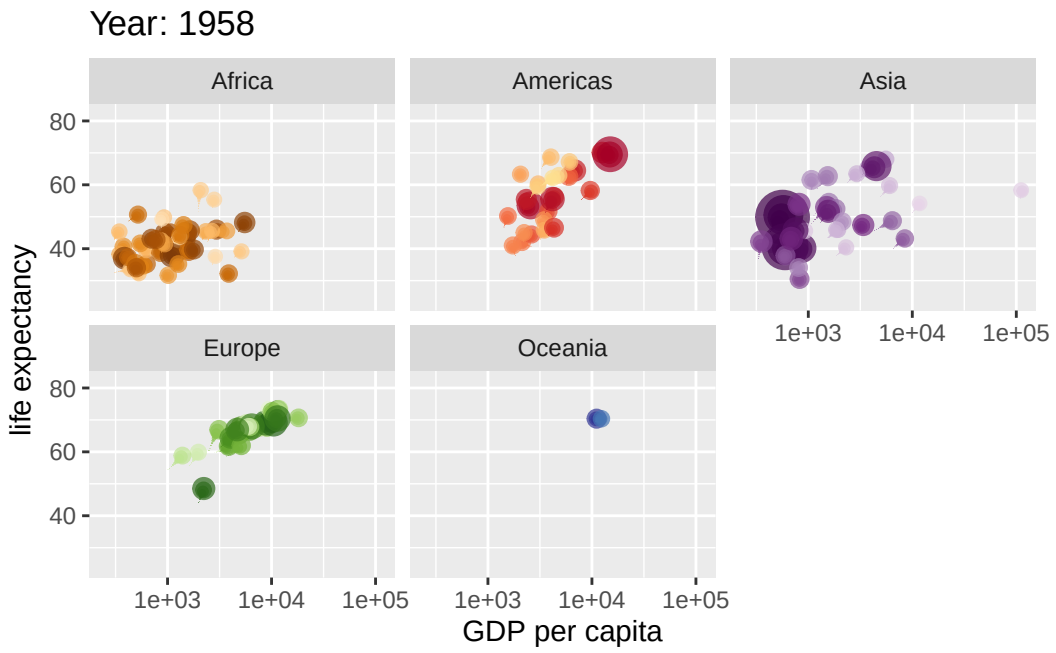
GDP per capita


Year: 1958
Africa
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Asia
Europe
Oceania
life expectancy
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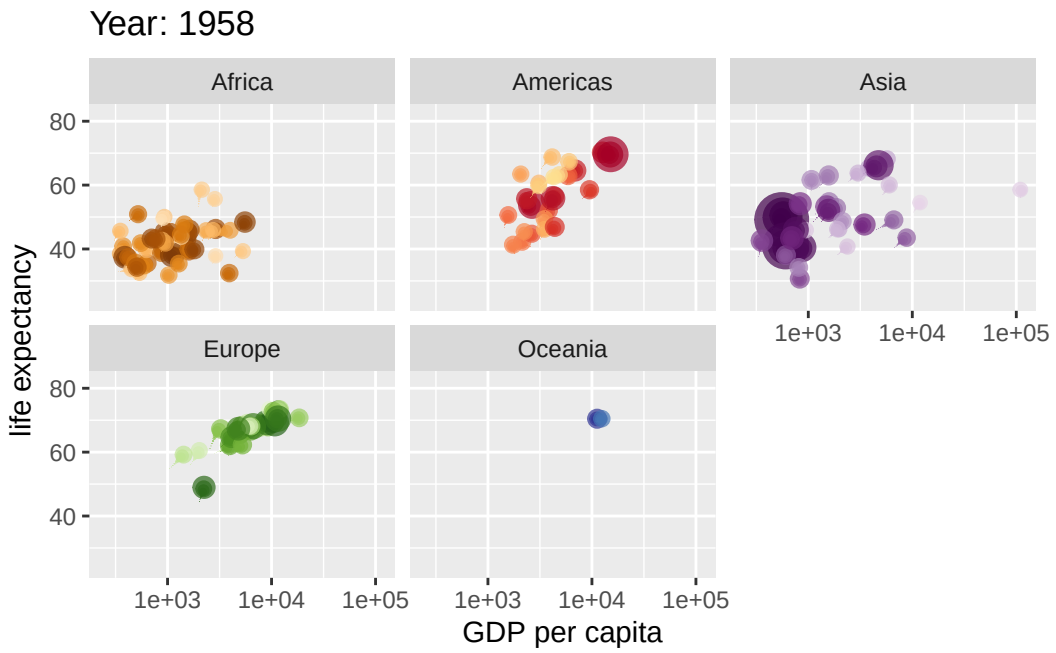
60
40
1e+03
1e+04
1e+05
GDP per capita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Africa
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Asia
Europe
Oceania
life expectancy
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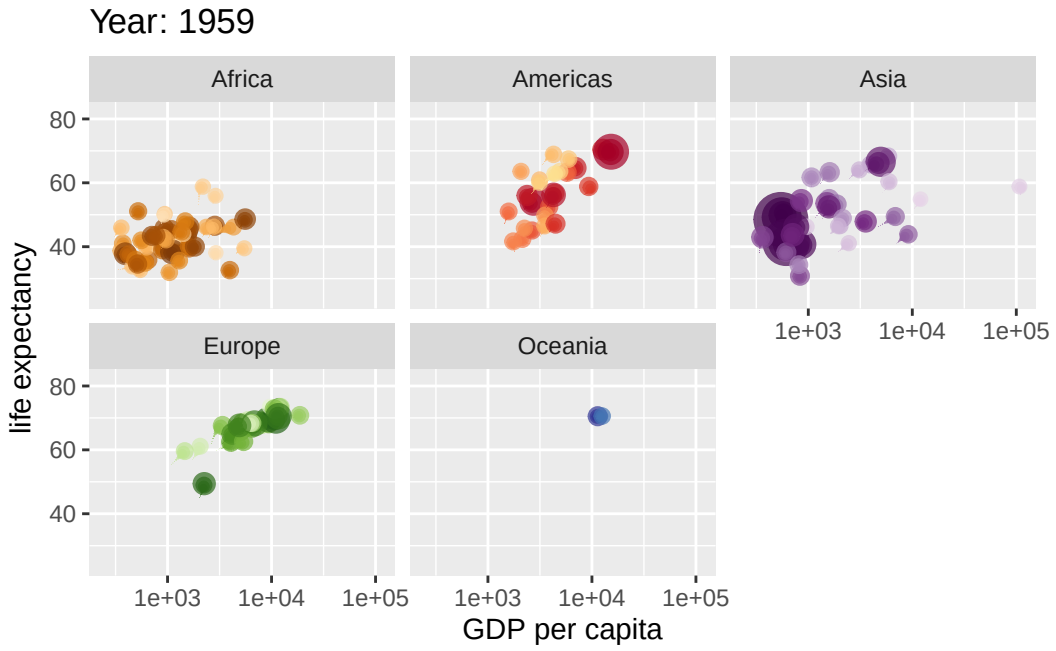
1e+03
1e+04
1e+05
GDP per capita


Year: 1959
Africa
Americas
Asia
Europe
Oceania
life expectancy
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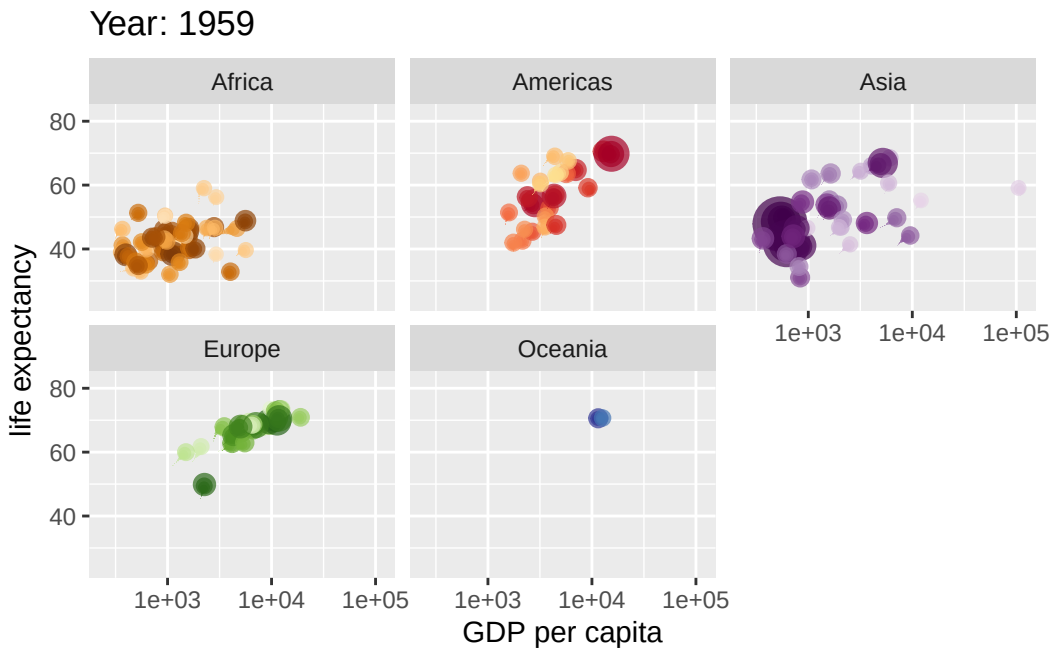
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1959
Africa
Americas
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Europe
Oceania
life expectancy
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1e+03
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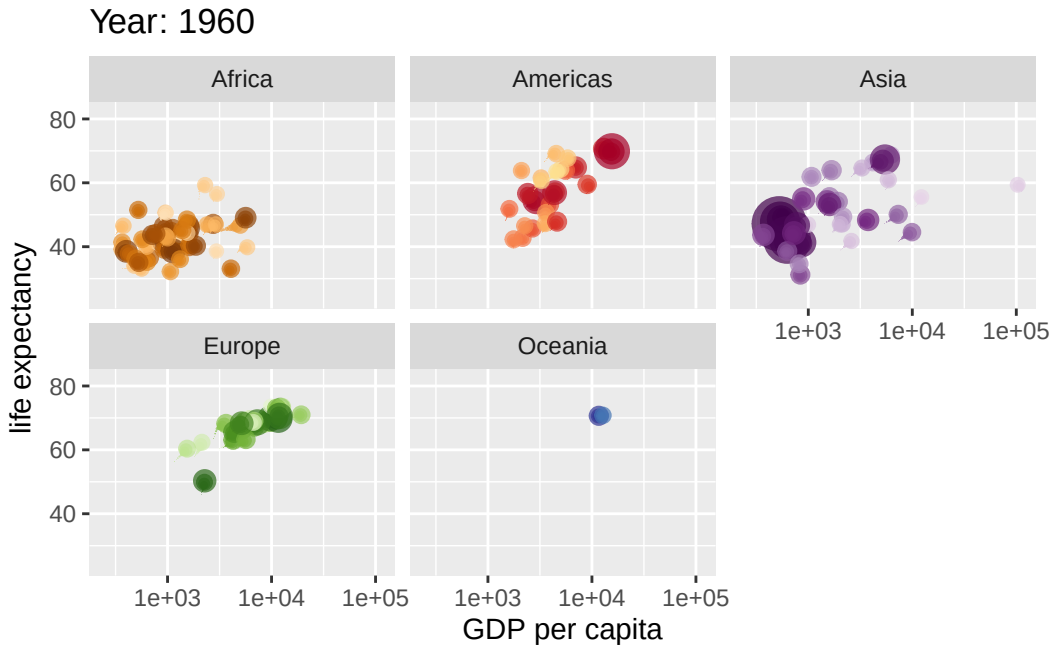
1e+05
GDP per capita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Africa
Americas
Asia
Europe
Oceania
life expectancy
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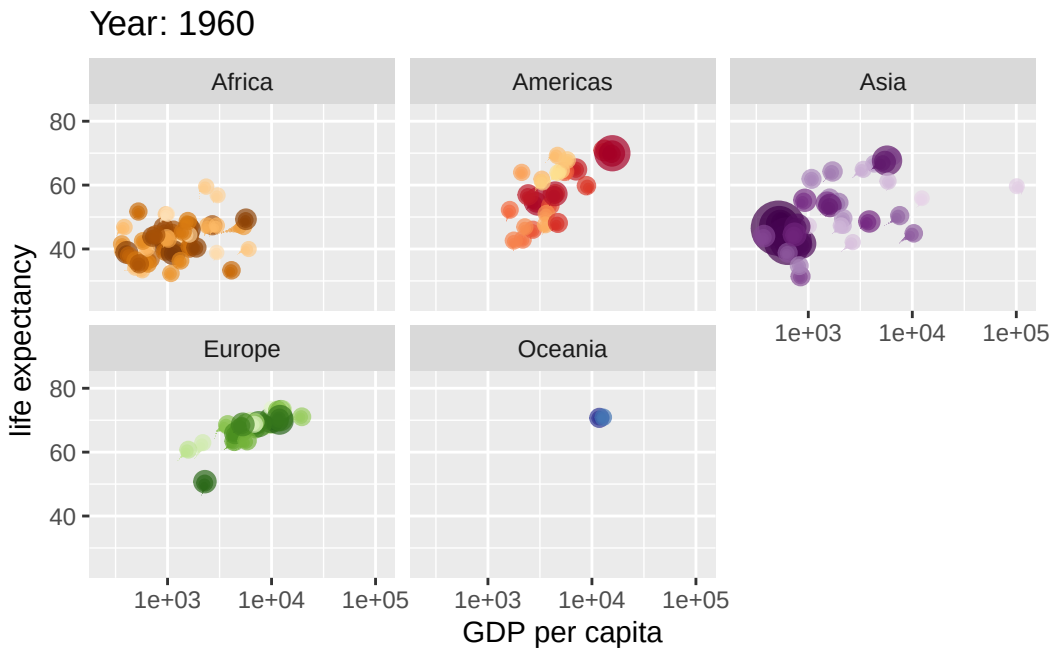
1e+03
1e+04
1e+05
GDP per capita


Year: 1960
Africa
Americas
Asia
Europe
Oceania
life expectancy
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1e+03
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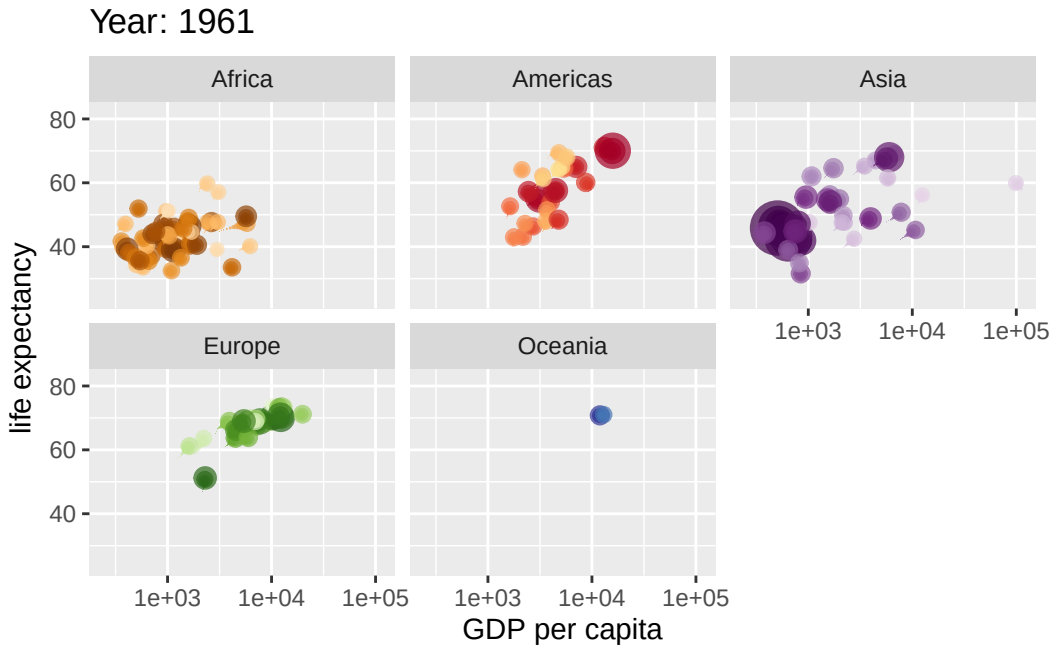
1e+05
GDP per capita


Year: 1961
Africa
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Asia
Europe
Oceania
life expectancy
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60
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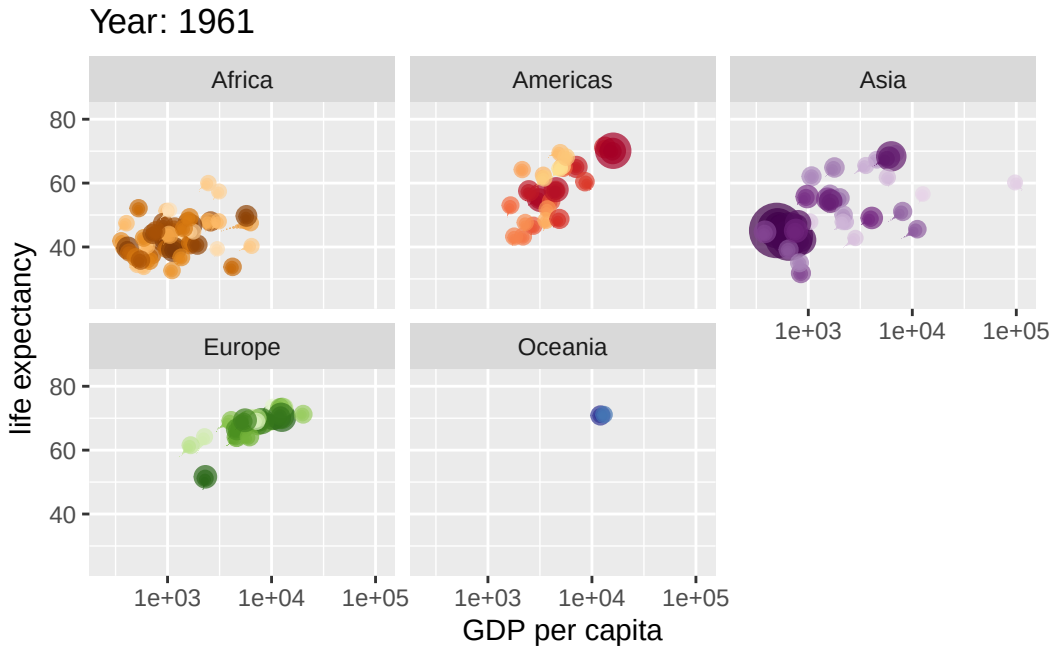
1e+03
1e+04
1e+05
GDP per capita


Year: 1961
Africa
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Oceania
life expectancy
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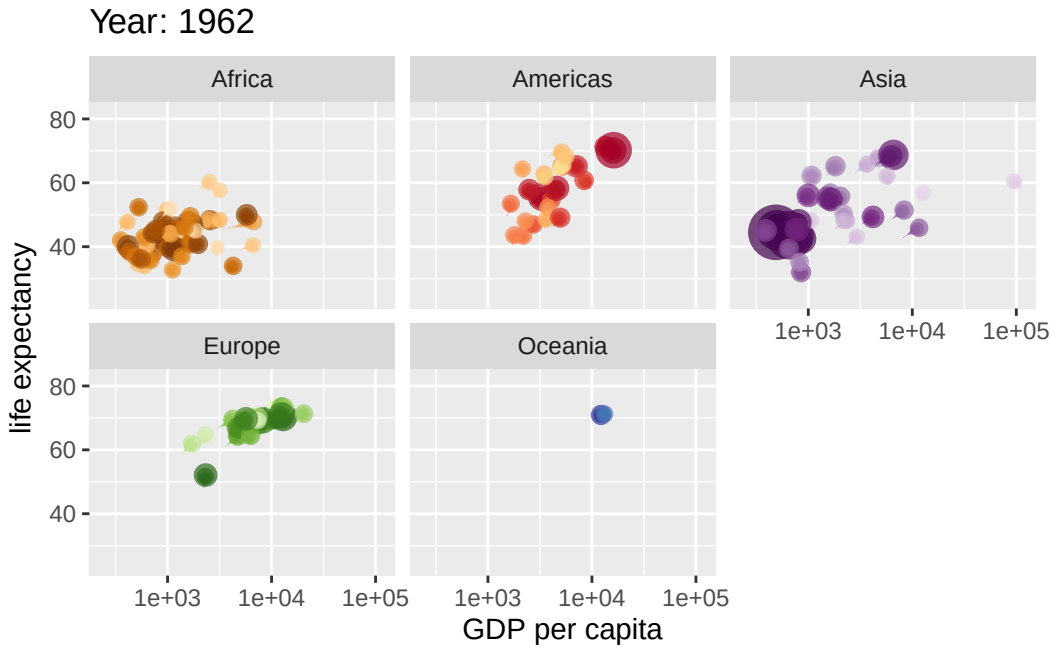
1e+05
GDP per capita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Africa
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Europe
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life expectancy
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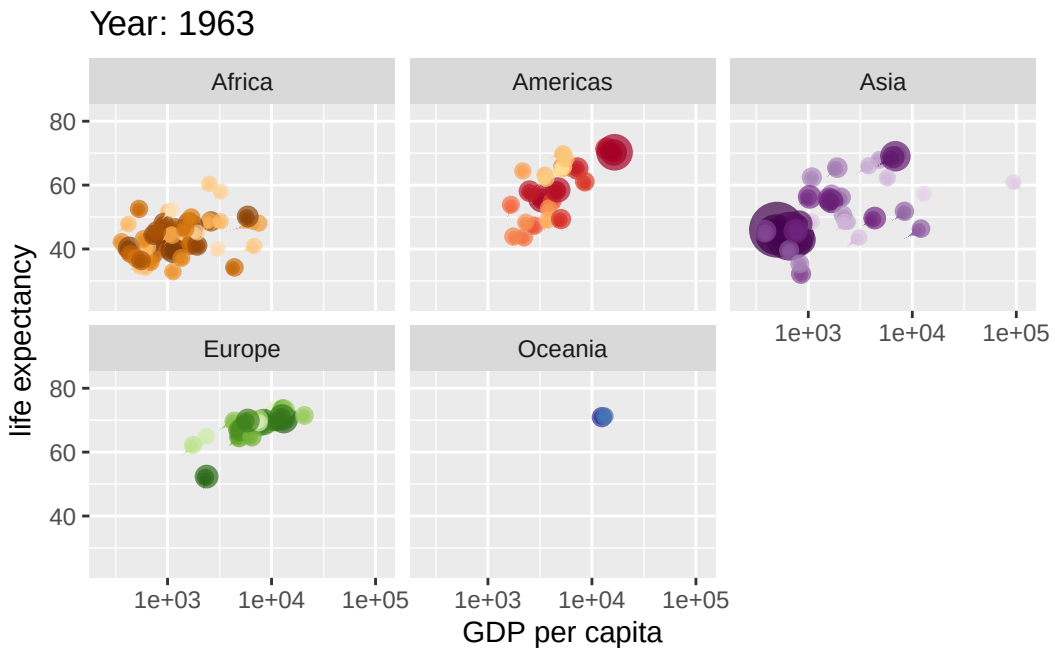
1e+05
GDP per capita


Year: 1963
Africa
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life expectancy
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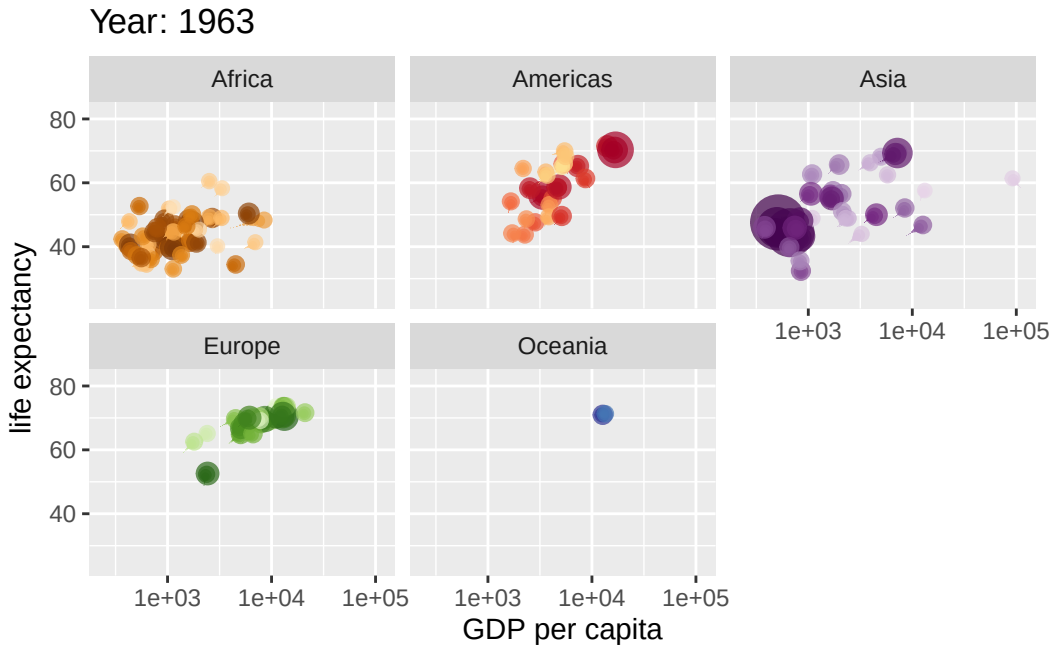
1e+05
GDP per capita


Year: 1963
Africa
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Europe
Oceania
life expectancy
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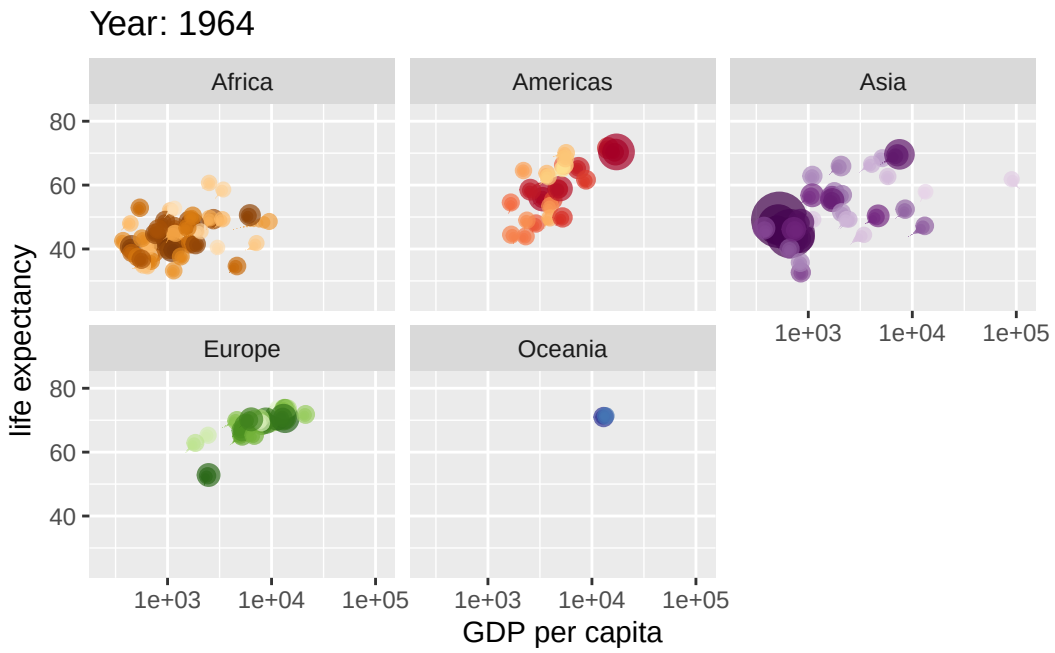
1e+03
1e+04
1e+05
GDP per capita


Year: 1964
Africa
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Asia
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Oceania
life expectancy
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60
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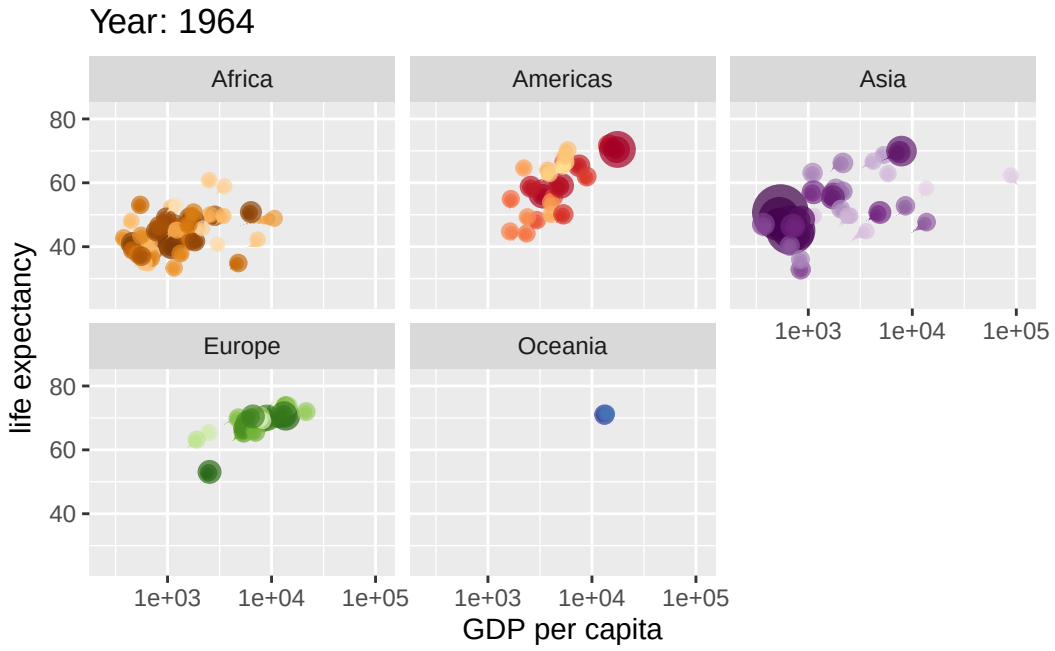
1e+03
1e+04
1e+05
GDP per capita


Year: 1964
Africa
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Asia
Europe
Oceania
life expectancy
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1e+03
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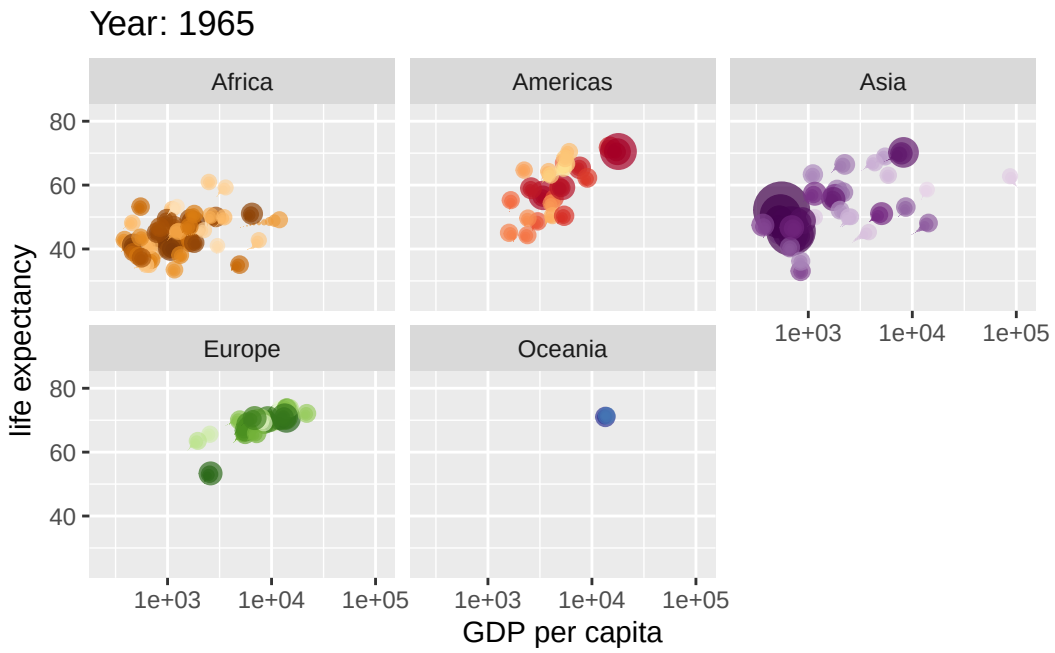
1e+05
GDP per capita


Year: 1965
Africa
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Asia
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Oceania
life expectancy
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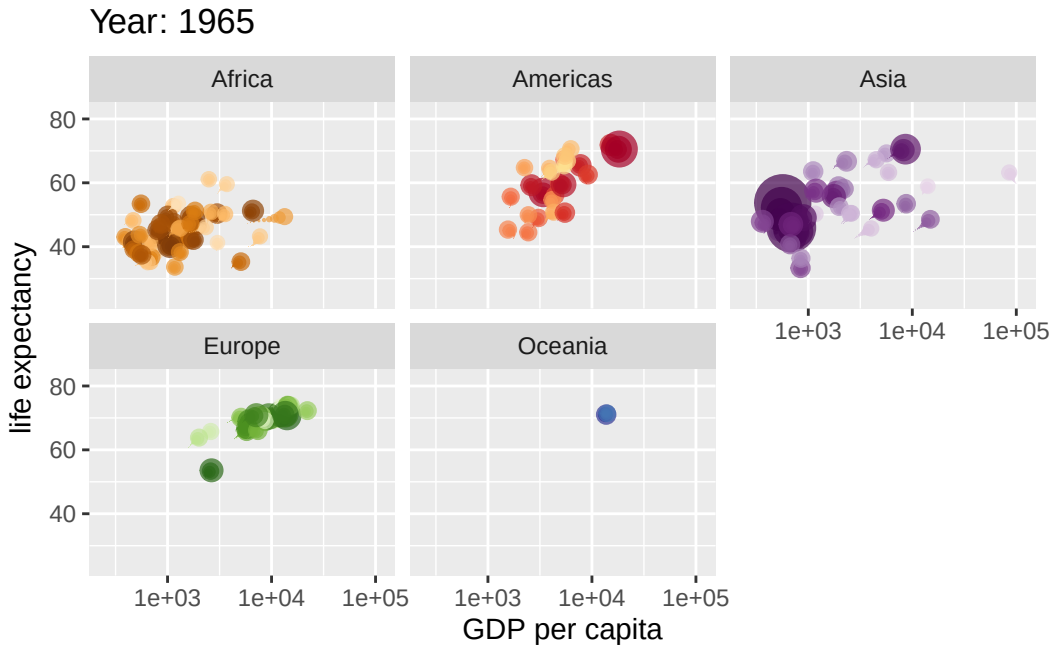
1e+03
1e+04
1e+05
GDP per capita


Year: 1965
Africa
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Europe
Oceania
life expectancy
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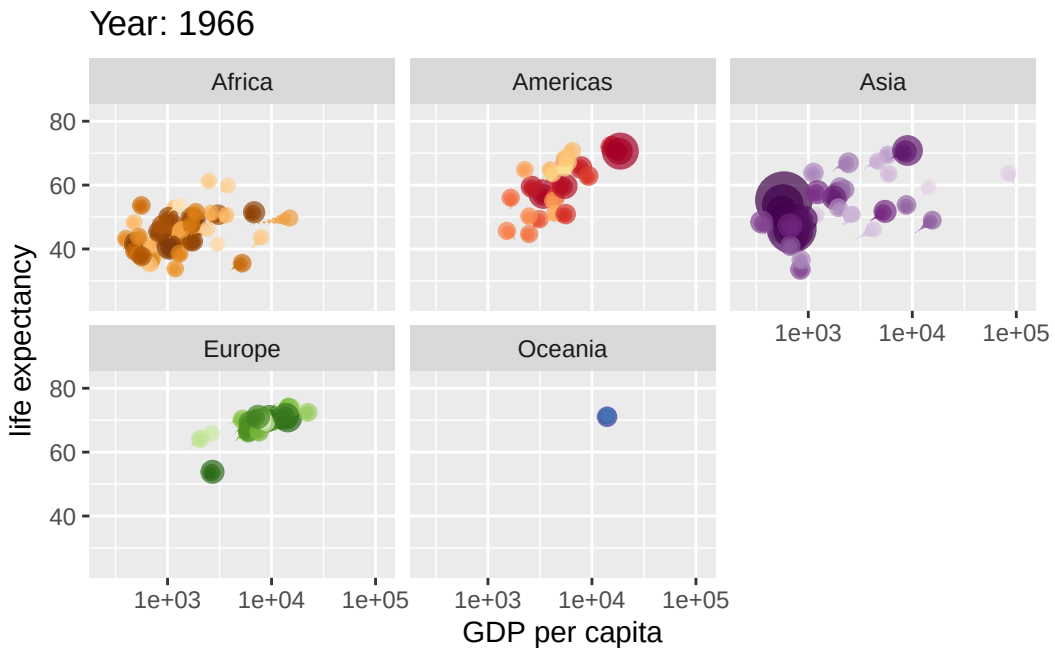
GDP per capita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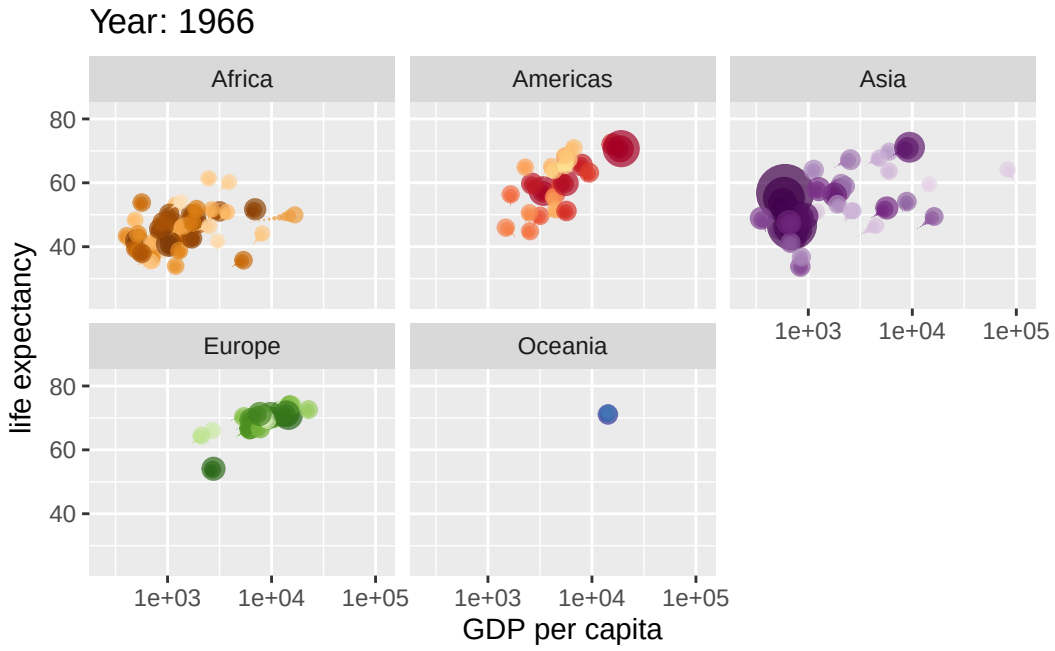
GDP per capita


Year: 1966
Africa
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life expectancy
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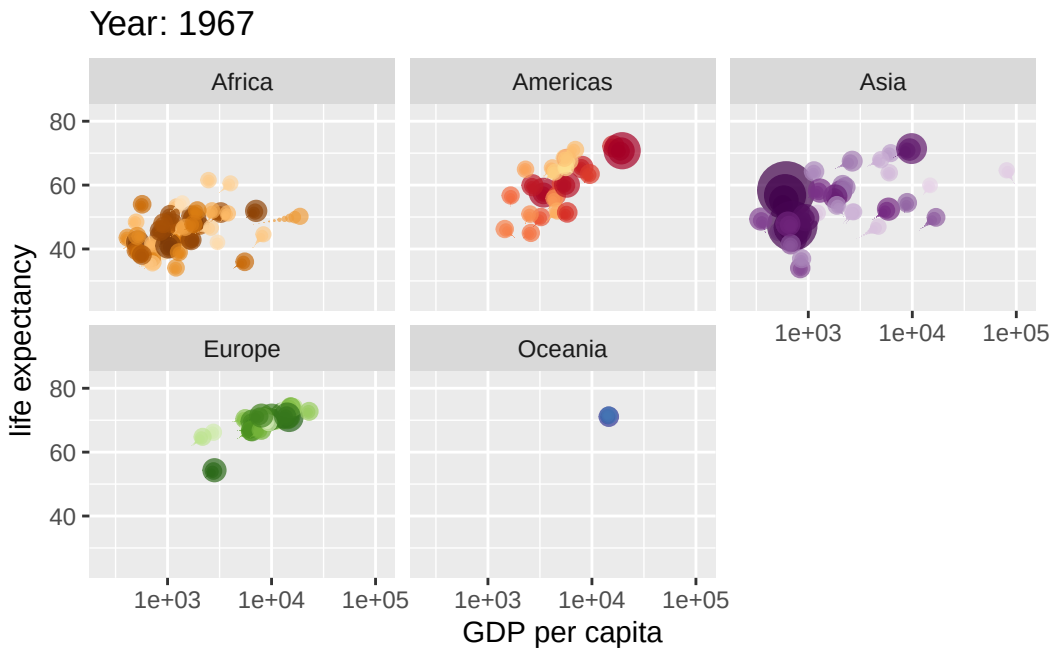
GDP per capita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life expectancy
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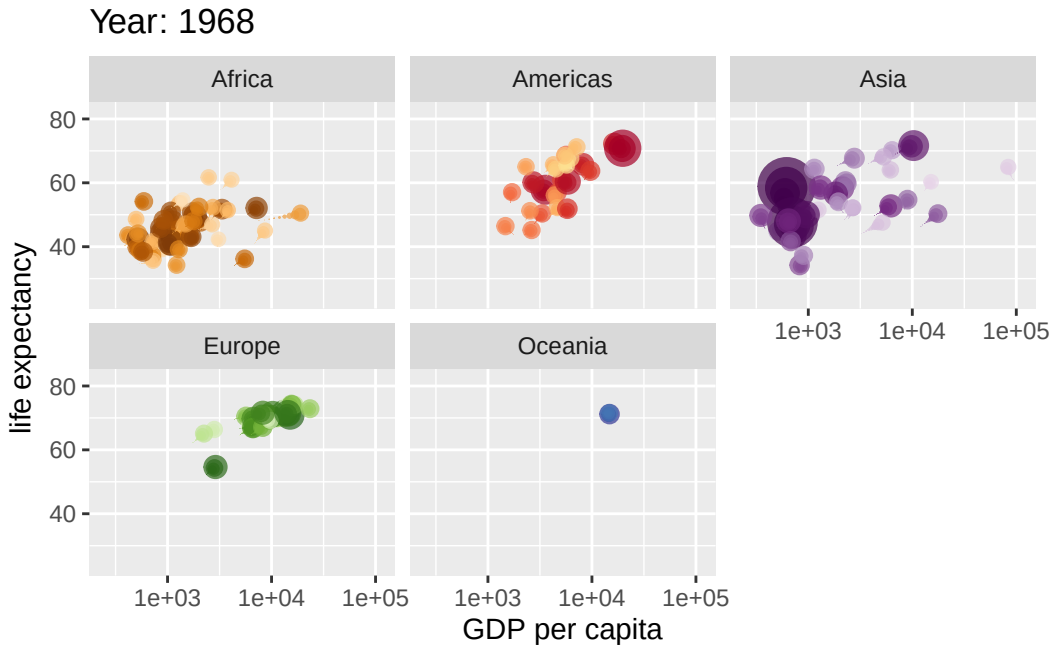
1e+05
GDP per capita


Year: 1968
Africa
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Oceania
life expectancy
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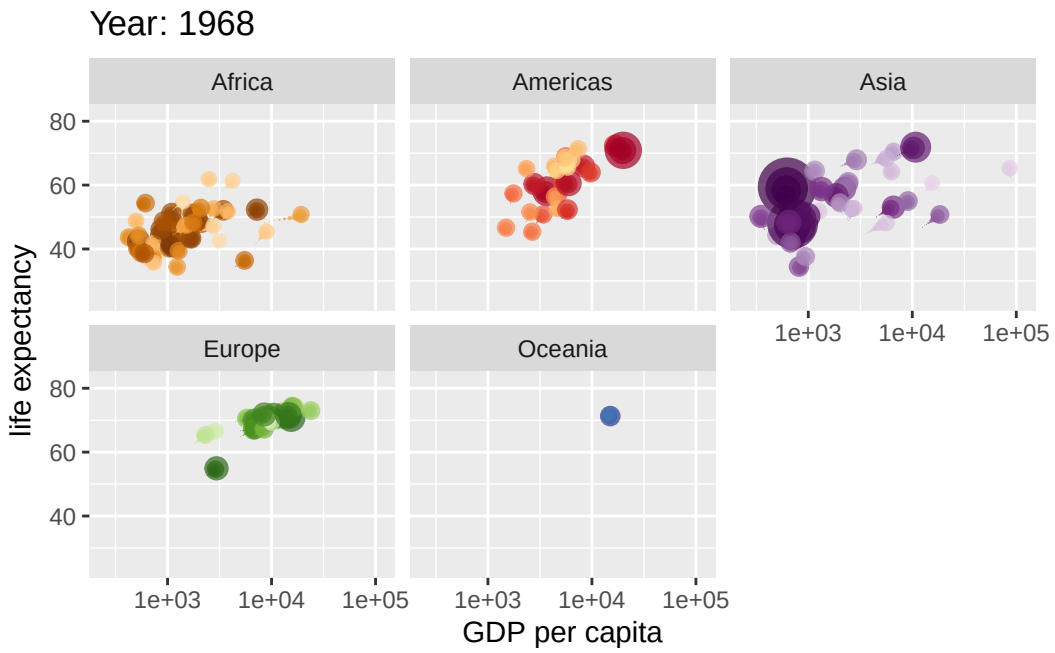
1e+05
GDP per capita


Year: 1968
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Europe
Oceania
life expectancy
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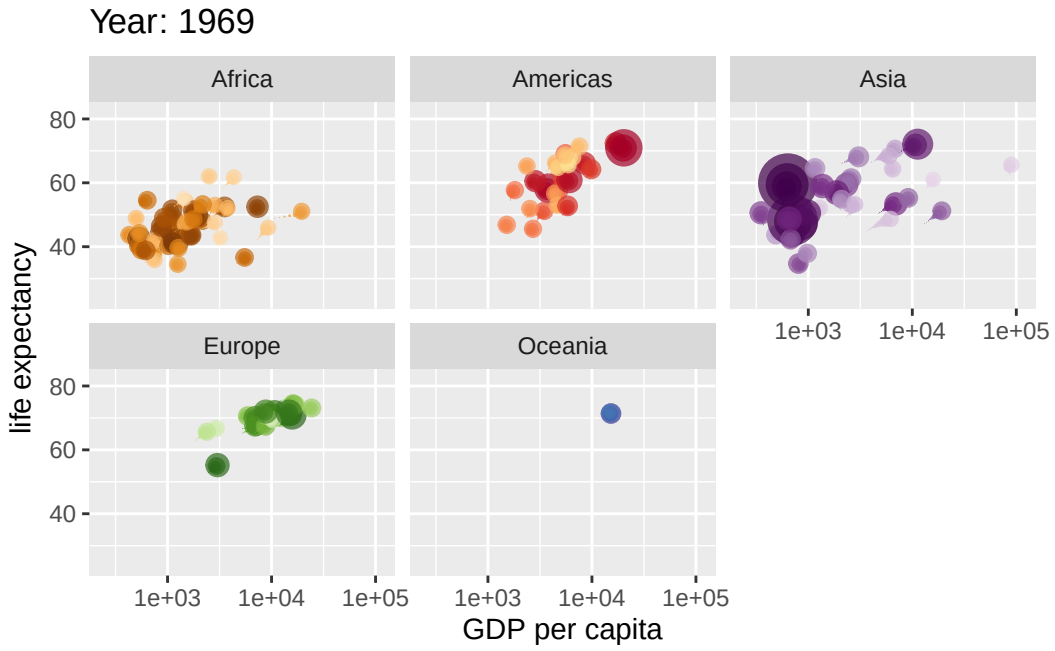
1e+04
1e+05
GDP per capita


Year: 1969
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life expectancy
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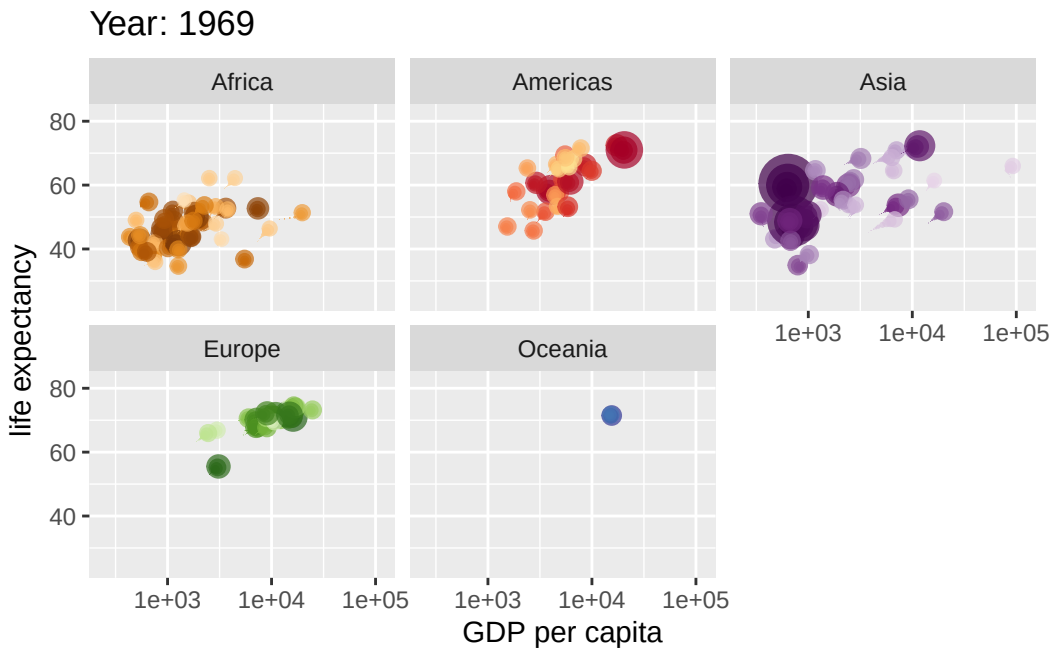
GDP per capita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life expectancy
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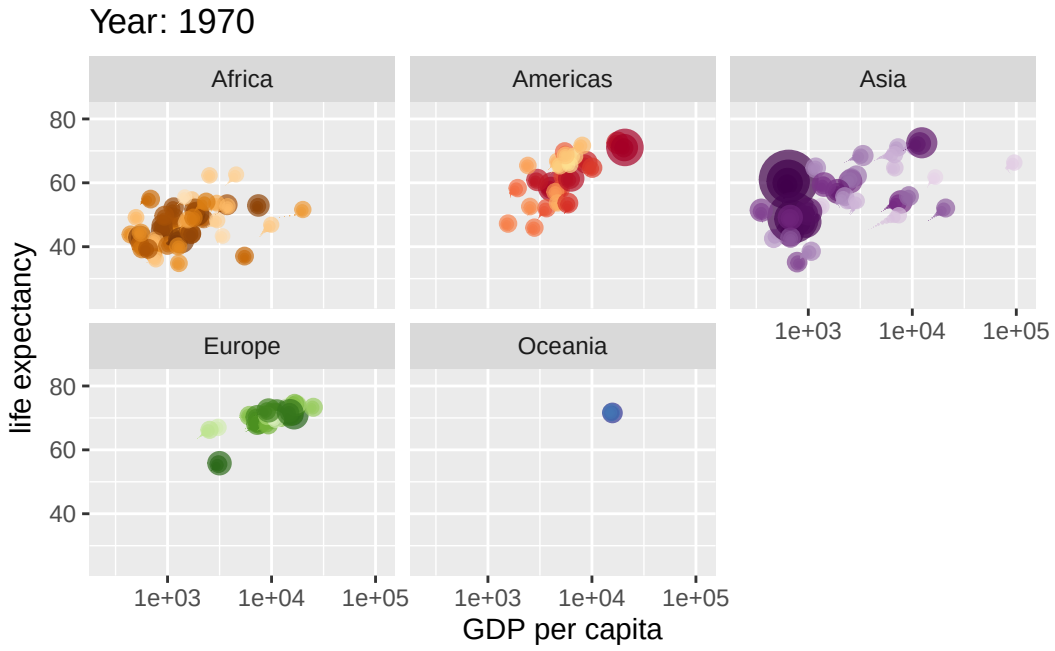
GDP per capita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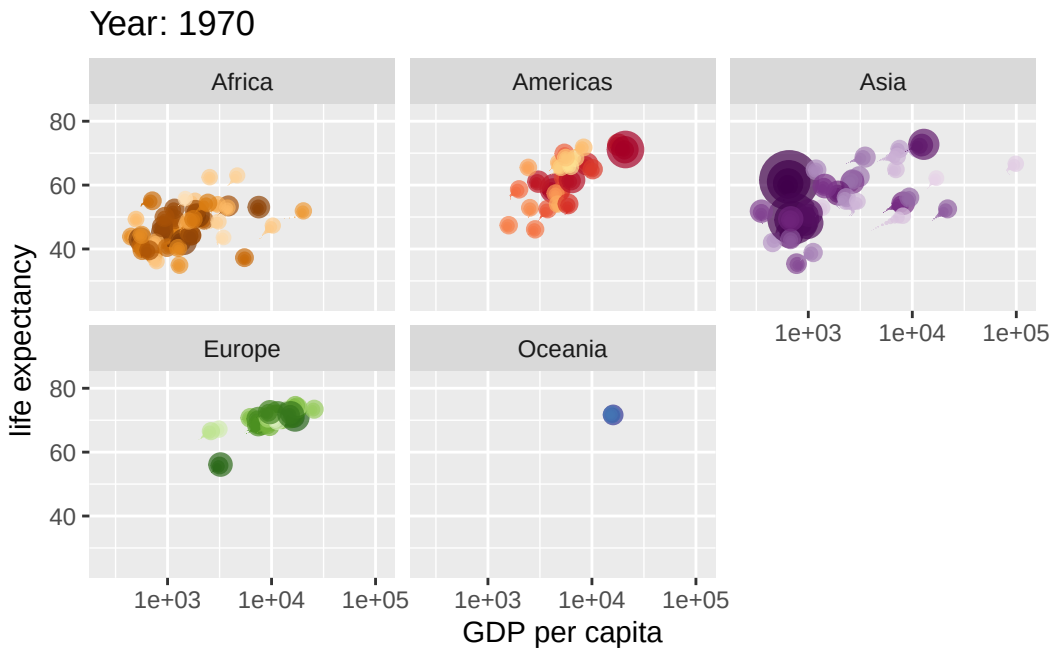
1e+05
GDP per capita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life expectancy
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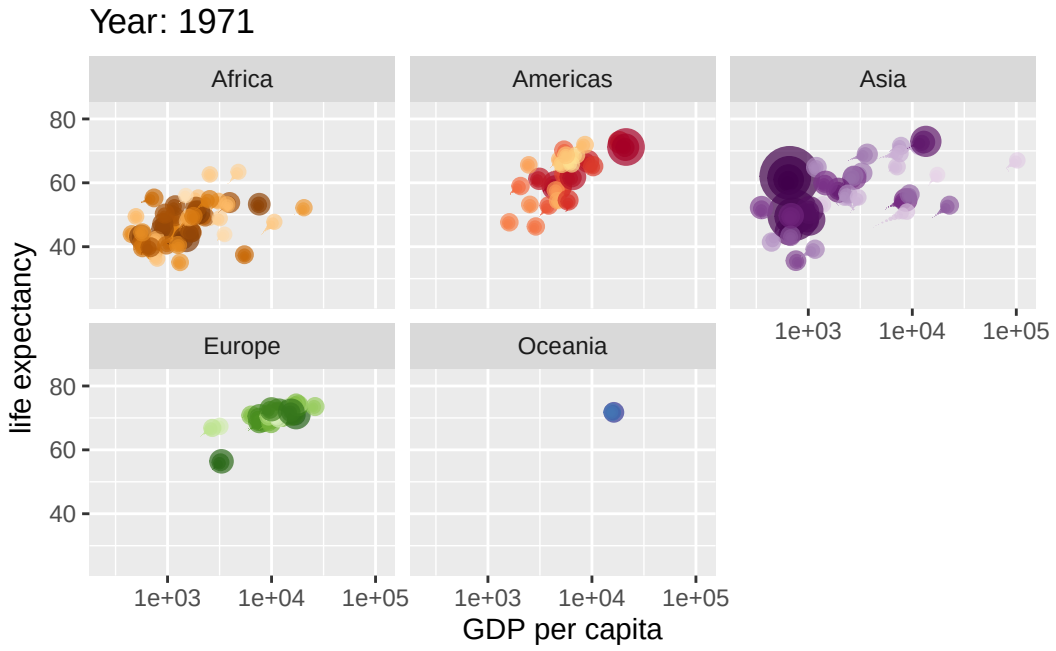
GDP per capita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Europe
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life expectancy
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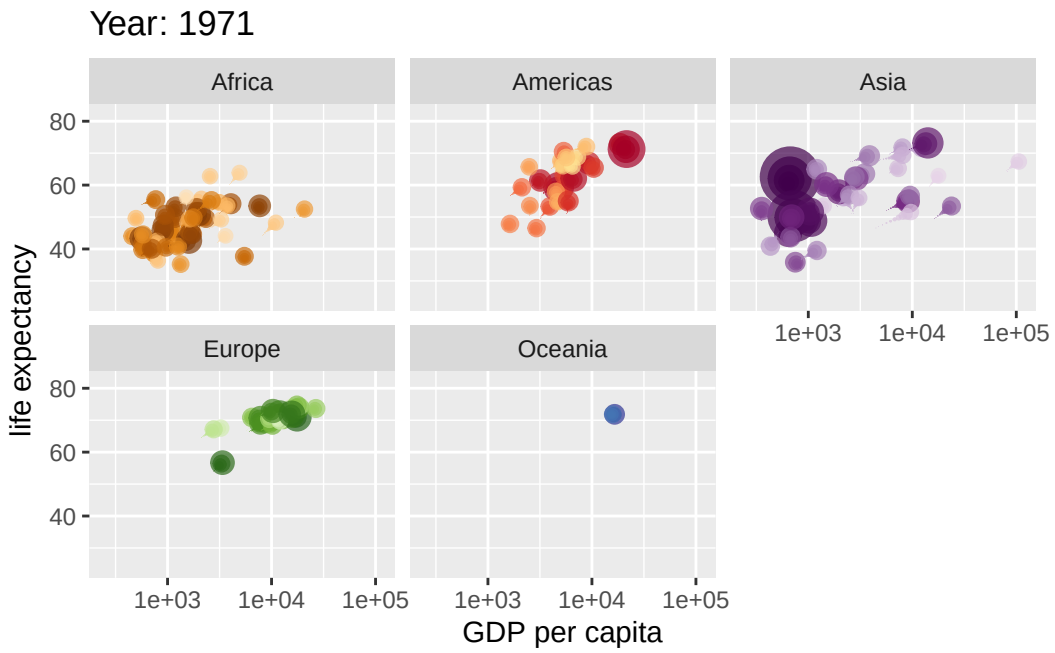
1e+05
GDP per capita


Year: 1971
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life expectancy
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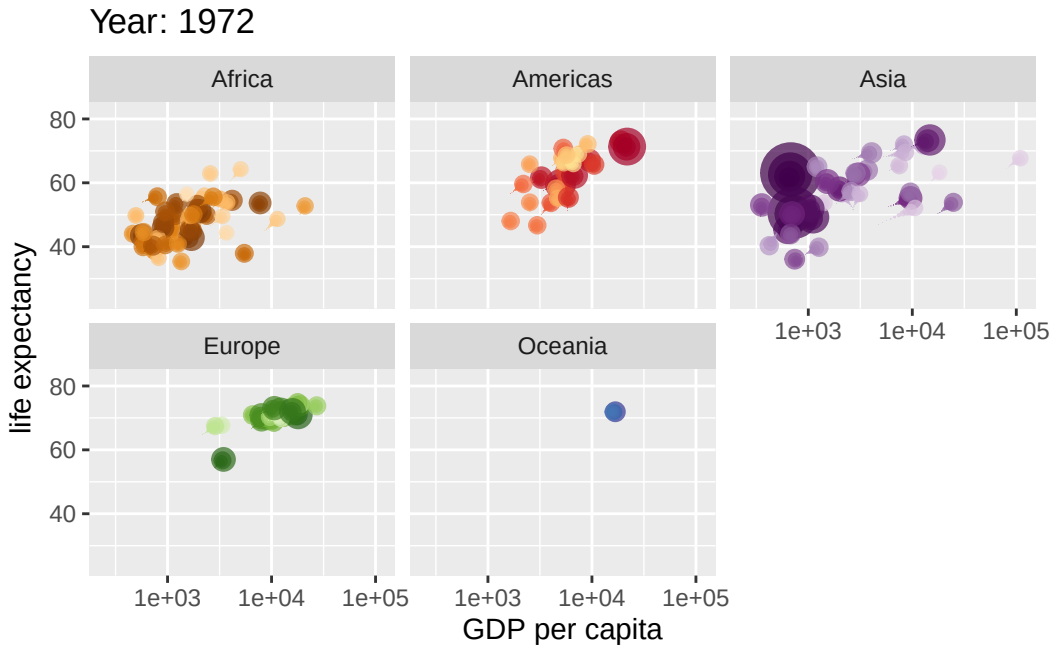
1e+05
GDP per capita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Oceania
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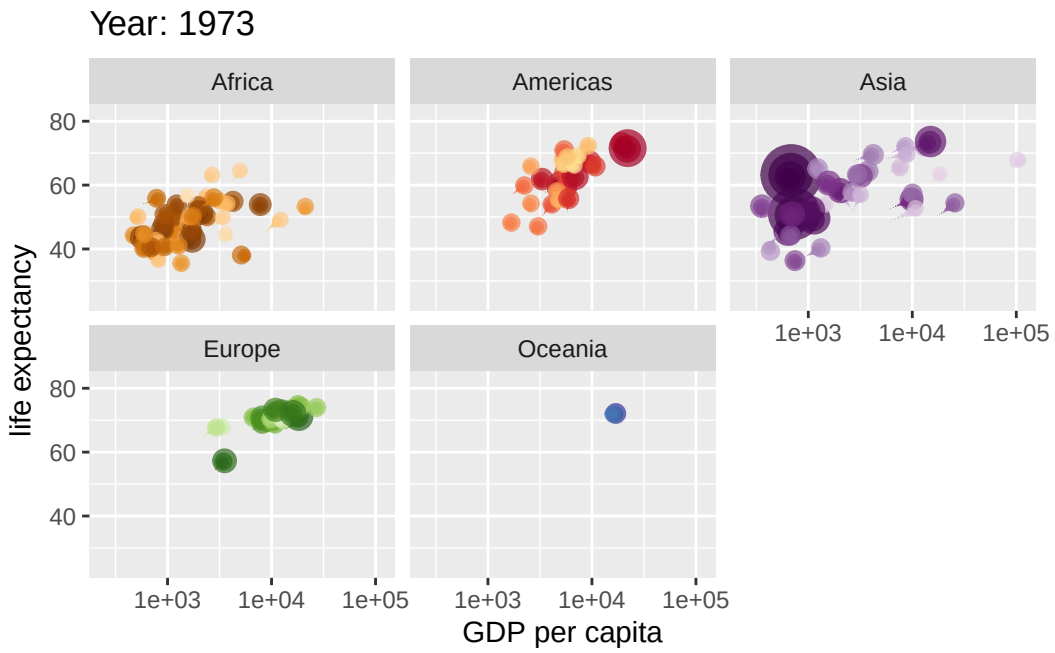
1e+05
GDP per capita


Year: 1973
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Oceania
life expectancy
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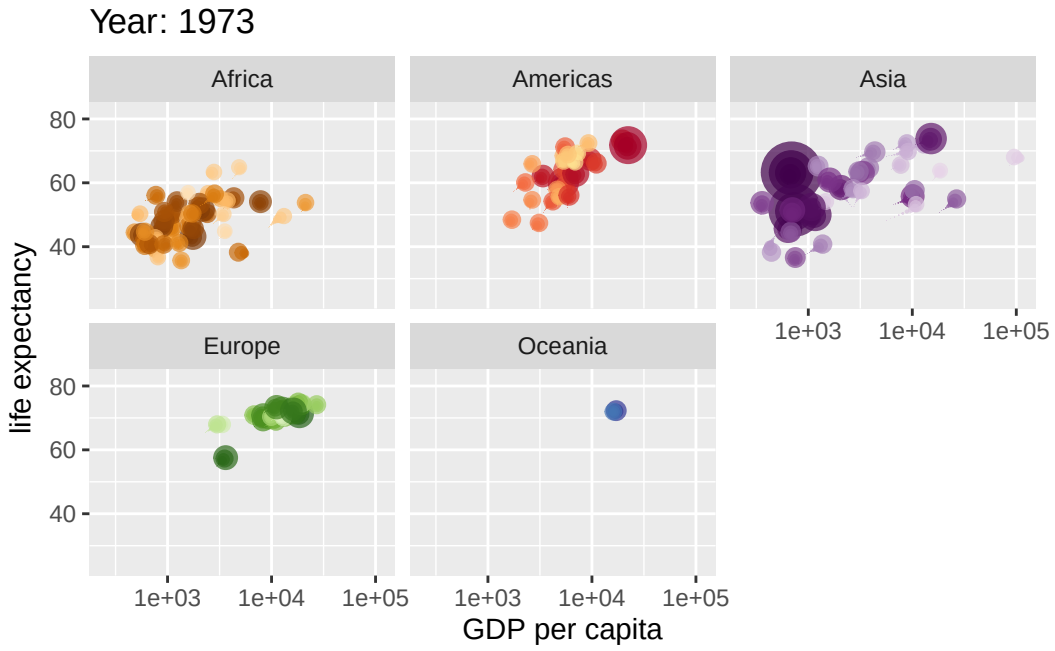
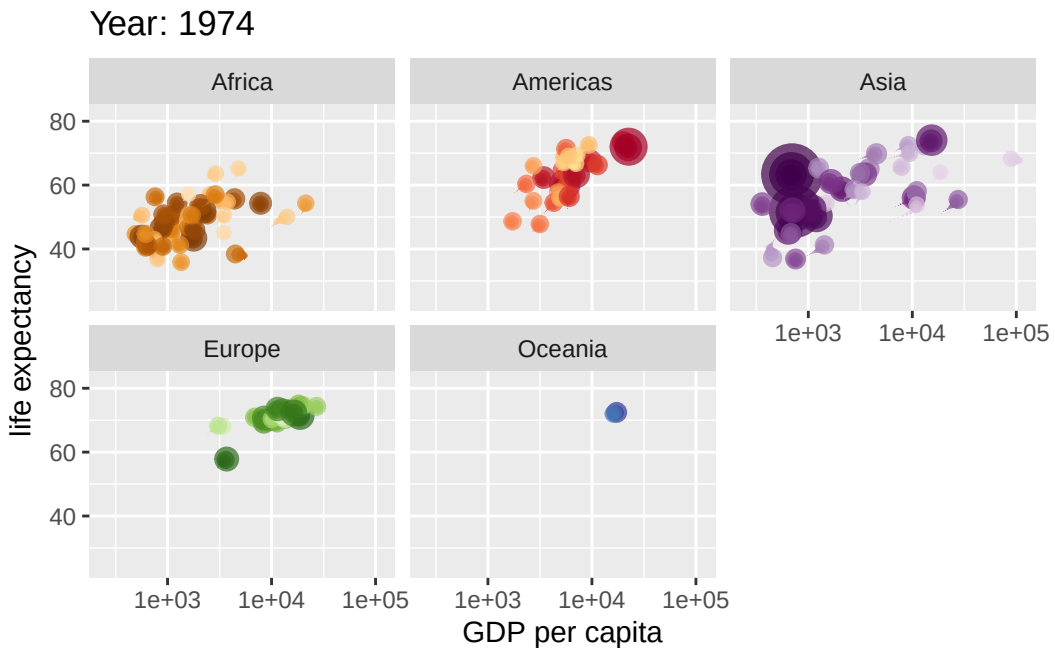
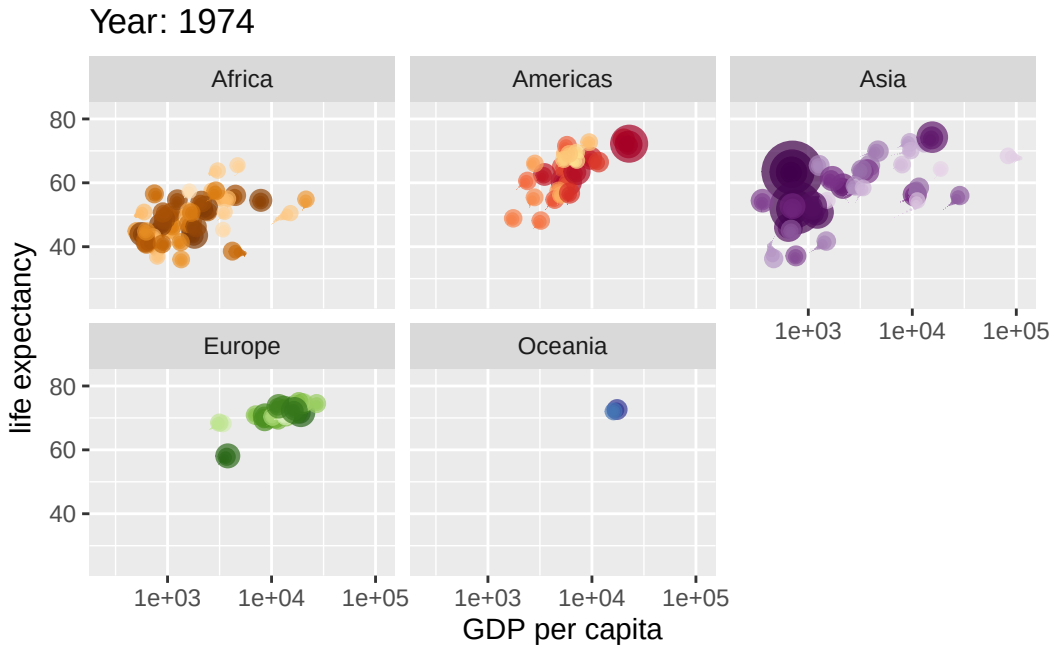
1e+04
1e+05
GDP per capita


Year: 1973
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1974
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1974
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
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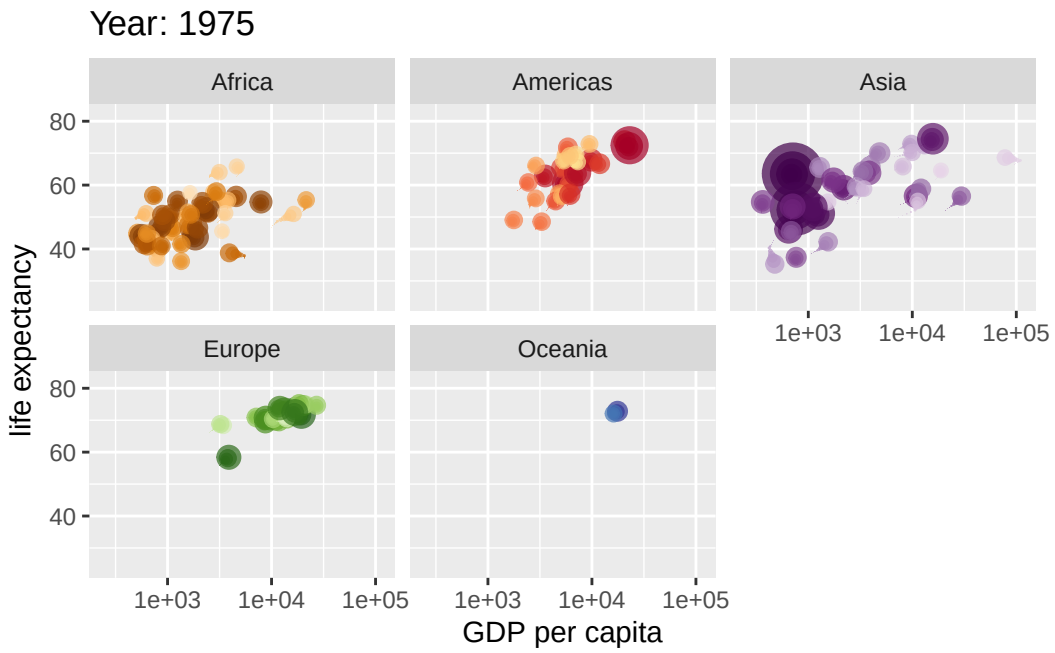
1e+05
GDP per capita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Africa
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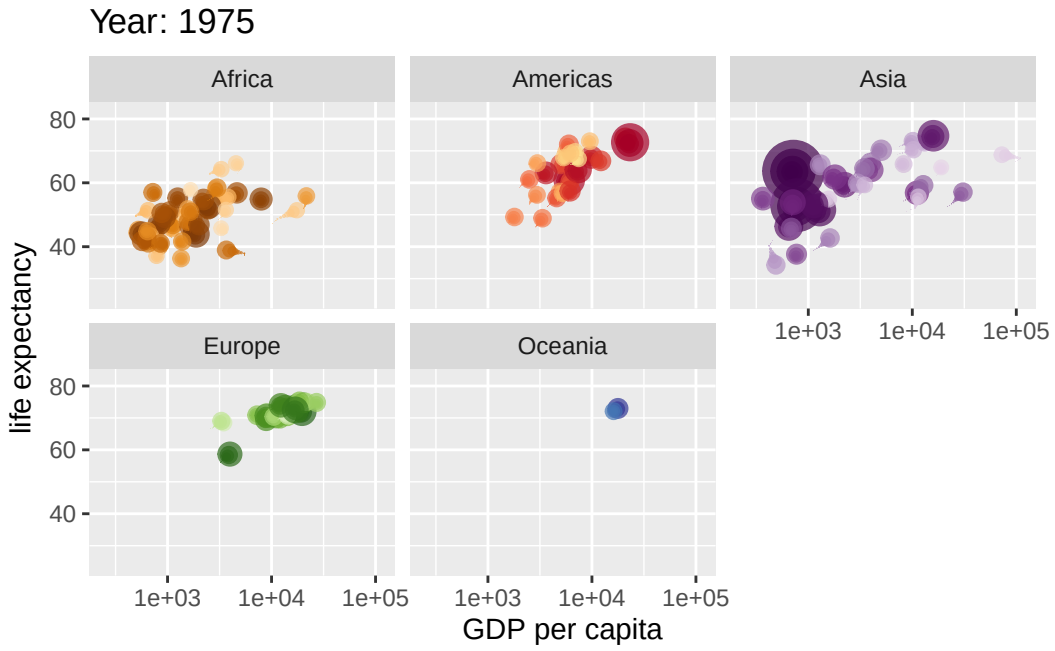
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1975
Africa
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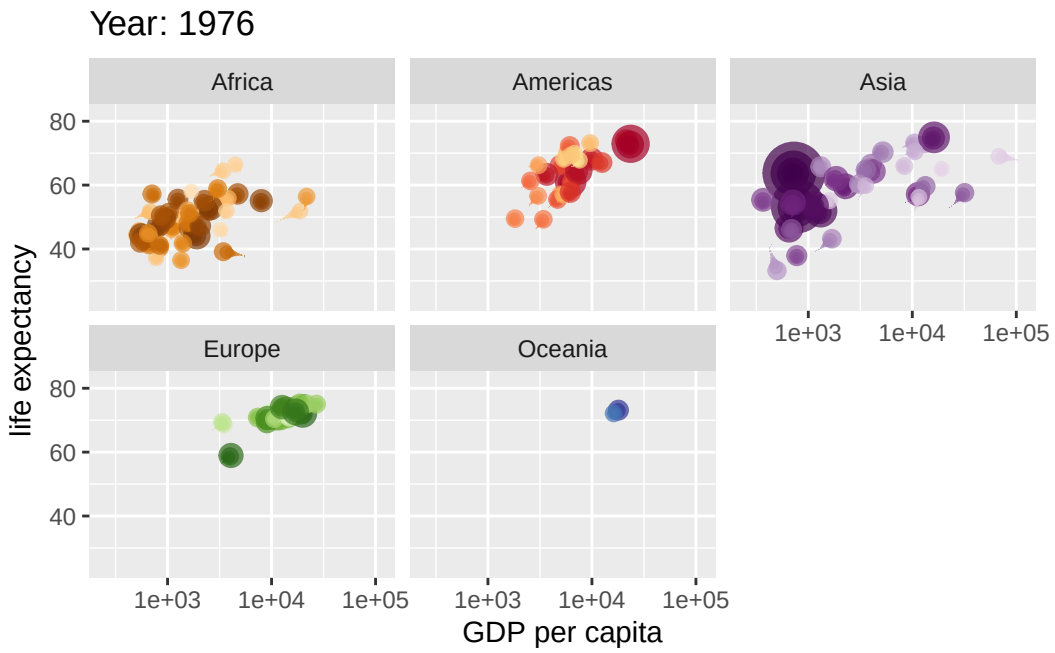
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1976
Africa
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Asia
Europe
Oceania
life expectancy
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60
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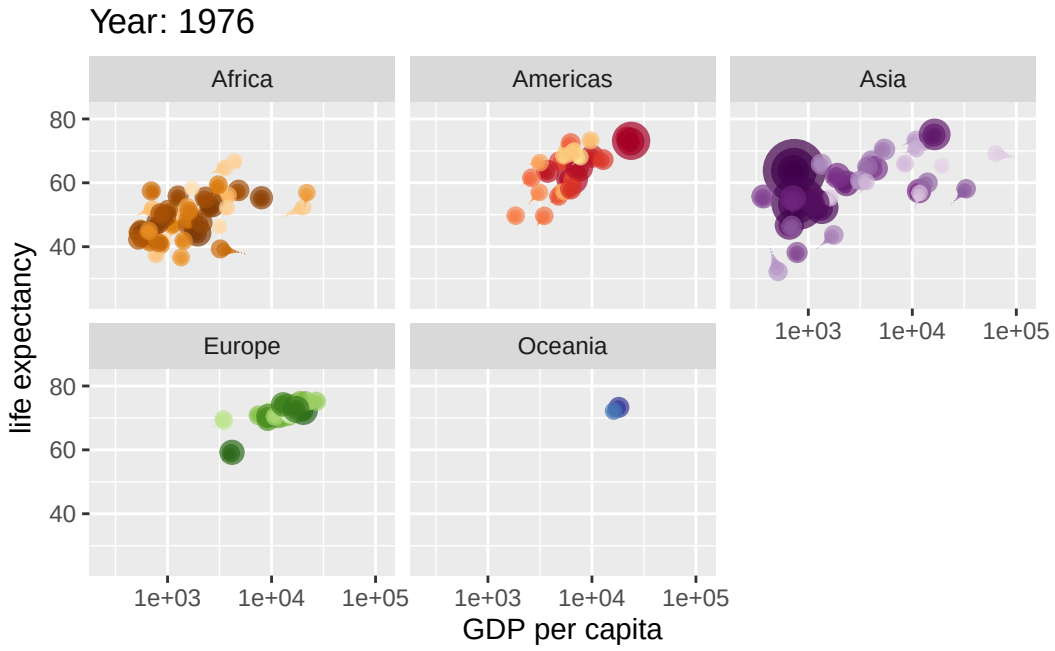
1e+04
1e+05
GDP per capita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Africa
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Asia
Europe
Oceania
life expectancy
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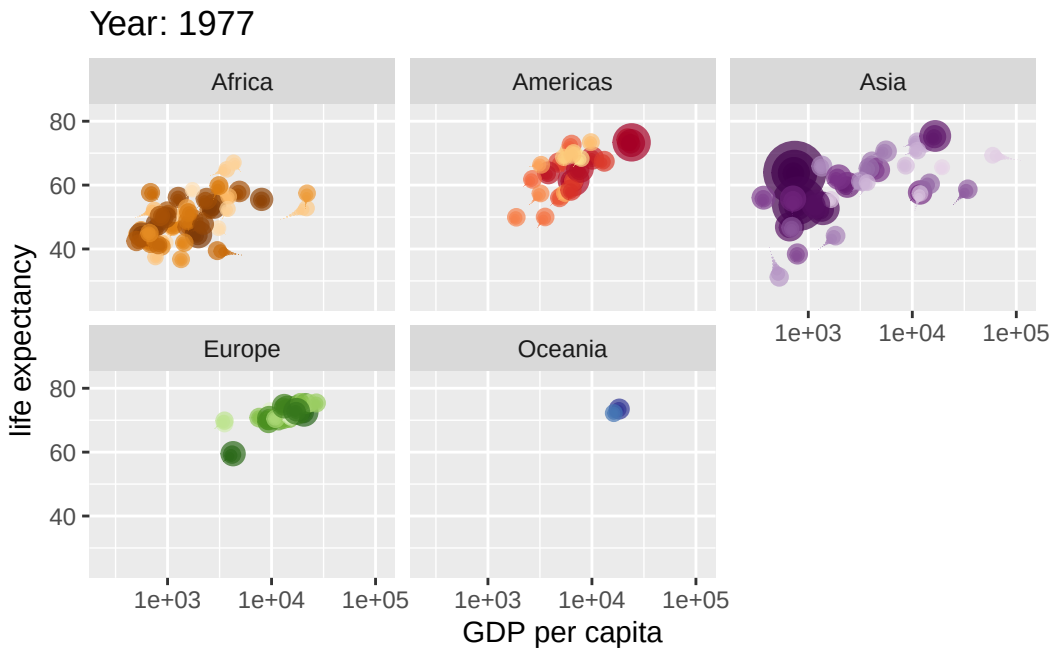
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1977
Africa
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Asia
Europe
Oceania
life expectancy
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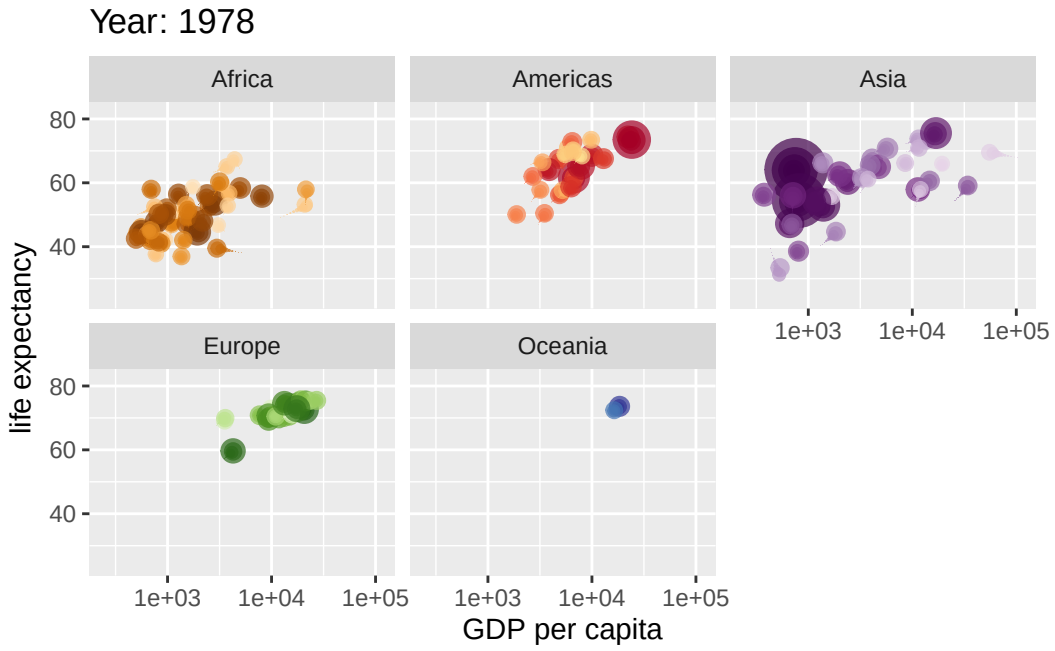
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1978
Africa
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Asia
Europe
Oceania
life expectancy
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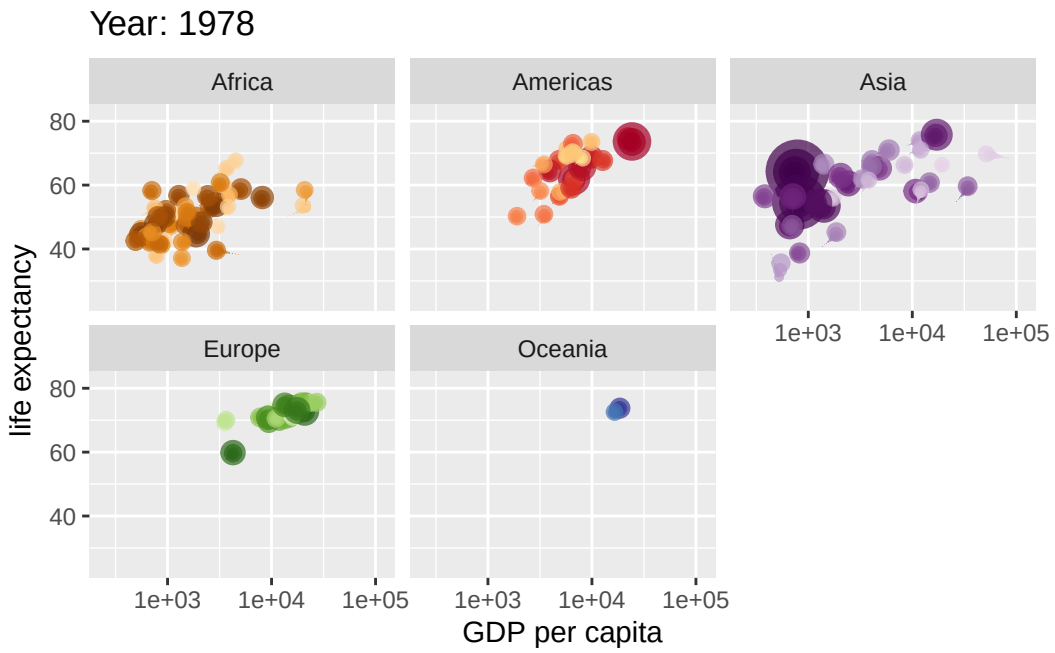
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1978
Africa
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Asia
Europe
Oceania
life expectancy
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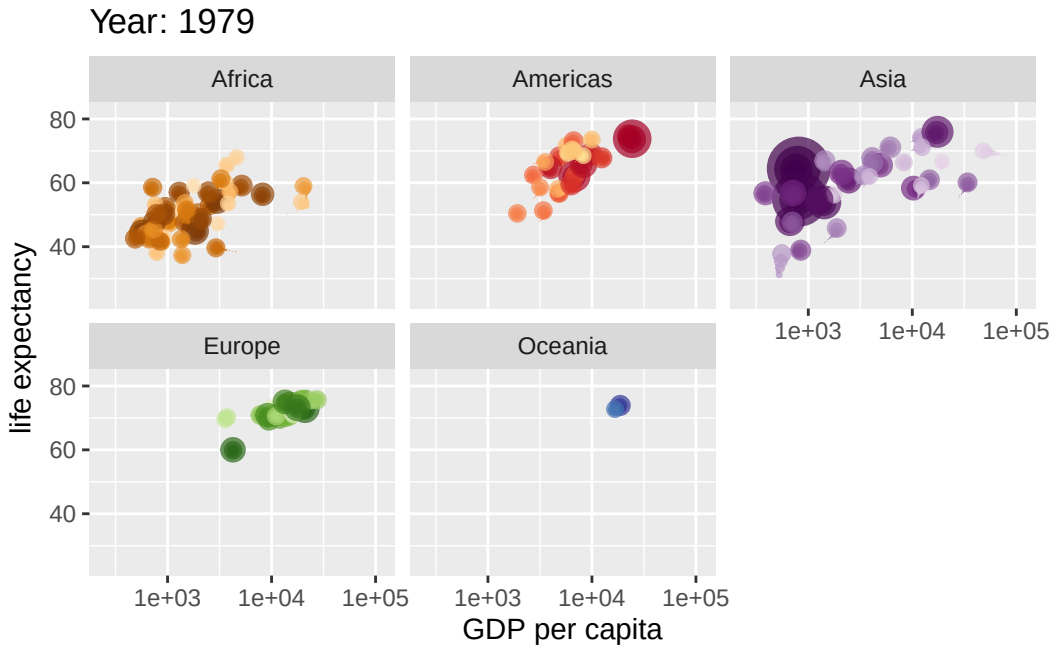
1e+03
1e+04
1e+05
GDP per capita


Year: 1979
Africa
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Asia
Europe
Oceania
life expectancy
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60
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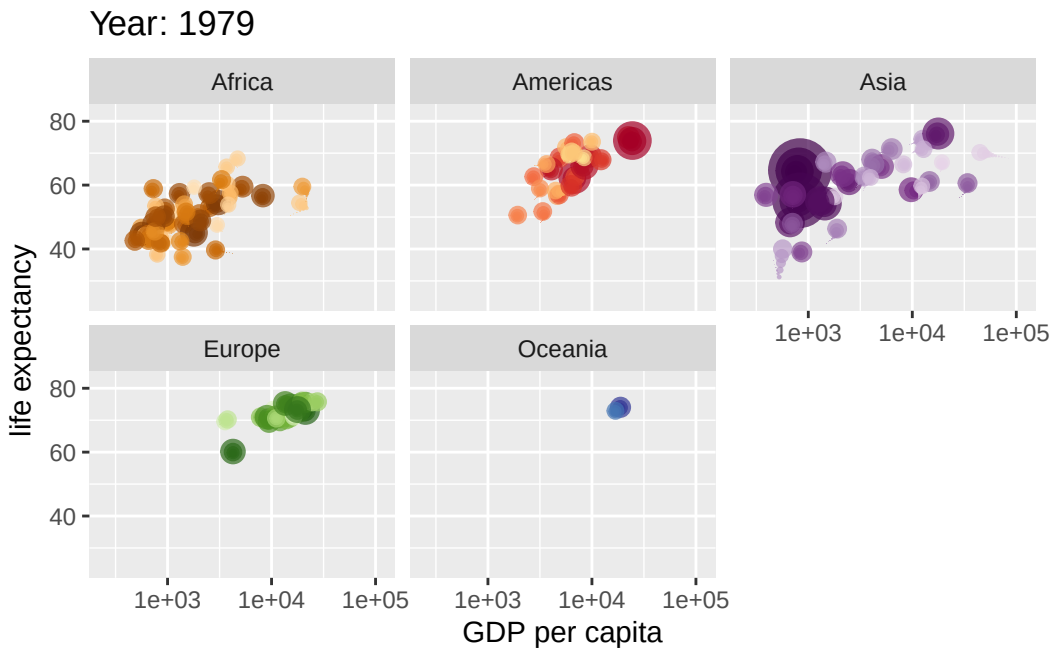
1e+03
1e+04
1e+05
GDP per capita


Year: 1979
Africa
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Asia
Europe
Oceania
life expectancy
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60
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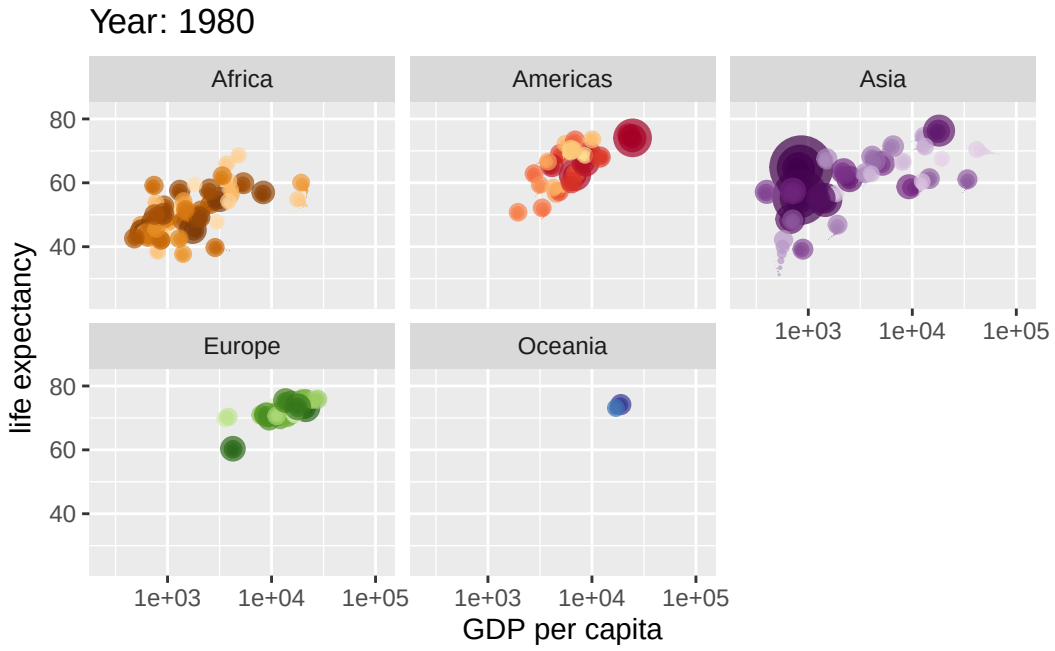
1e+03
1e+04
1e+05
GDP per capita


Year: 1980
Africa
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Asia
Europe
Oceania
life expectancy
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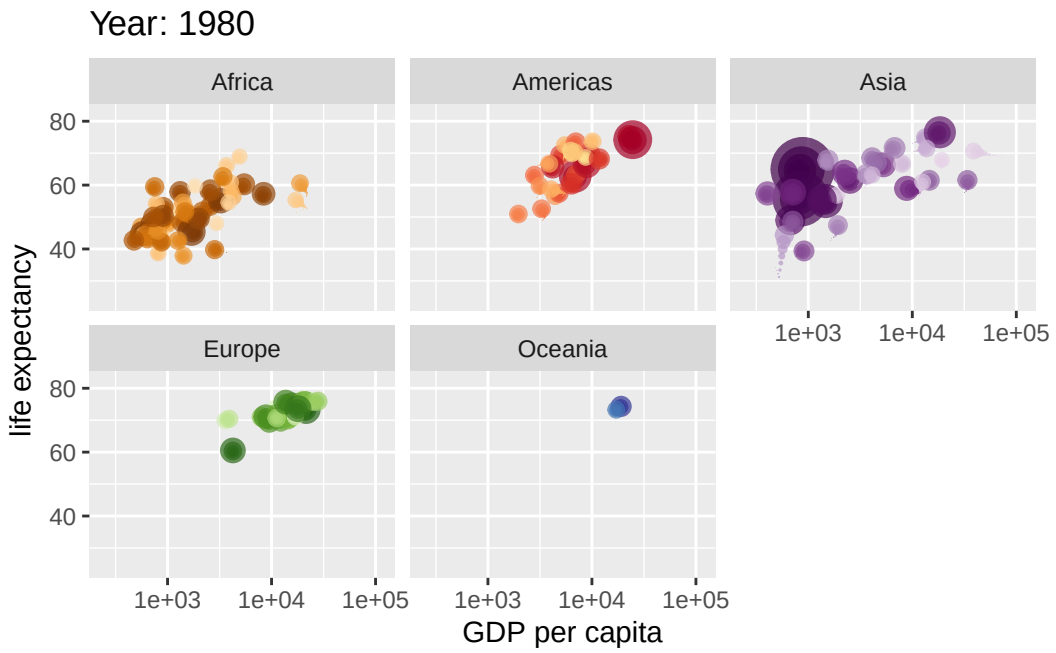
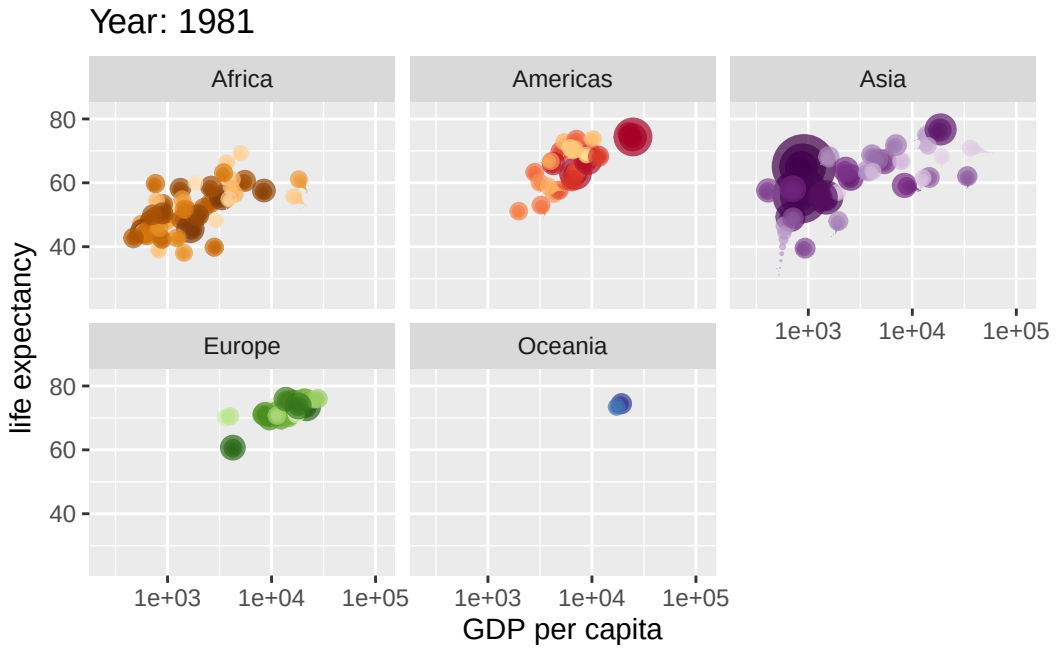
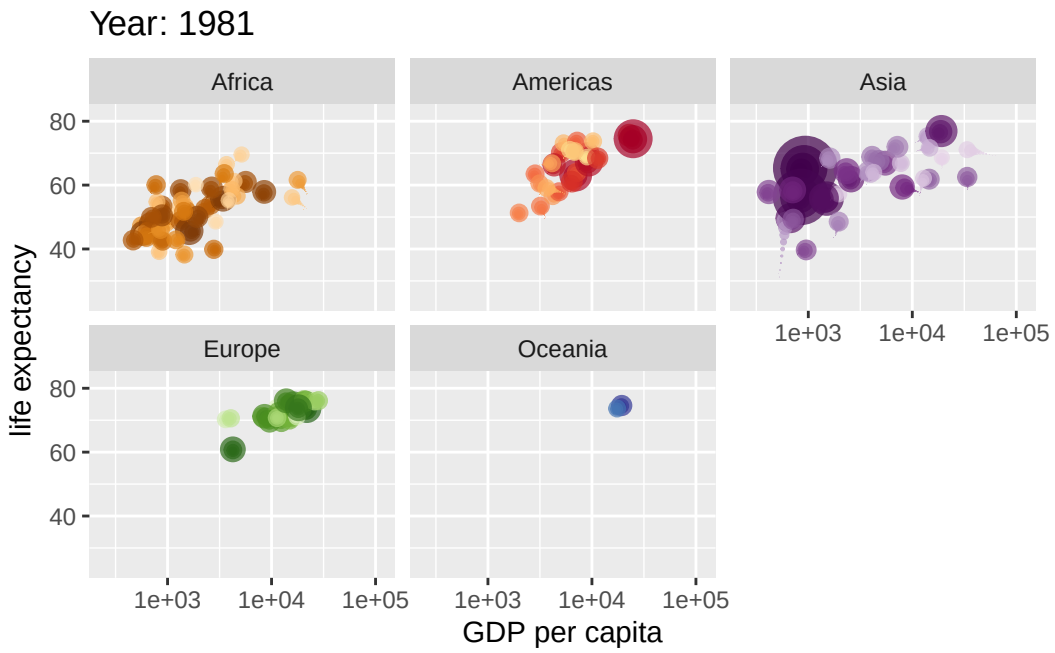
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1980
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1981
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1981
Africa
Americas
Asia
Europe
Oceania
life expectancy
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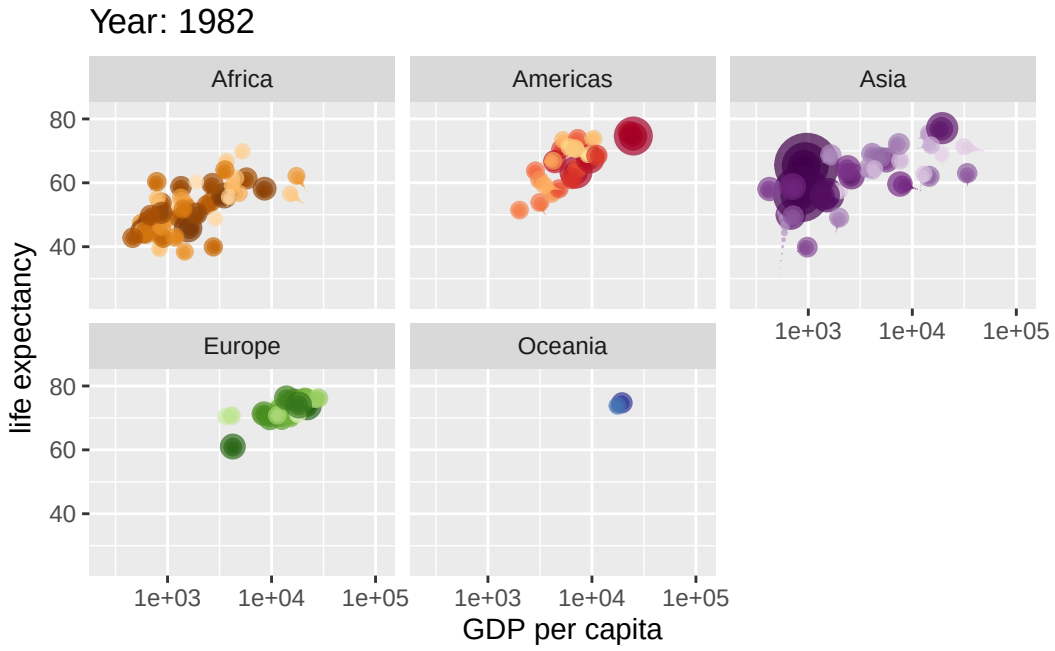
1e+03
1e+04
1e+05
GDP per capita


Year: 1982
Africa
Americas
Asia
Europe
Oceania
life expectancy
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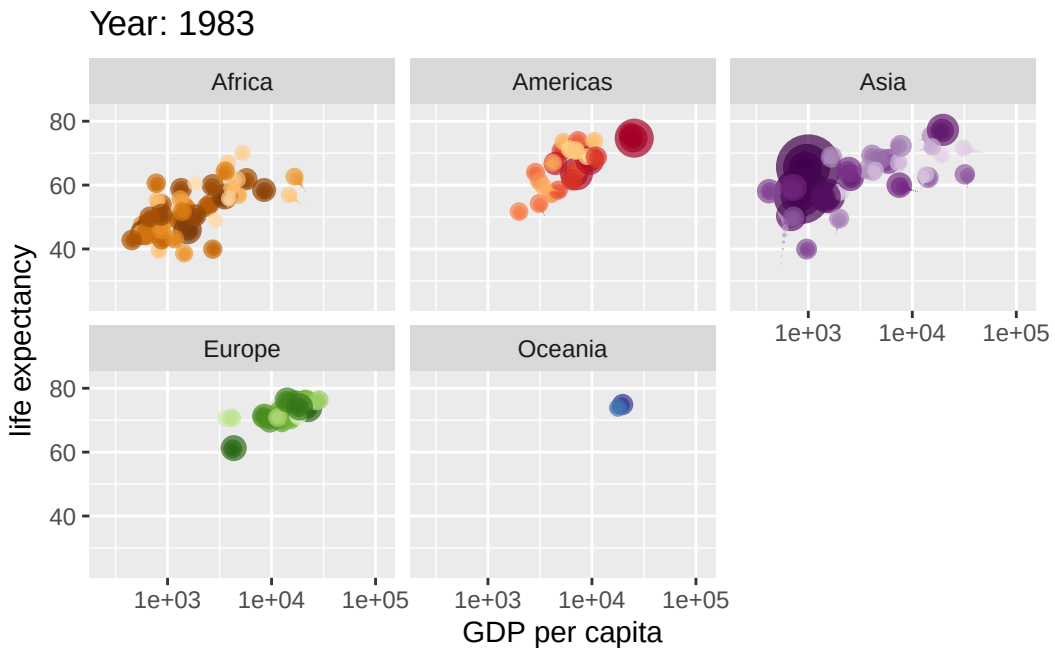
1e+03
1e+04
1e+05
GDP per capita


Year: 1983
Africa
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Asia
Europe
Oceania
life expectancy
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60
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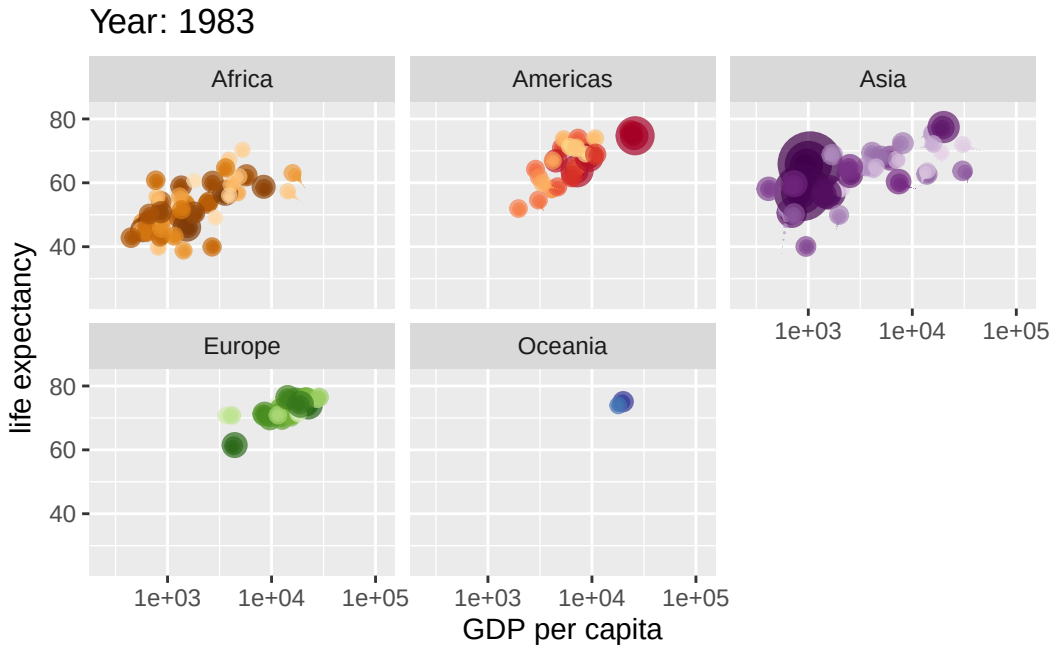
1e+03
1e+04
1e+05
GDP per capita


Year: 1983
Africa
Americas
Asia
Europe
Oceania
life expectancy
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60
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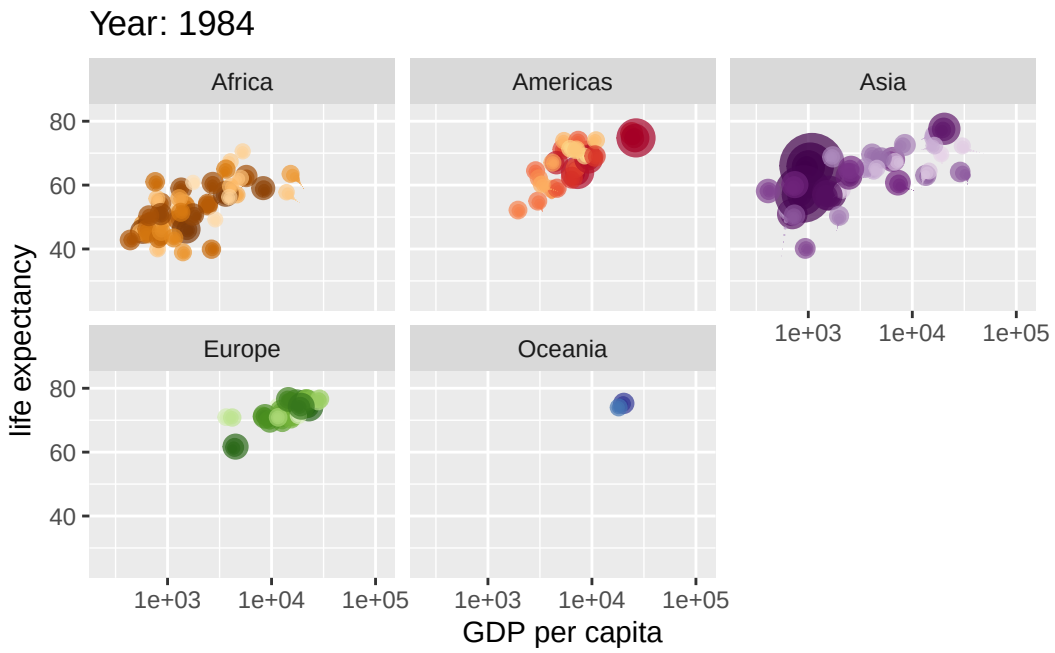
1e+03
1e+04
1e+05
GDP per capita


Year: 1984
Africa
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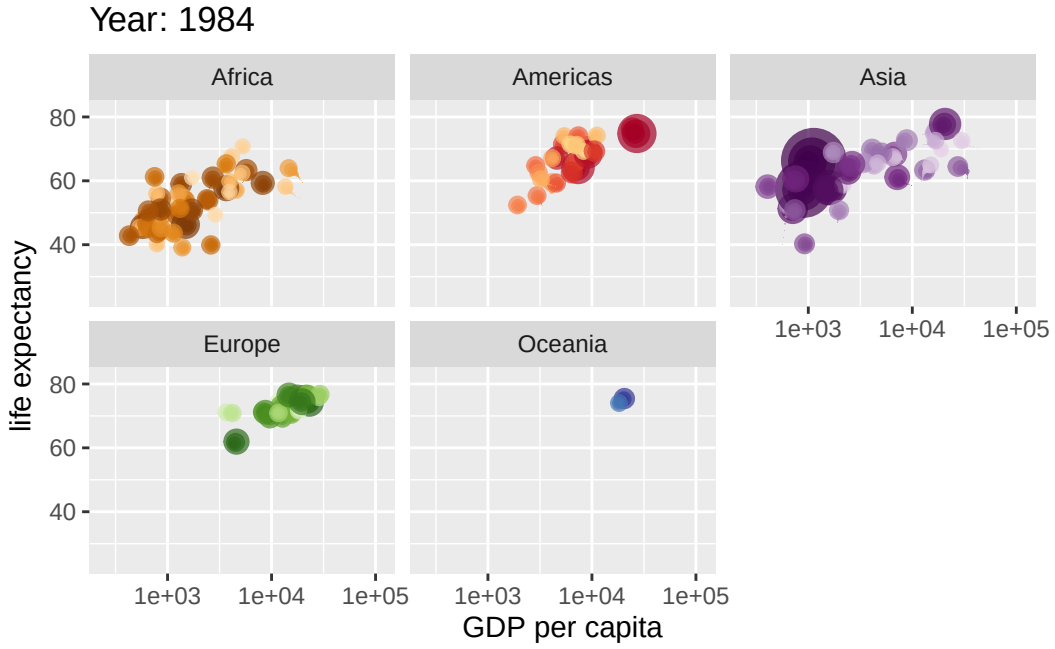
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1984
Africa
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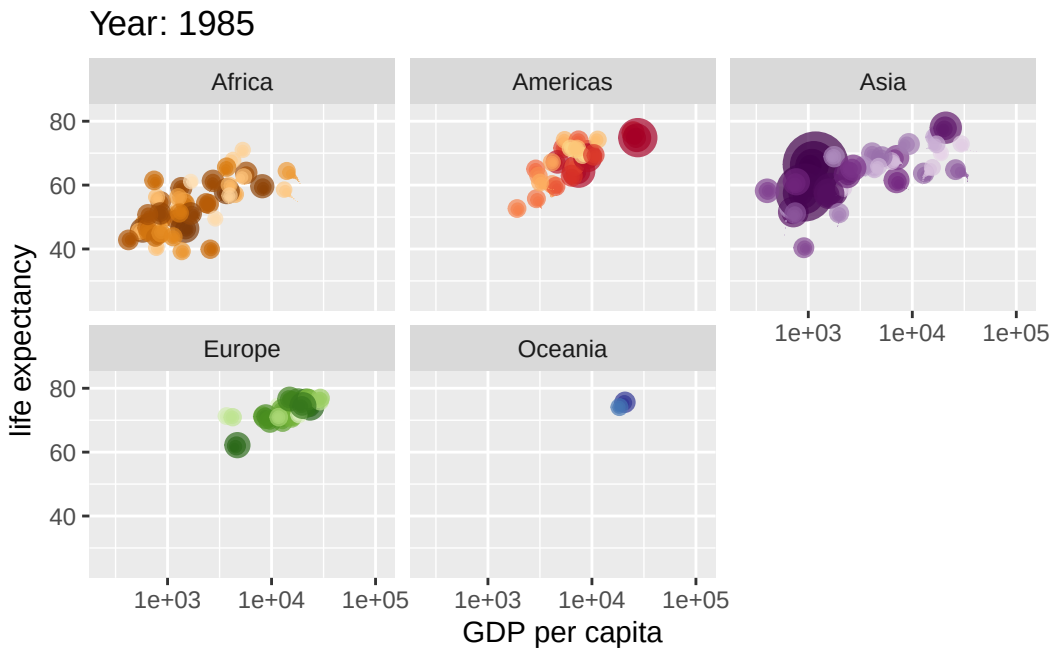
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1985
Africa
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Asia
Europe
Oceania
life expectancy
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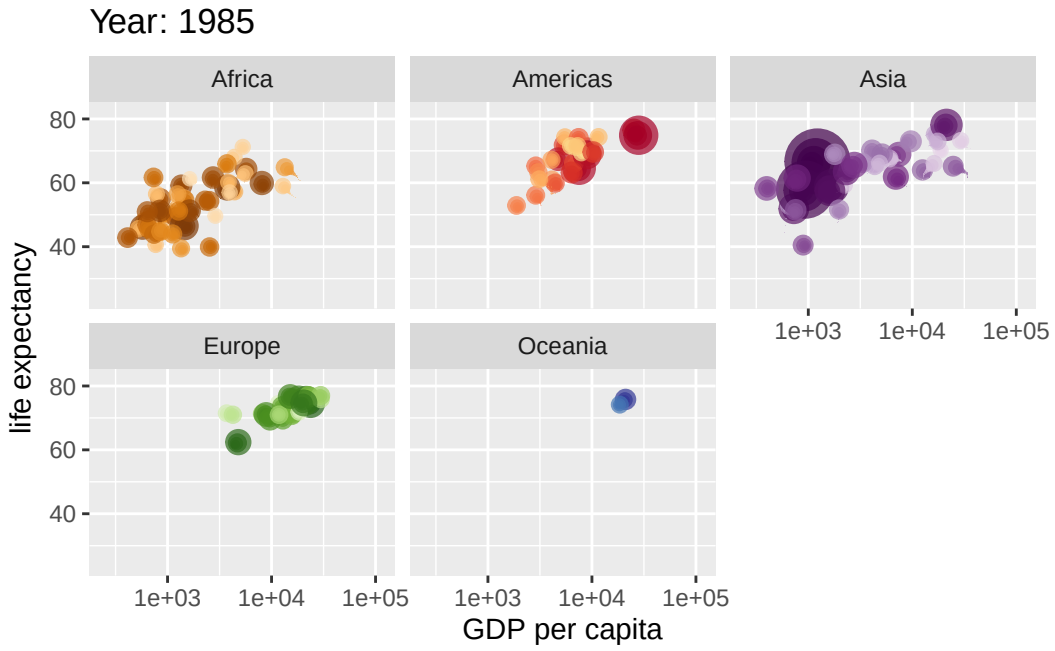
1e+03
1e+04
1e+05
GDP per capita


Year: 1985
Africa
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Asia
Europe
Oceania
life expectancy
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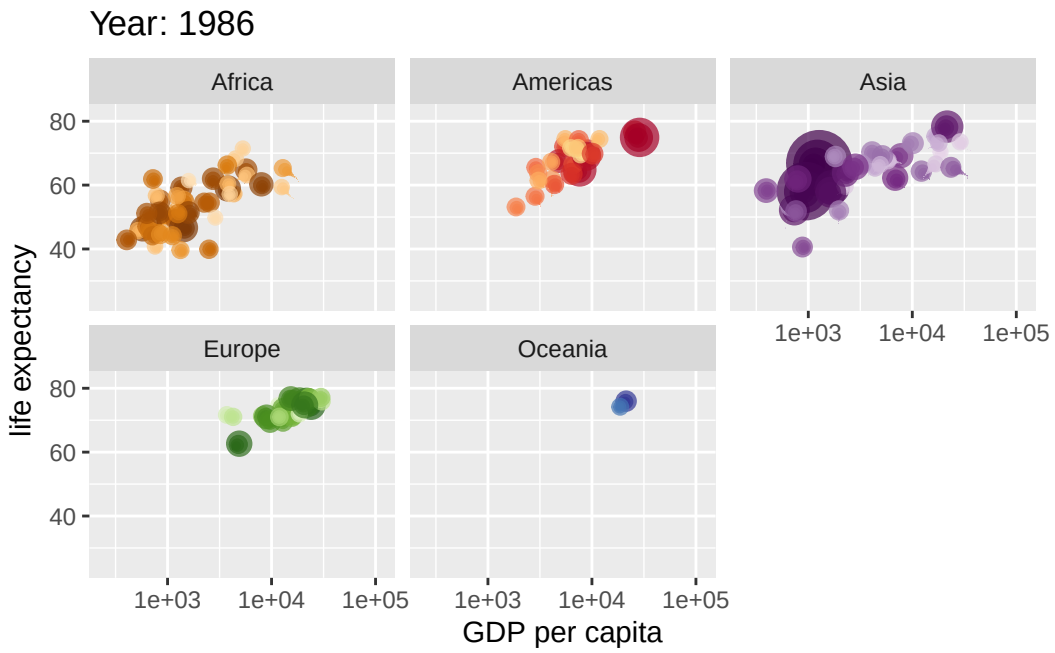
1e+03
1e+04
1e+05
GDP per capita


Year: 1986
Africa
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Asia
Europe
Oceania
life expectancy
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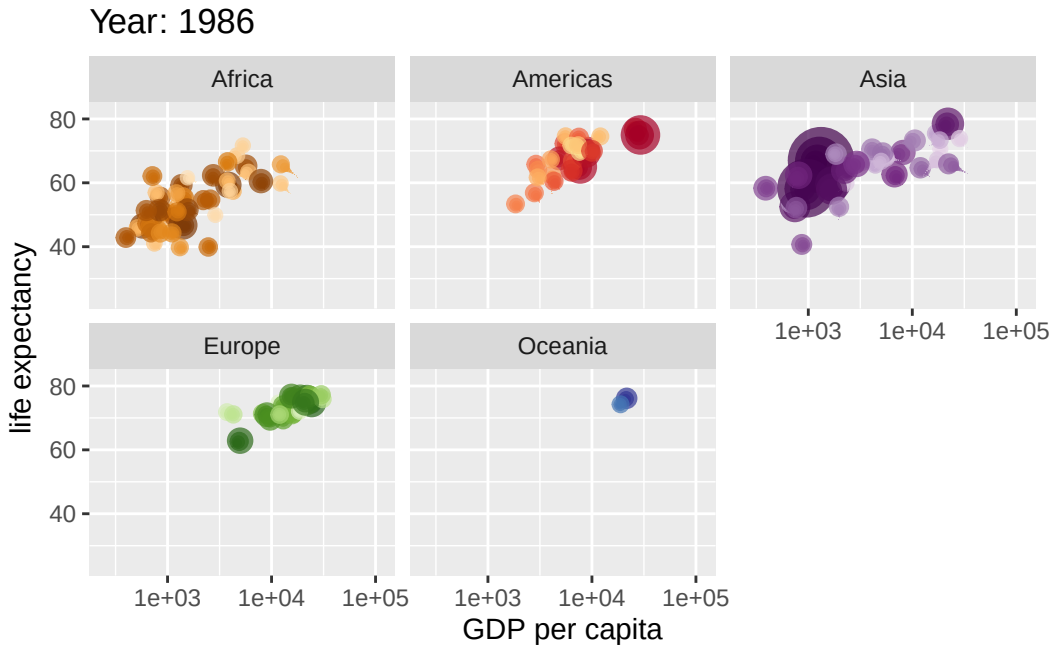
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1986
Africa
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Asia
Europe
Oceania
life expectancy
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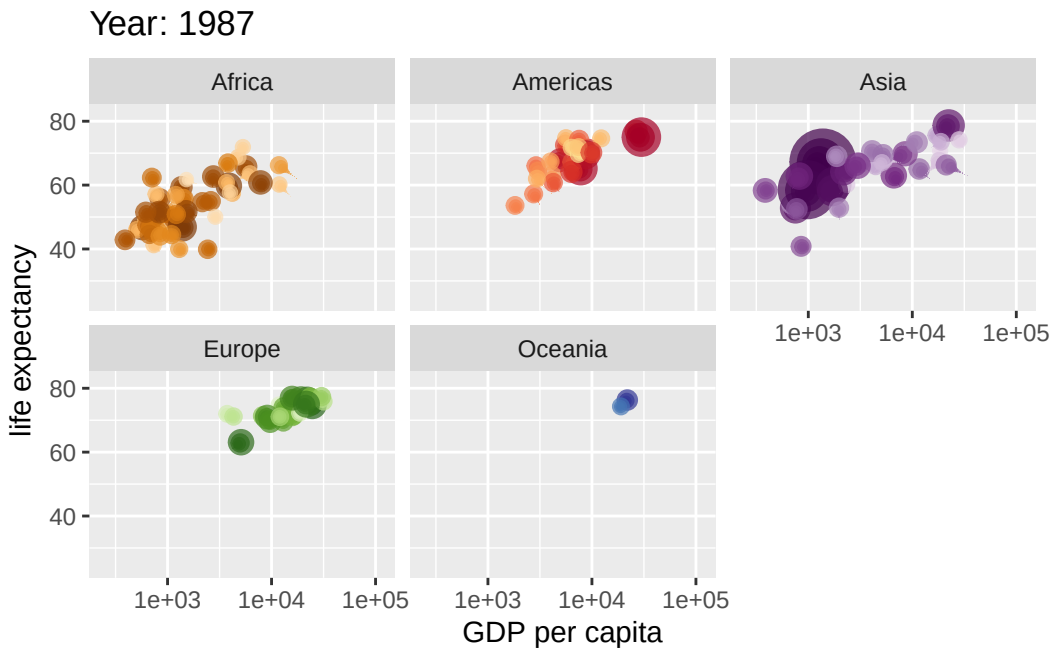
1e+03
1e+04
1e+05
GDP per capita


Year: 1987
Africa
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Oceania
life expectancy
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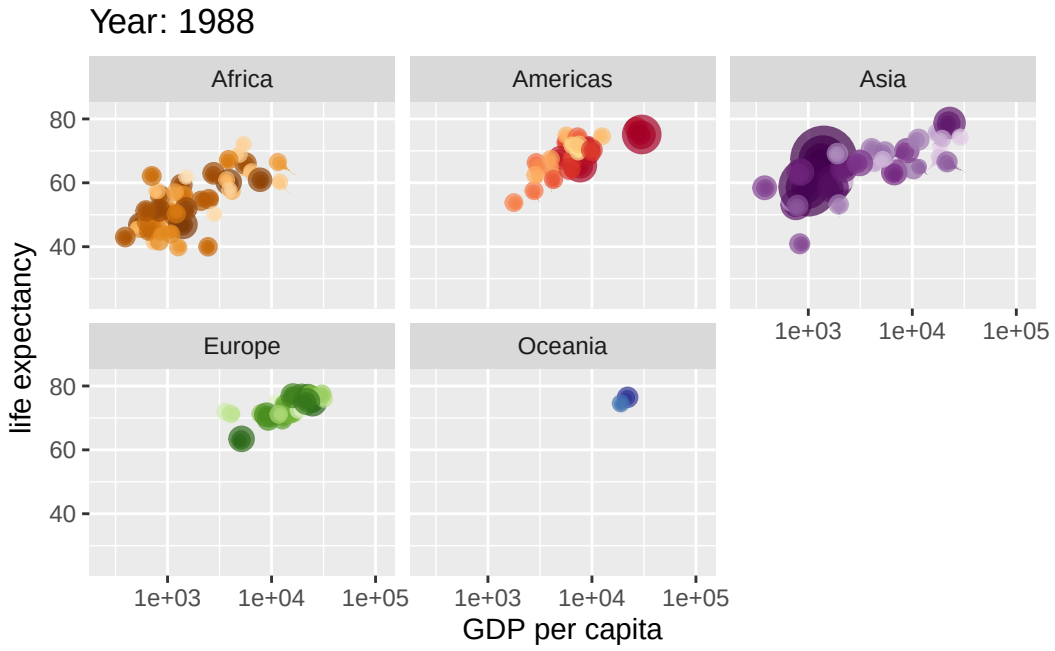
1e+05
GDP per capita


Year: 1988
Africa
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Asia
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Oceania
life expectancy
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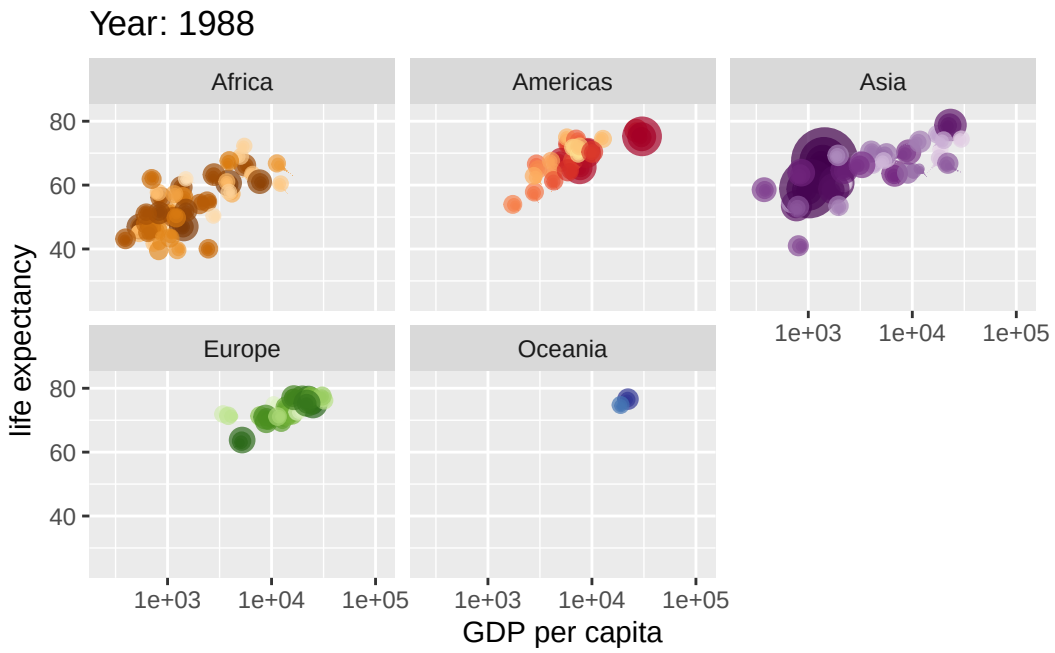
1e+03
1e+04
1e+05
GDP per capita


Year: 1988
Africa
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Asia
Europe
Oceania
life expectancy
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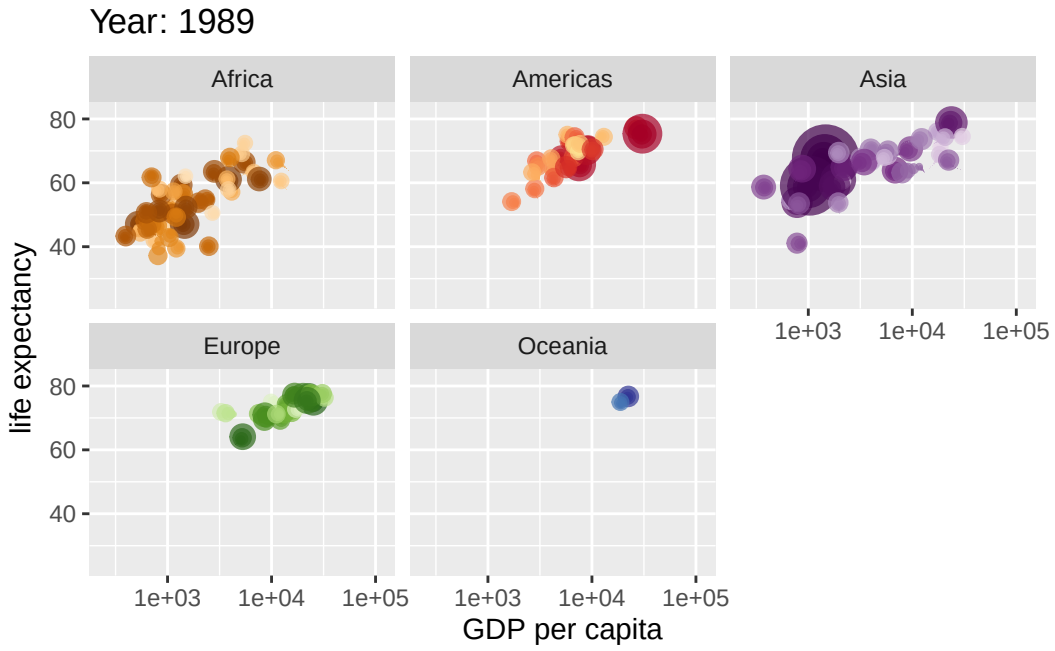
1e+04
1e+05
GDP per capita


Year: 1989
Africa
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Asia
Europe
Oceania
life expectancy
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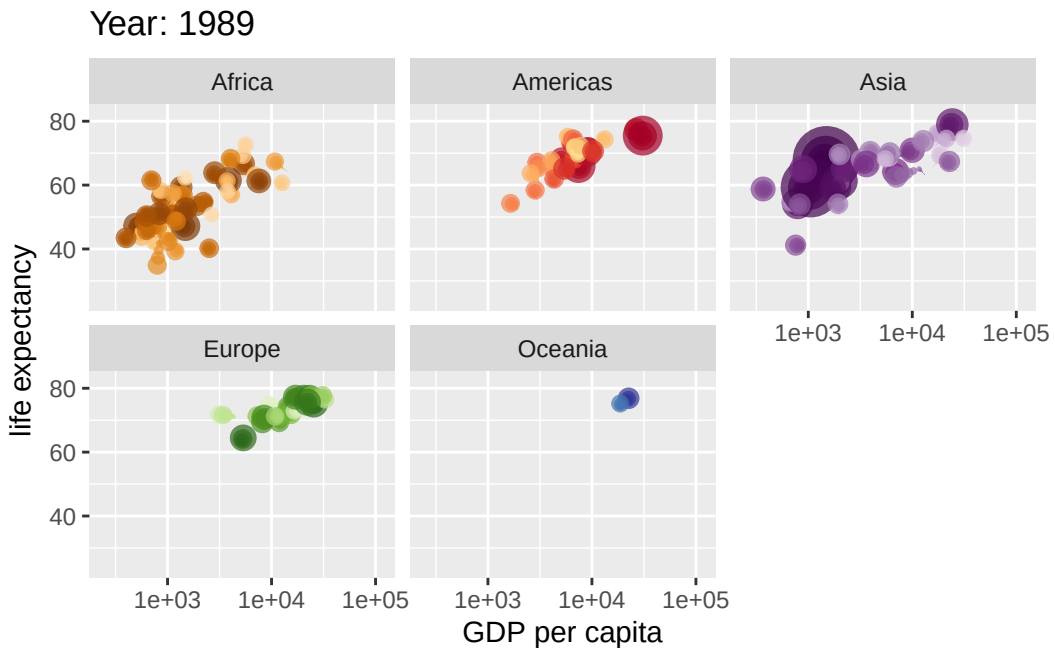
1e+03
1e+04
1e+05
GDP per capita


Year: 1989
Africa
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Asia
Europe
Oceania
life expectancy
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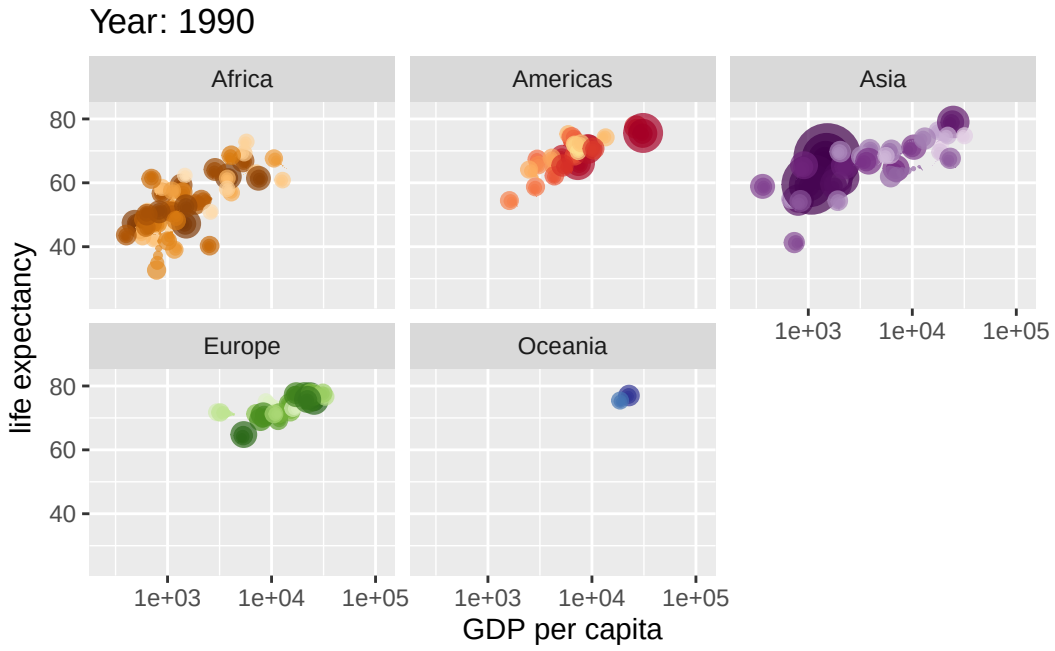
1e+03
1e+04
1e+05
GDP per capita


Year: 1990
Africa
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Europe
Oceania
life expectancy
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60
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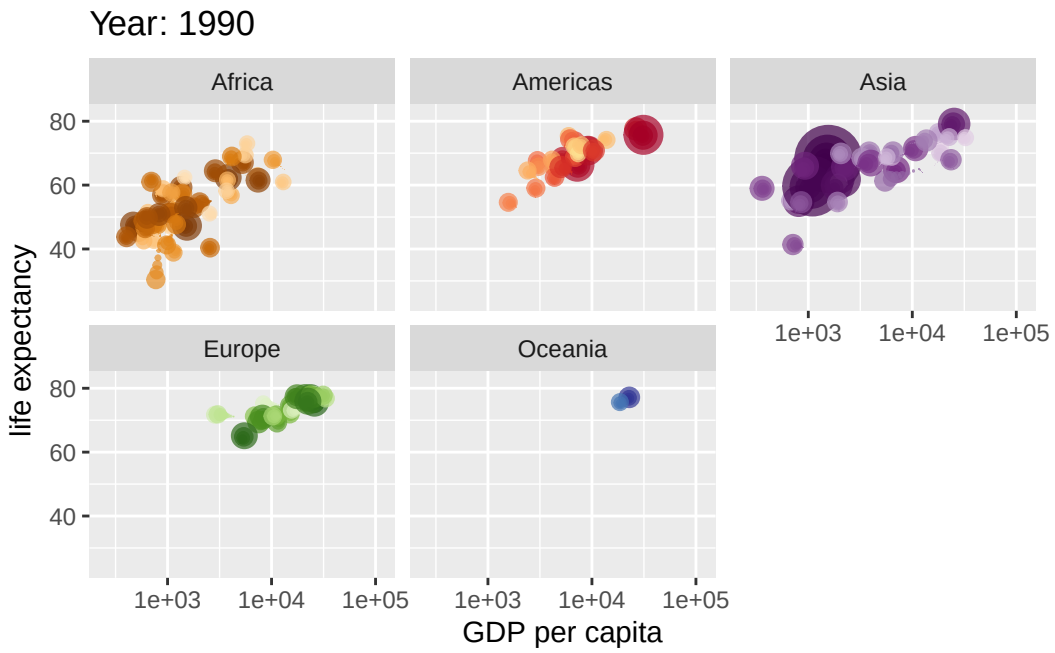
1e+03
1e+04
1e+05
GDP per capita


Year: 1990
Africa
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life expectancy
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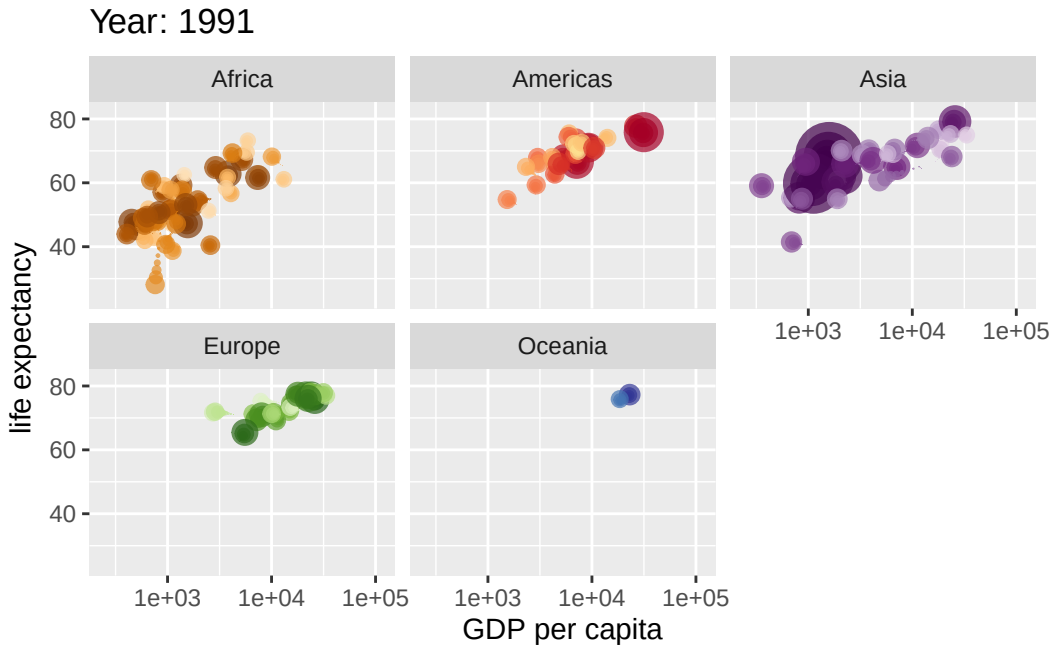
1e+04
1e+05
GDP per capita


Year: 1991
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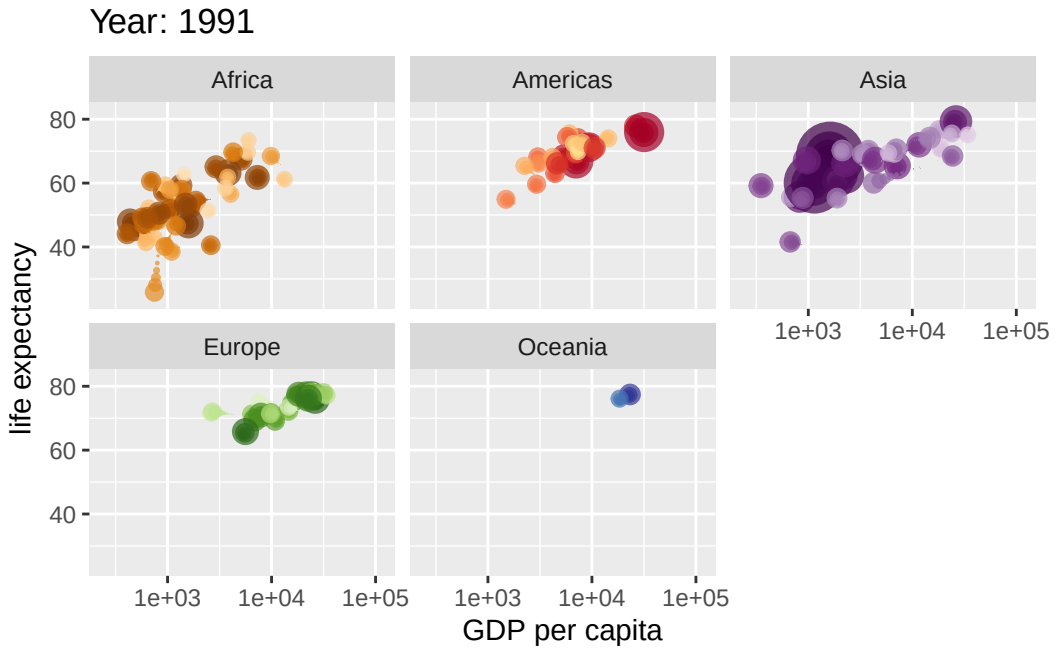
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1991
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Oceania
life expectancy
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1e+03
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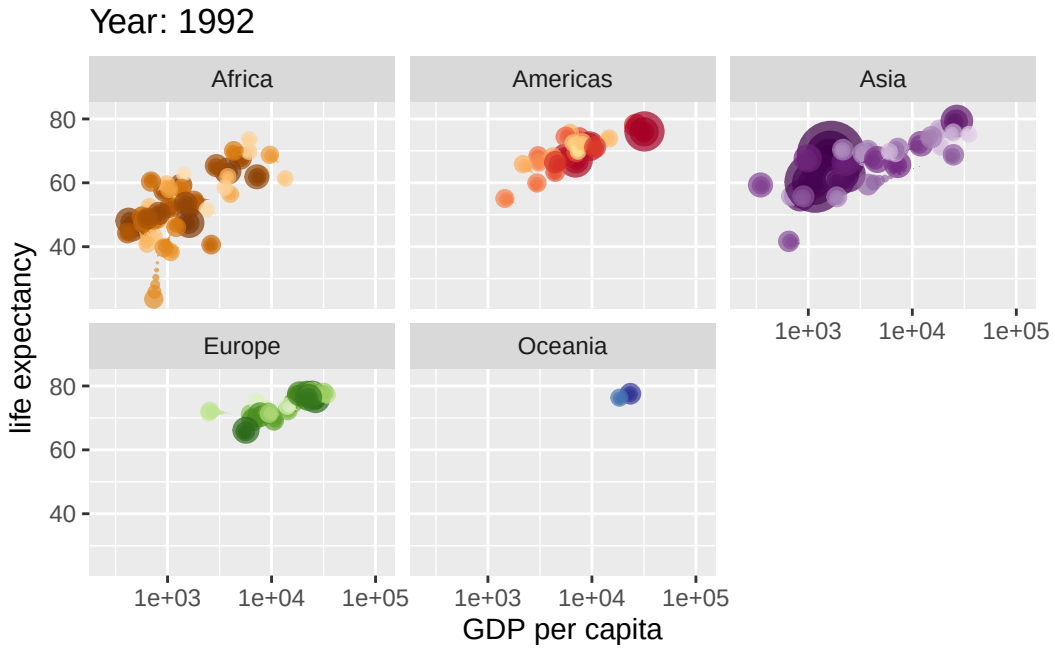
GDP per capita


Year: 1992
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Europe
Oceania
life expectancy
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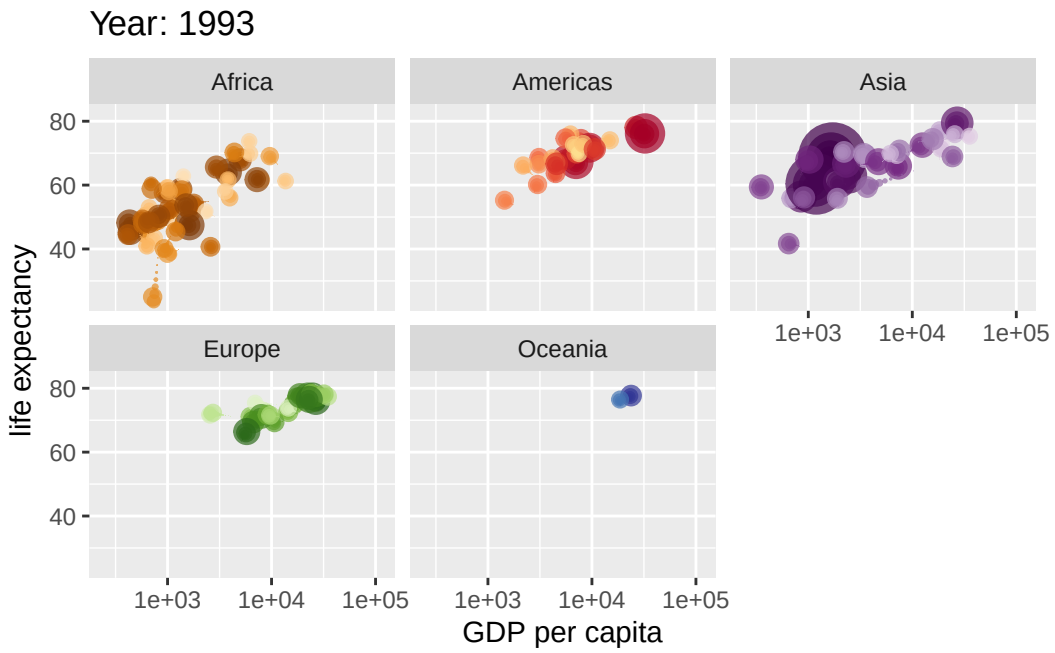
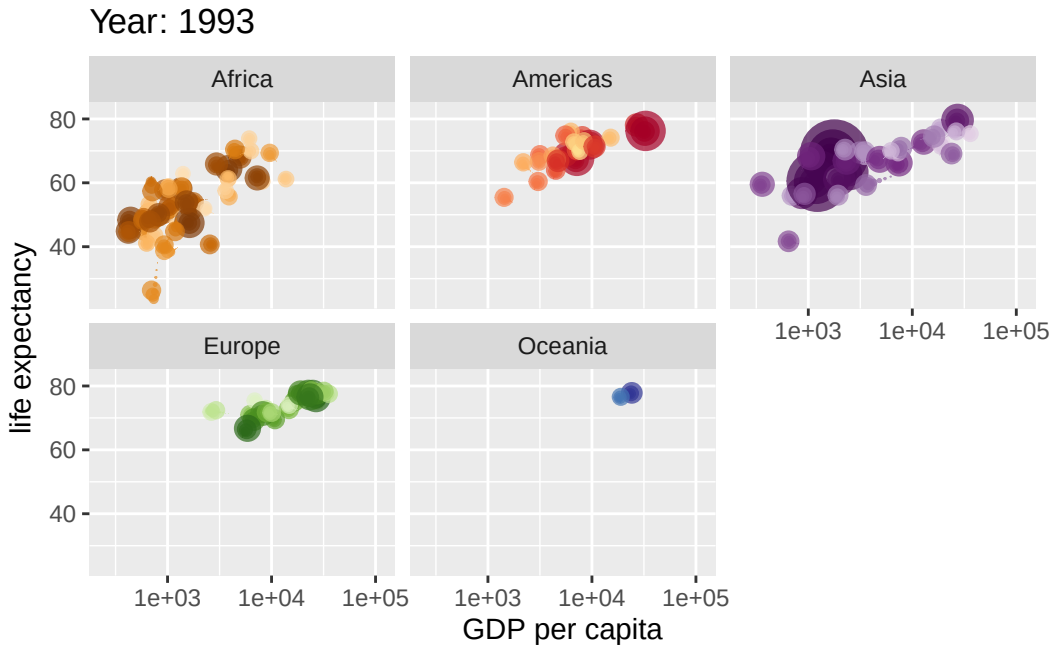
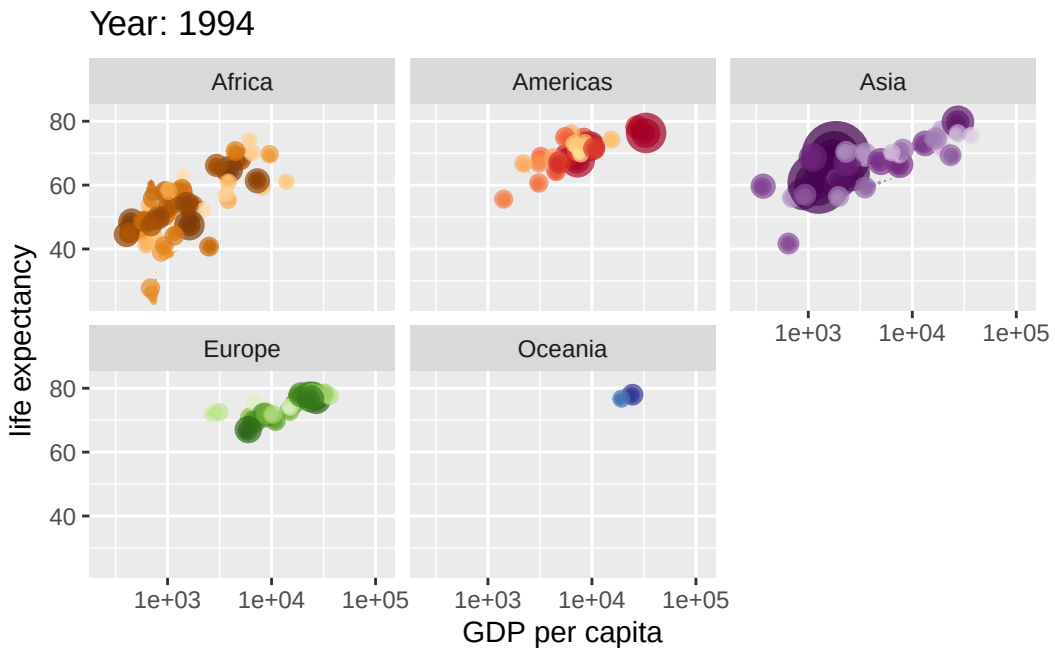
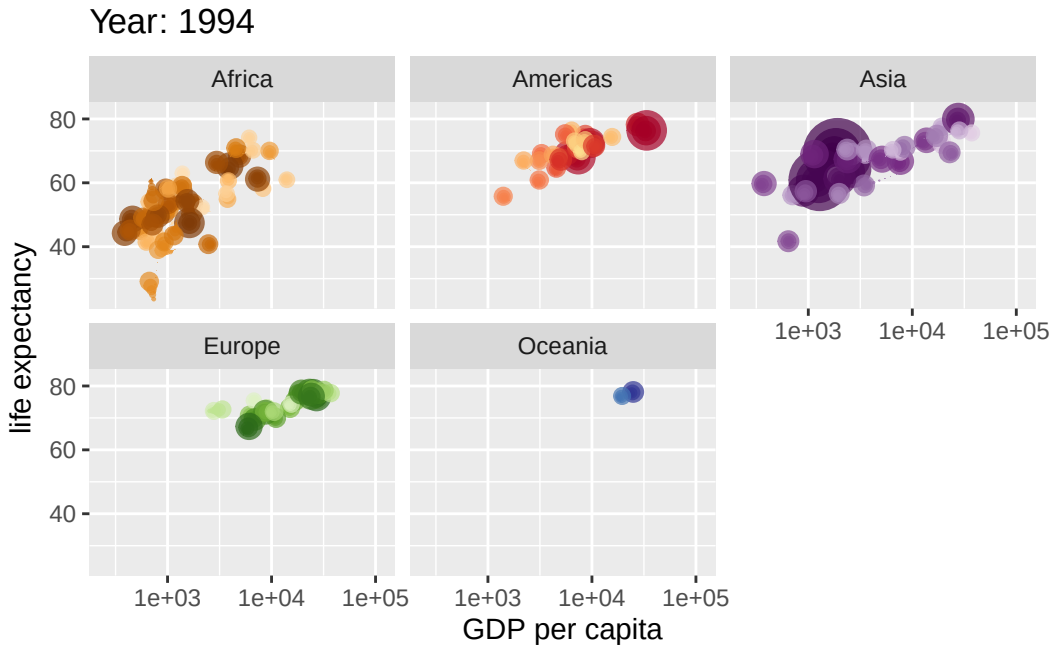
1e+03
1e+04
1e+05
GDP per capita


Year: 1993
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1993
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1994
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 1994
Africa
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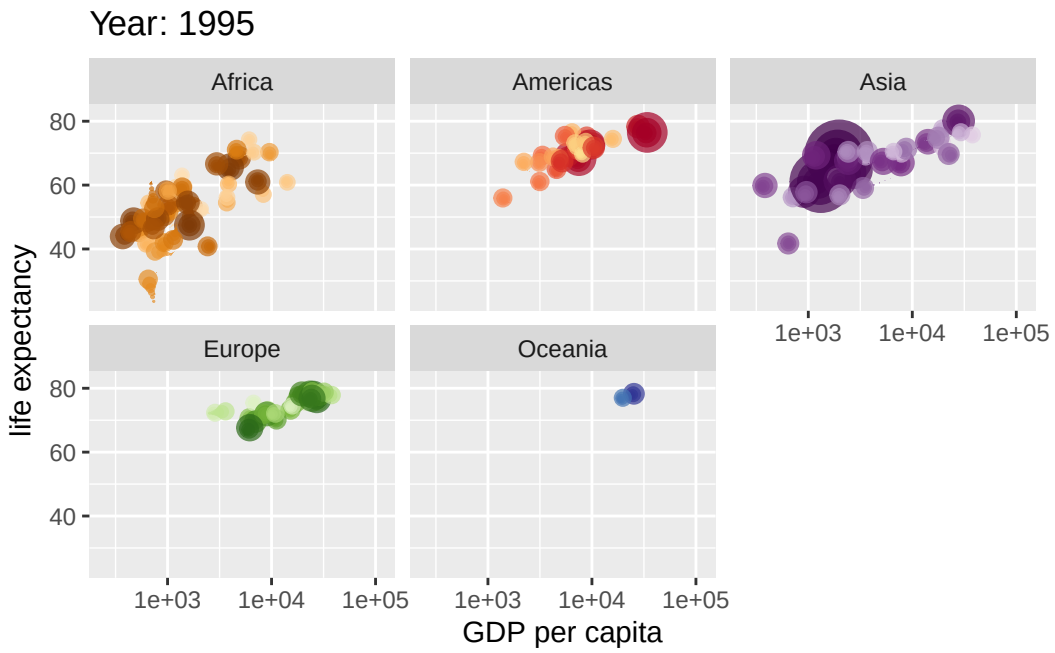
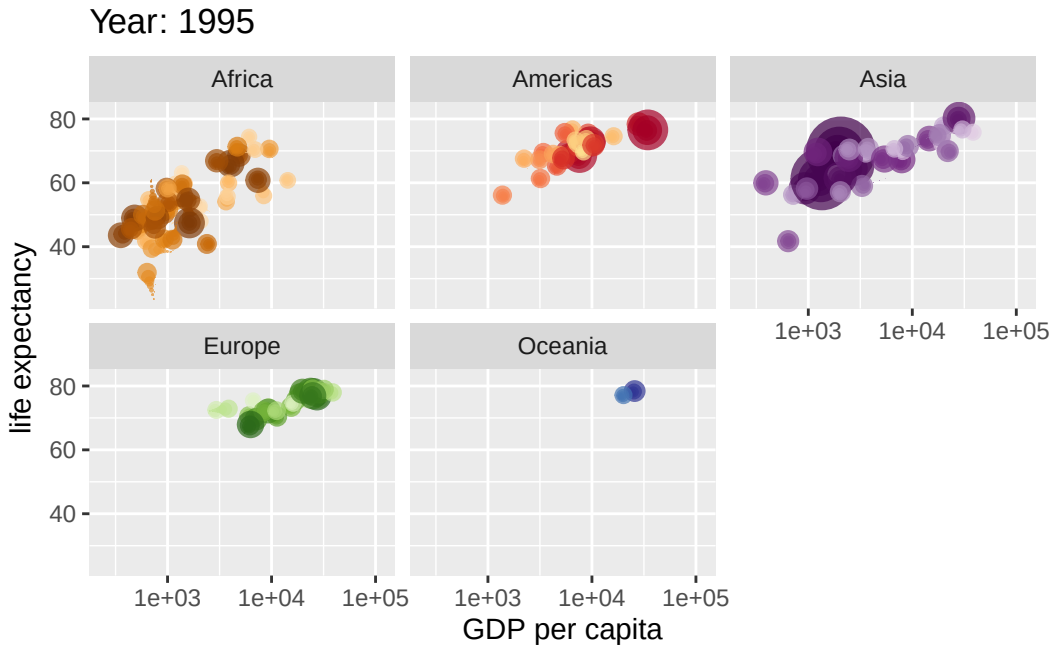
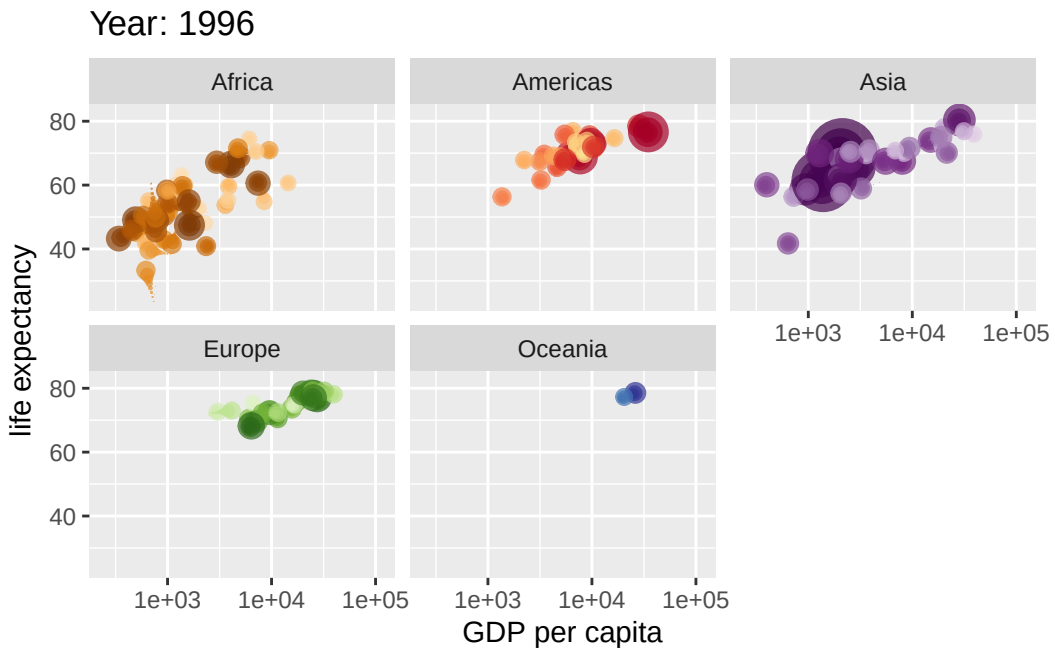
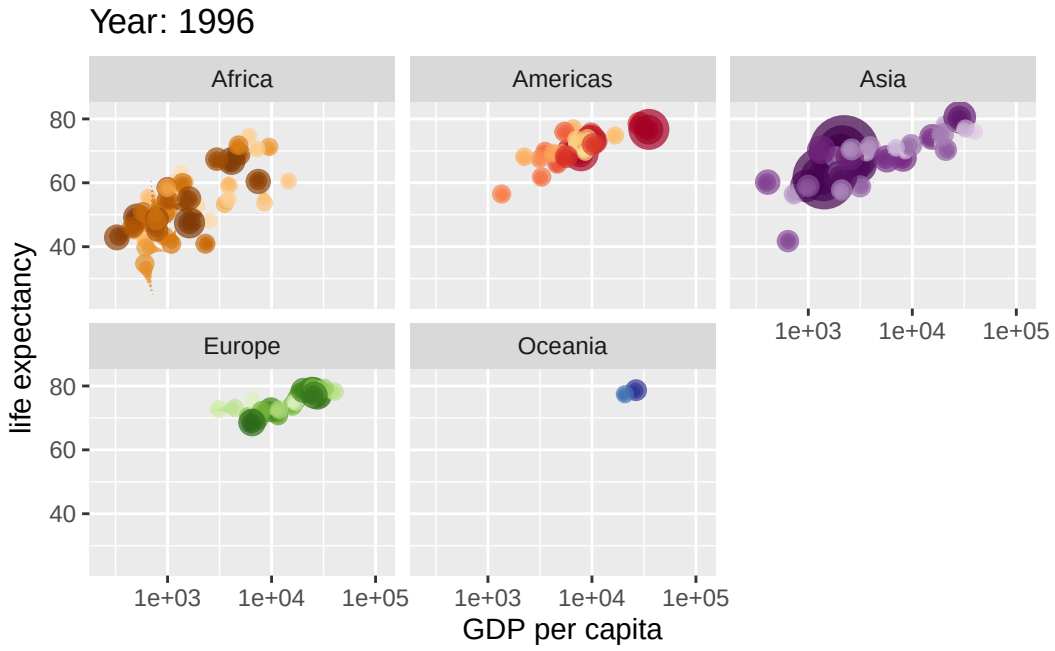
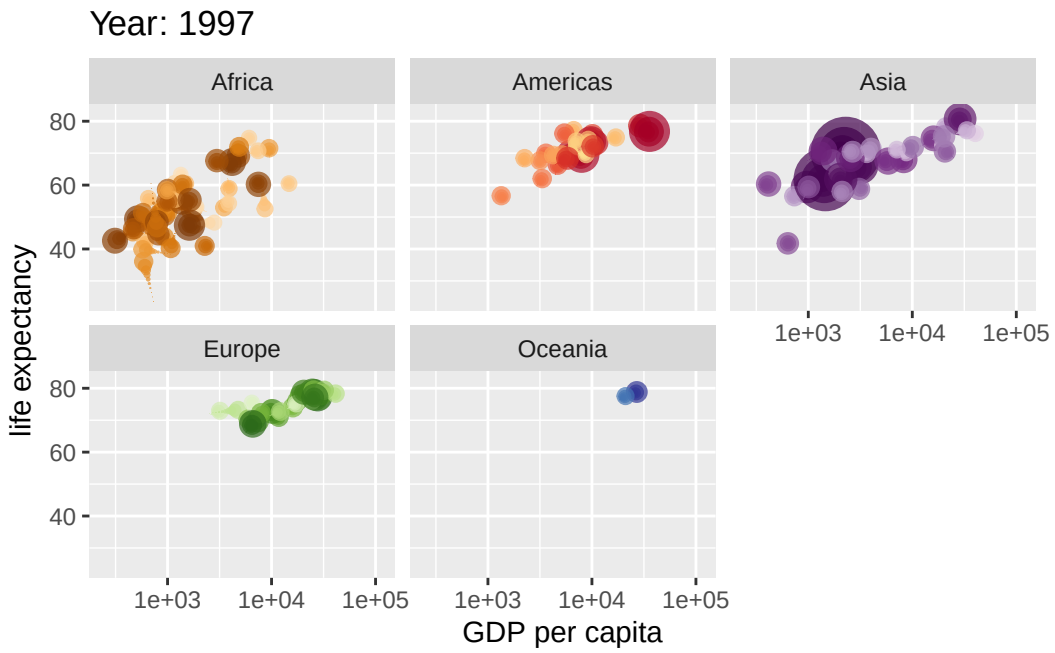
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

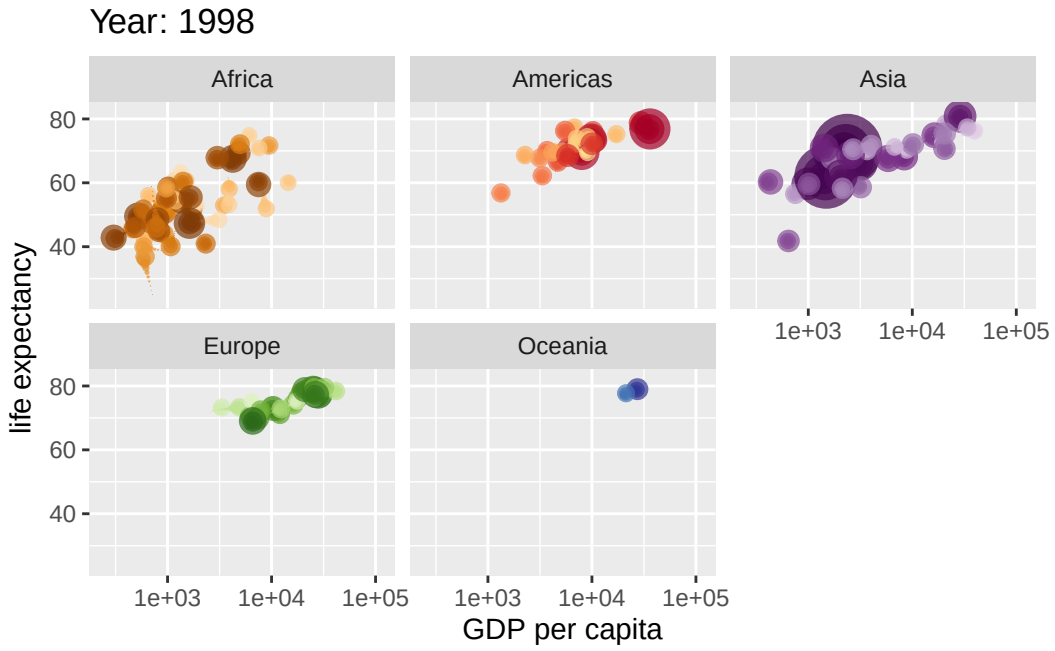

Year: 1995
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

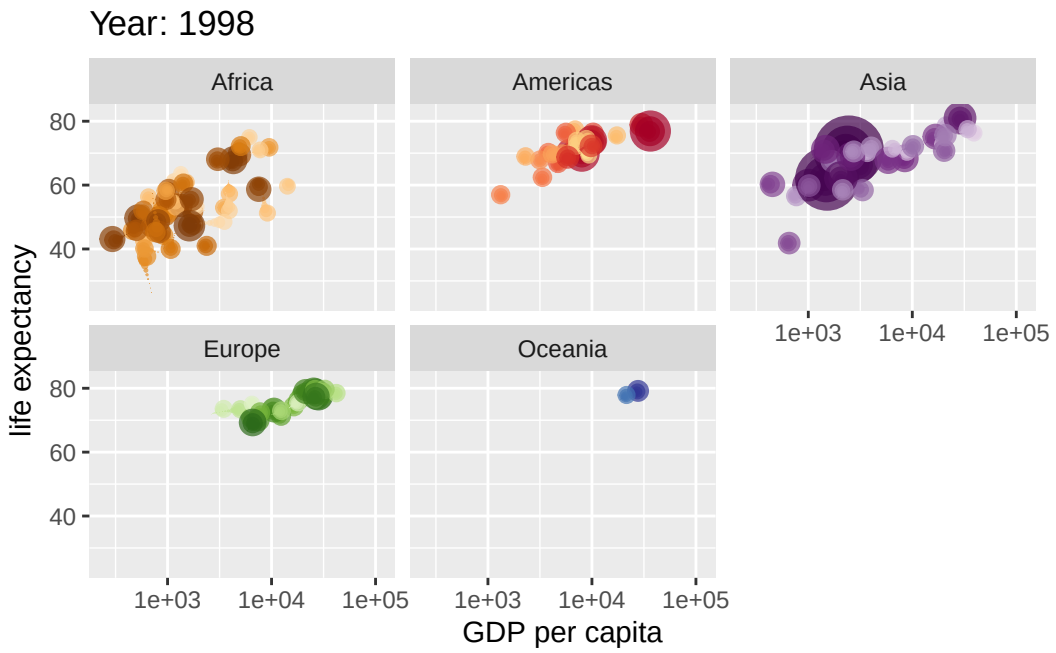

Year: 1995
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

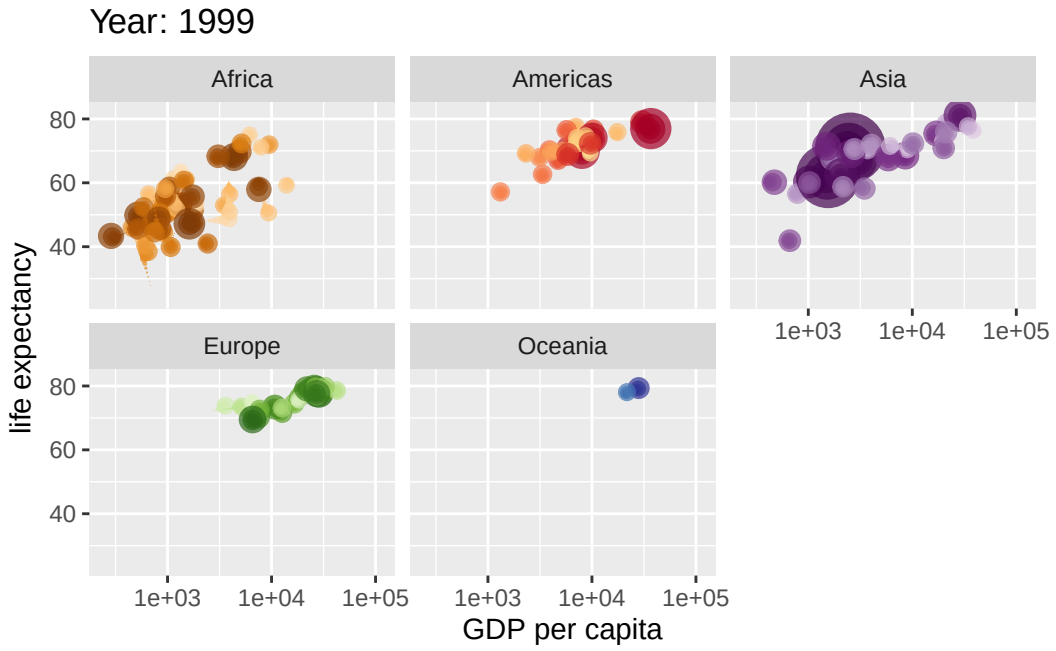

Year: 1996
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

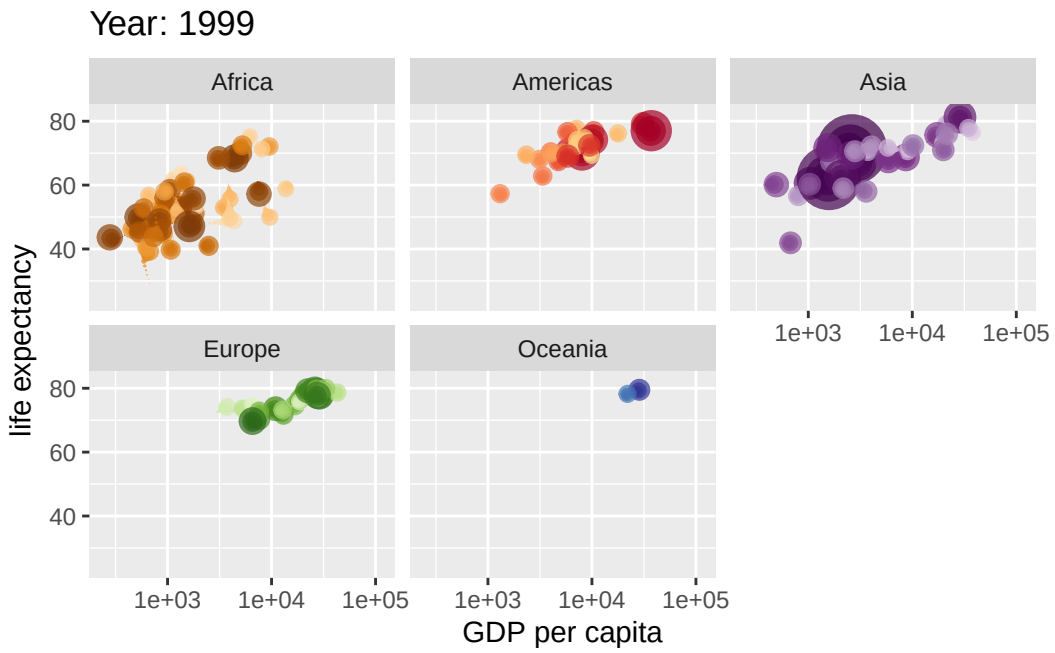

Year: 1996
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

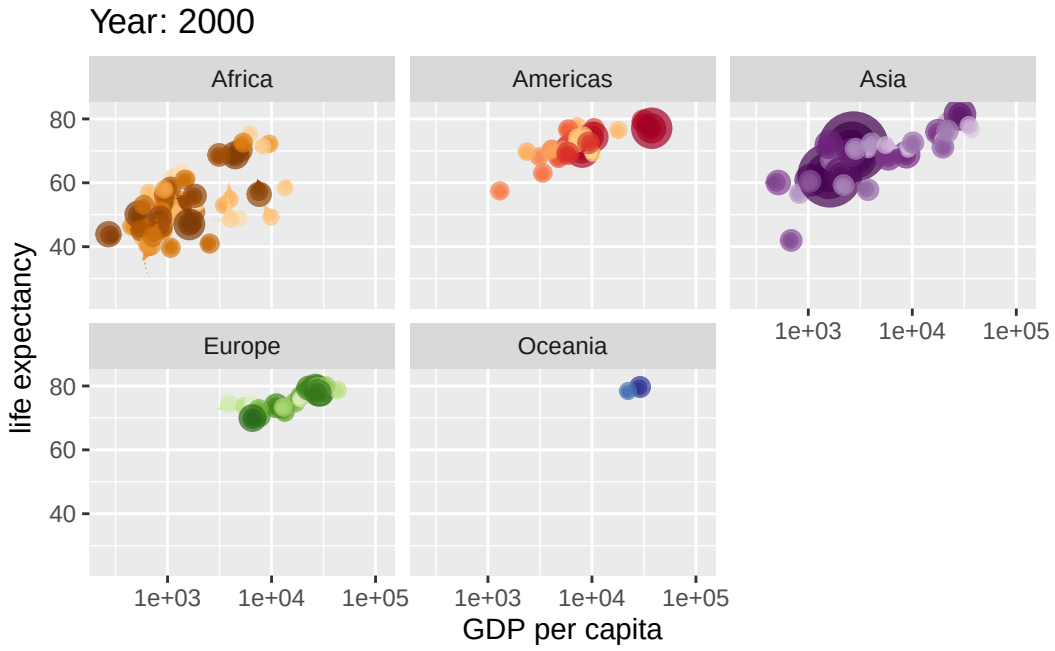

Year: 1997
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

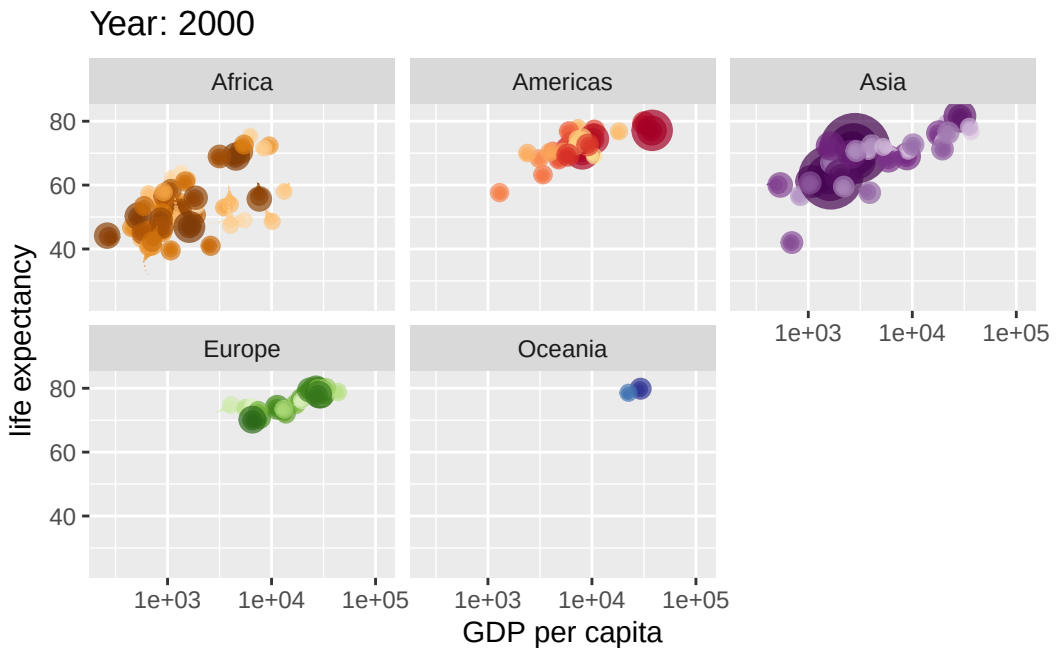

Year: 1998
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

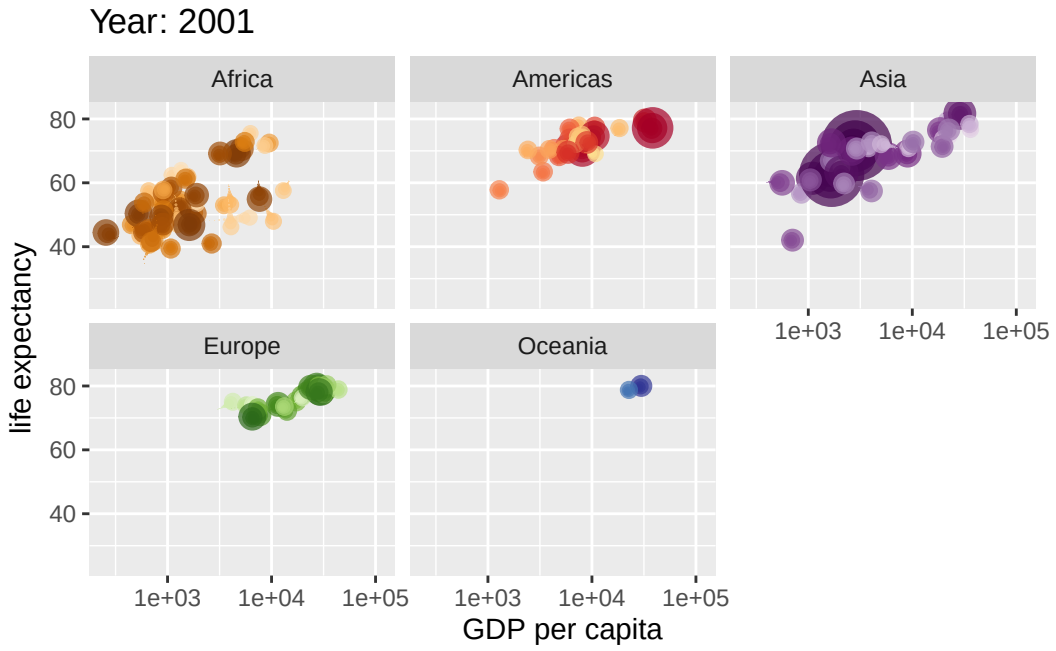

Year: 1998
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

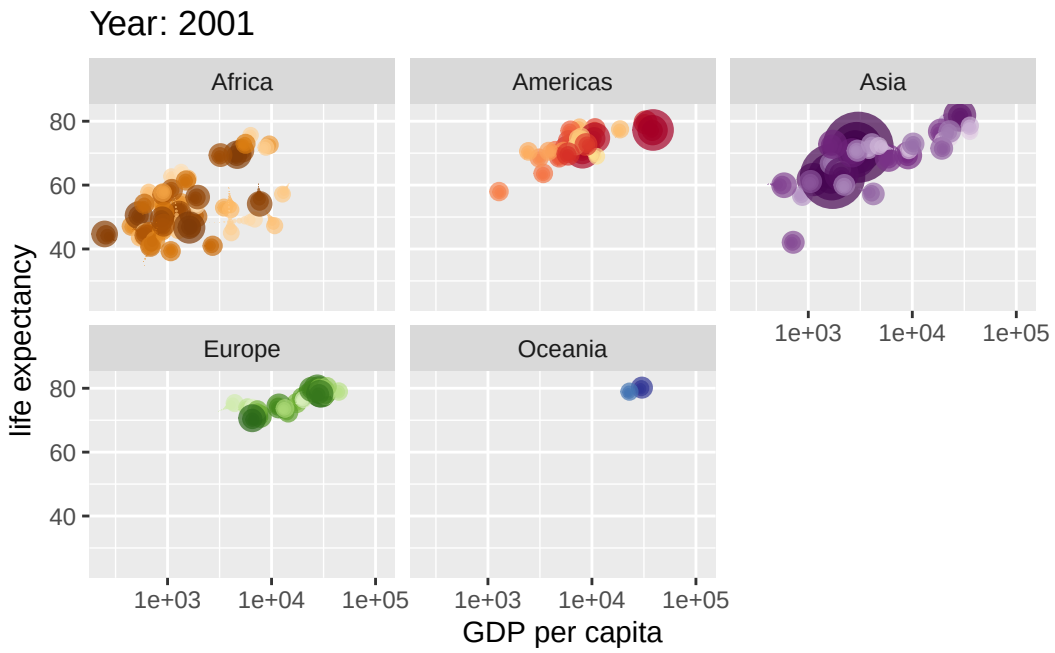

Year: 1999
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

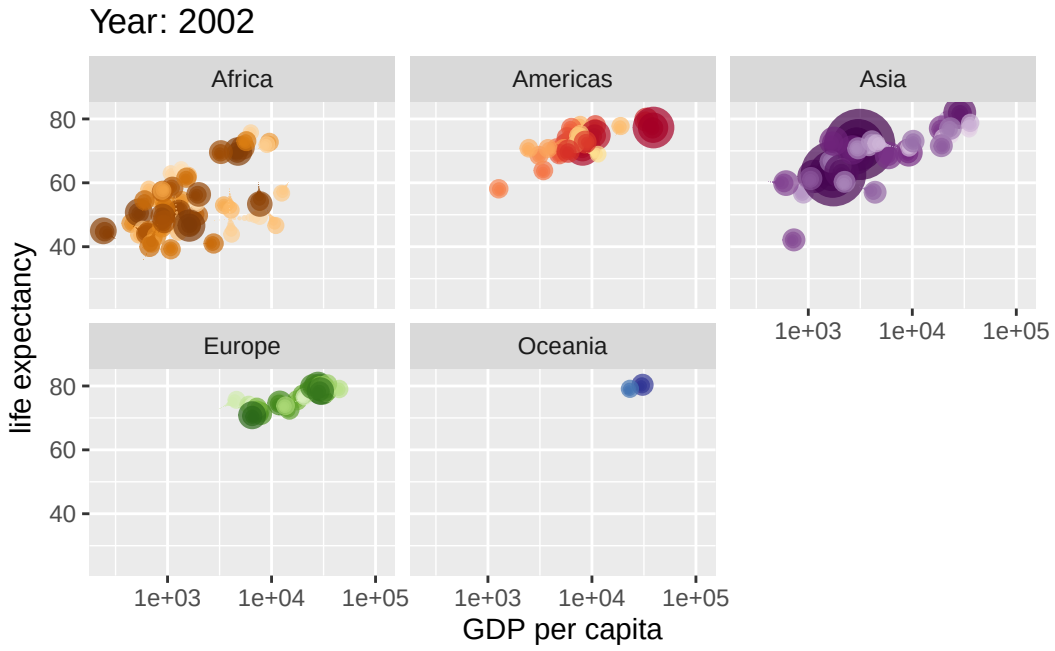

Year: 1999
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

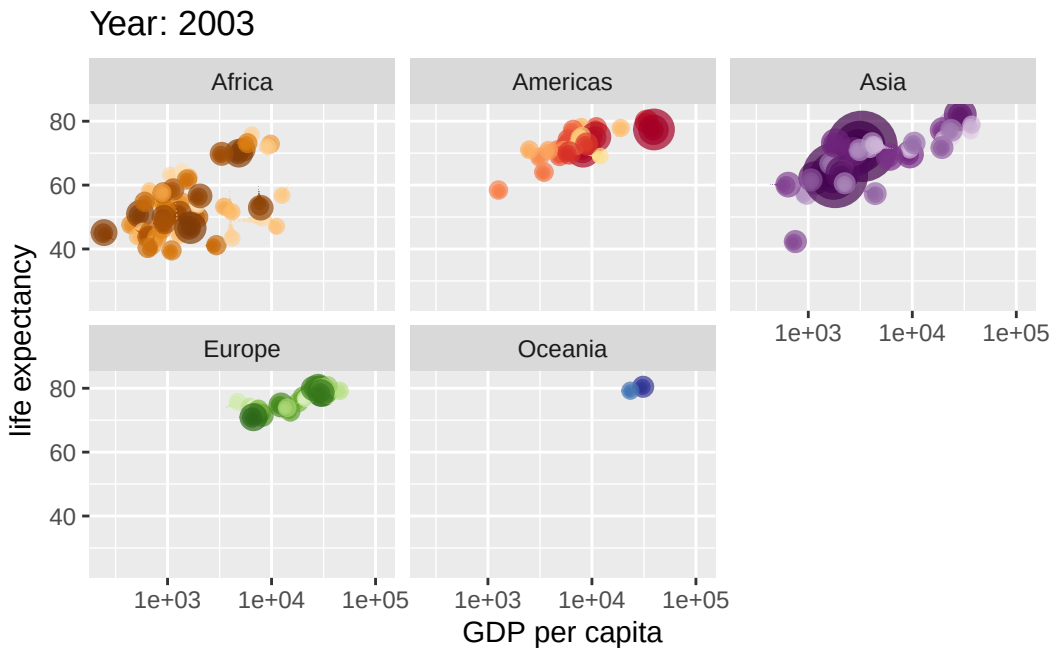

Year: 2000
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

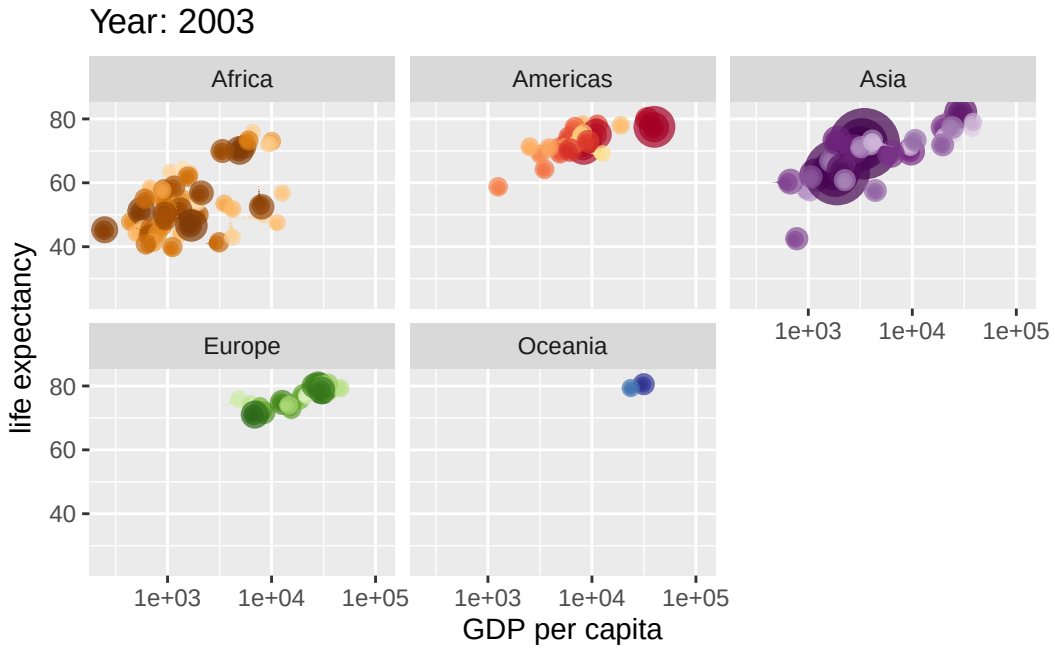

Year: 2000
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

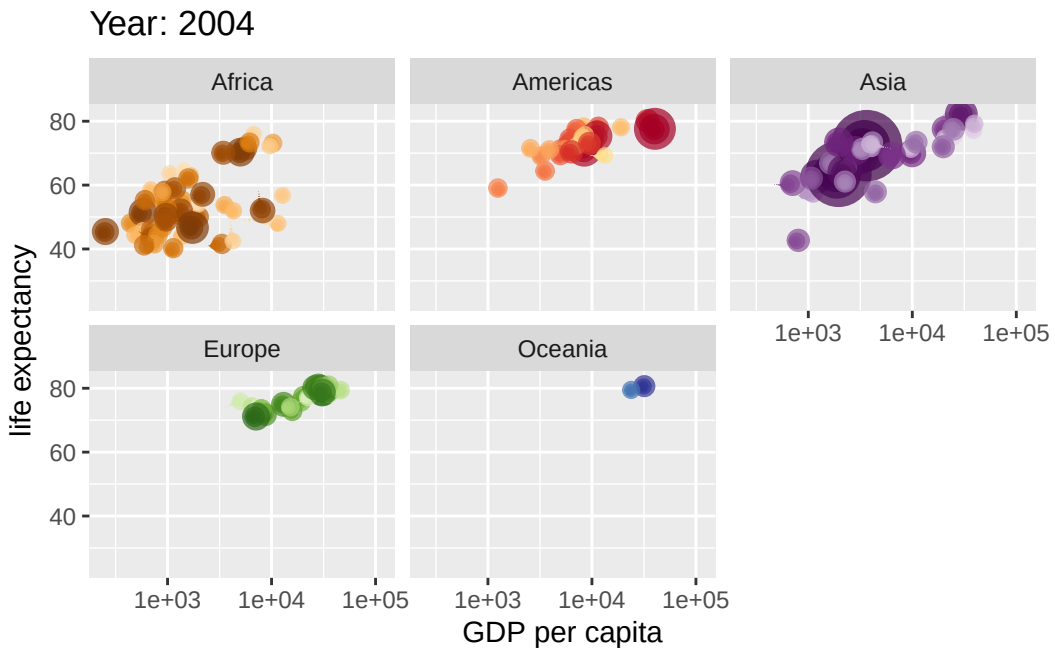

Year: 2001
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

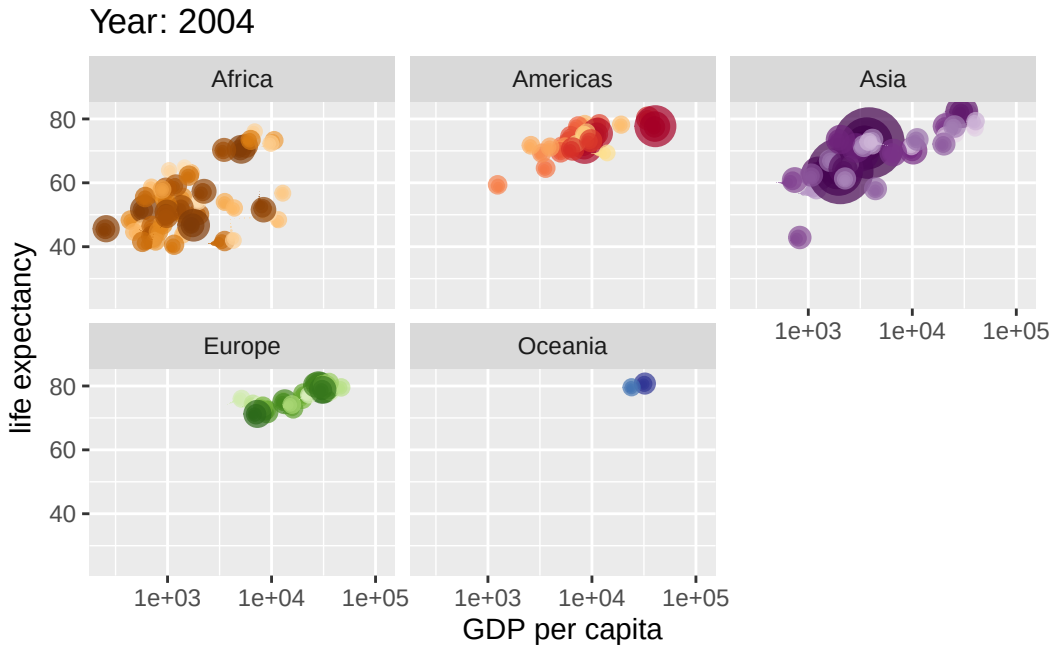

Year: 2001
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

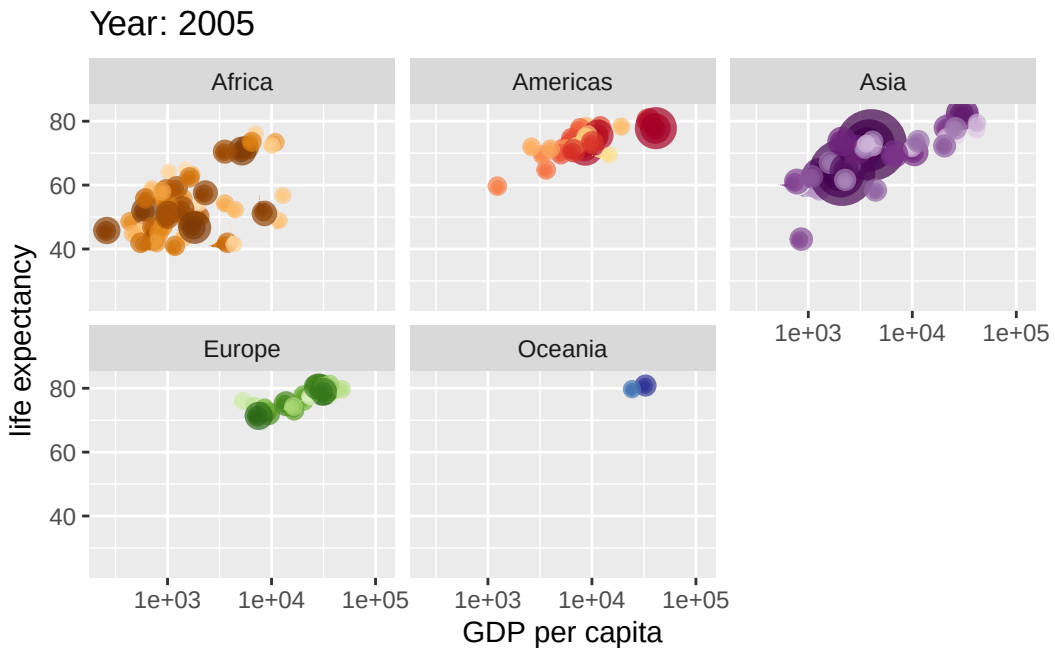

Year: 2002
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

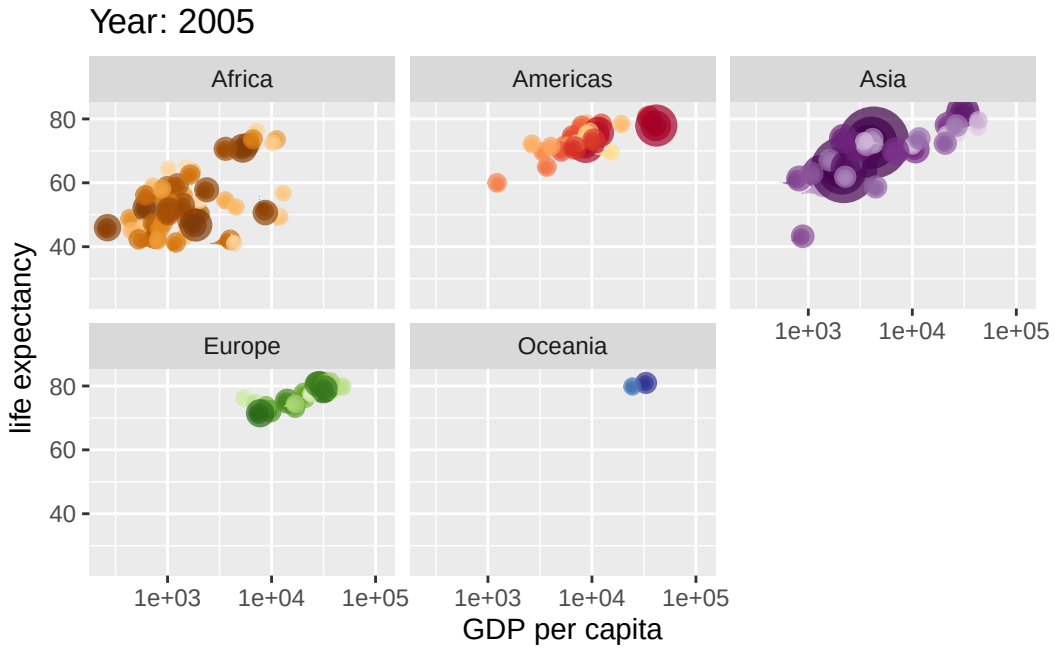

Year: 2003
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

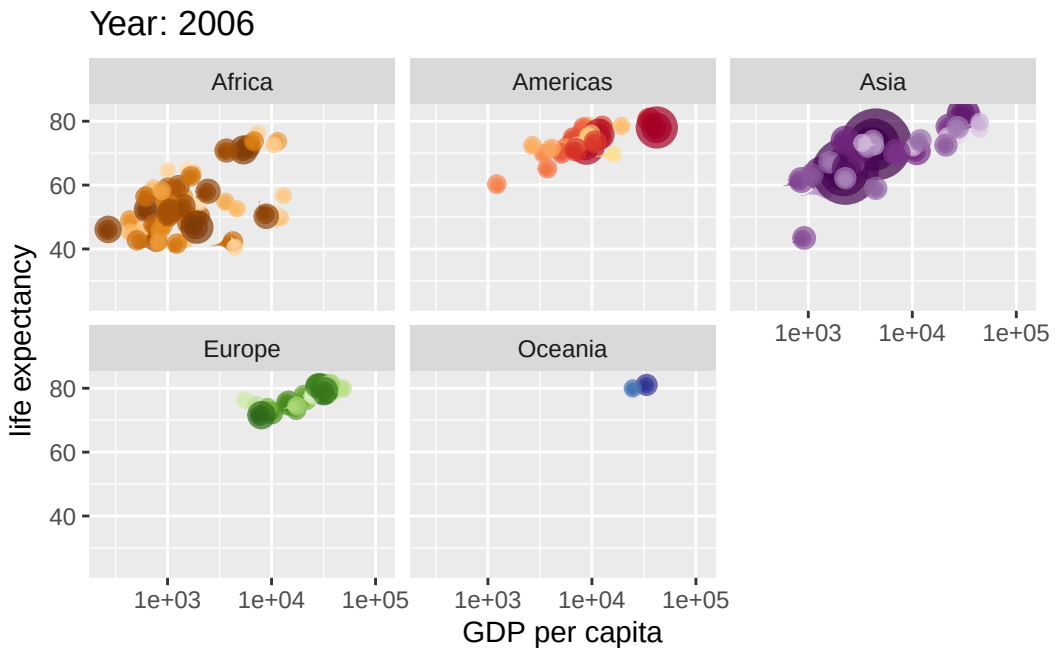

Year: 2003
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

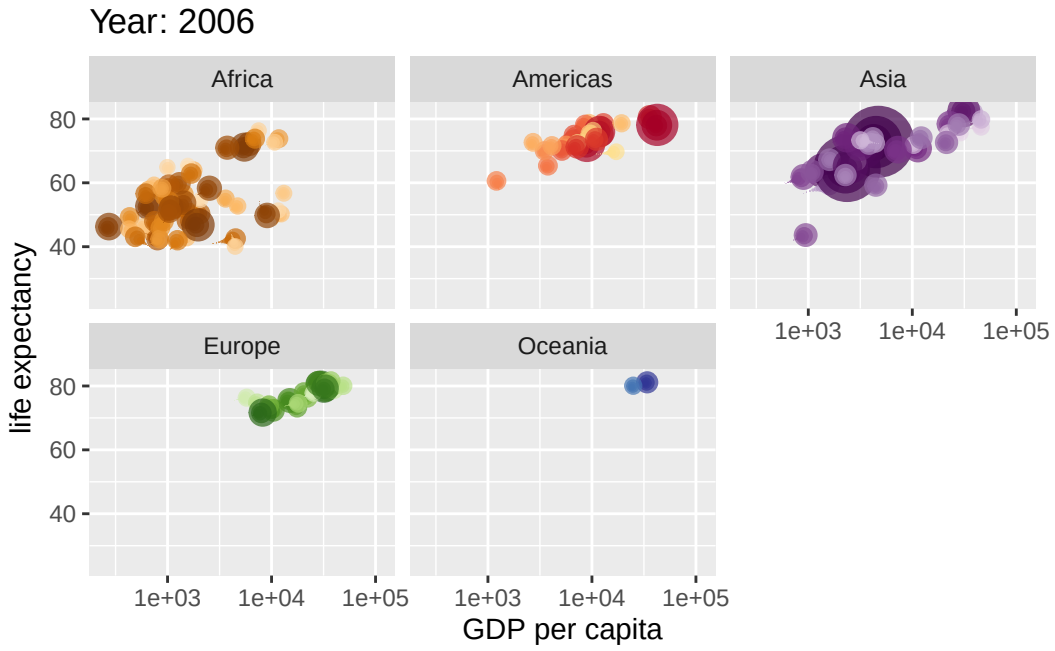

Year: 2004
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita

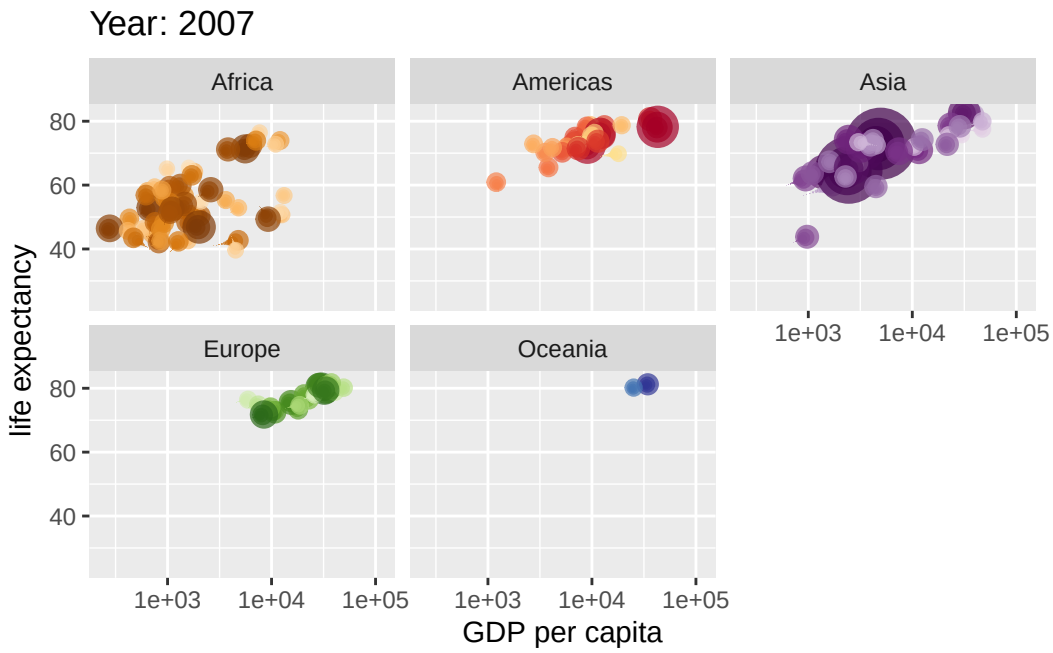

Year: 2004
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 2005
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 2005
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 2006
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita


Year: 2006
Africa
Americas
Asia
Europe
Oceania
life expectancy
80
60
40
1e+03
1e+04
1e+05
GDP per capita



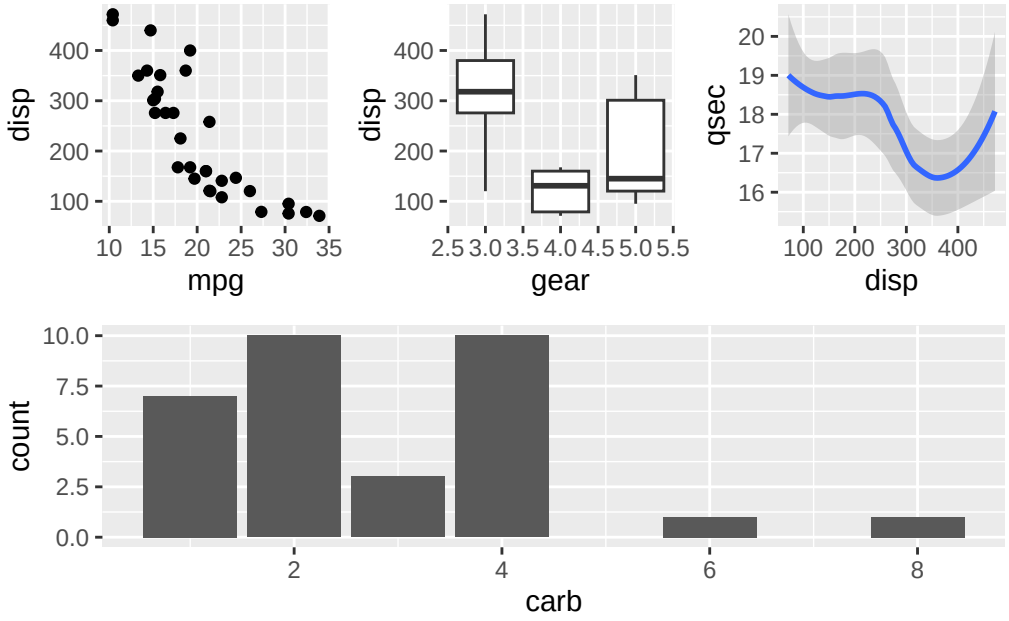

**Part 10: Combining Plots**

```
library(patchwork)

# Setup some example plots
p1 <- ggplot(mtcars) + geom_point(aes(mpg, disp))
p2 <- ggplot(mtcars) + geom_boxplot(aes(gear, disp, group = gear))
p3 <- ggplot(mtcars) + geom_smooth(aes(disp, qsec))
p4 <- ggplot(mtcars) + geom_bar(aes(carb))

# Use patchwork to combine them here:
(p1 | p2 | p3) /
      p4
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```


disp
400
300
200
100
10 15 20 25 30 35
mpg
disp
400
300
200
100
2.5 3.0 3.5 4.0 4.5 5.0 5.5
gear
qsec
20
19
18
17
16
100 200 300 400
disp
count
10.0
7.5
5.0
2.5
0.0
2 4 6 8
carb